

Climate-Smart Agriculture Investment Plan for Burkina Faso

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FOREWORD

Forthcoming

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ACRONYMS

AAA	Adaption of African Agriculture
AEZ	Agro-Ecological zones
AFOLU	Agriculture, Forest and Land-Use sector
AfDB	African Development Bank
AFR100	African Forest Landscape Restoration Initiative
BCR	Benefit Cost Ratio
CAADP	Comprehensive Africa Agriculture Development Program
CCAFS	Climate Change, Agriculture, and Food Security (CGIAR Research programme)
CGIAR	Consultative Group on International Agricultural Research
CIAT	International Center for Tropical Agriculture
CSA	climate-smart agriculture
CSAIP	climate-smart agriculture investment plan
CSI	Coping Strategy Index
ECOWAS	Economic Community of West African States
EU	European Union
FAO	Food and Agriculture Organisation
GDP	Gross Domestic Product
GEF	Global Environment Facility
GHG	greenhouse gas
FIE	Fond d'intervention pour l'environnement
HFIAS	Household Food Insecurity and Access Scale
IFAD	International Fund for Agricultural Development
IMPACT	International Model for Policy Analysis of Agricultural Commodities and Trade
IUCN	International Union for Conservation of Nature
KPI	with key performance indicators
MAAH	Ministry of Agriculture and Water Resources
MEA	Ministry of Water and Sanitation
MEVCC	Ministry of Environment, Green Economy and Climate Change
M&E	Monitoring and Evaluation
NAMA	Nationally Appropriate Mitigation Actions
NAP	National Adaptation Plan
NGO	non-governmental organization
NDC	nationally determined contributions
NPV	net present value
NTFP	non-timber forest products
PIF	Forest Investment Program
PNDES	National Economic and Social Development Plan
PO	policy objectives
PRP	Priorities Resilience Pays
PSNR	National Rural Sector Program
RCP	representative concentration pathway
REDD	Reducing Emissions from Deforestation and Forest Degradation Program
ROI	return on investment
RIMA	Resilience Index Measurement and Analysis
R&D	Research and Development

SDG	sustainable development goal
SE/CSNA	Executive Secretariat of the National Council for Food Security
SNV	Netherlands Development Organisation
SP/CPSA	Permanent Secretariat for the Coordination of Agricultural Sector Policies
SSP	shared socioeconomic pathway
TOC	Theory of Change
UNCCD	United Nations Convention to Combat Desertification
UNEP	United Nations Environment Programme
UNFCCC	United Nations Framework Convention on Climate Change
USAID	United States Agency for International Development
WASCAL	West African Science Service Center on Climate Change and Adapted Land Use
WB	The World Bank

EXECUTIVE SUMMARY

1. **Burkina Faso's economy is growing, yet much remains to be done. Burkina Faso has made significant progress in poverty reduction in the last few decades, but it faces high social and economic challenges. The country is one of the fastest growing in the world; the annual population growth rate of 3.1% is expected to drive the population to 42 million by 2050¹.** The 2018 national Gross Domestic Product (GDP) was US\$14.4 billion, or US\$731 per capita; this is below the sub-Saharan African average of US\$1,574, and far below the world average of US\$11,300². There are 19.75 million inhabitants of Burkina Faso, and the population density is relatively high--72/ km², versus the global average of 60/ km²³. While the poverty rate fell between 2003 and 2014, and GDP growth was 6% or higher during that period, Burkina Faso remains a highly impoverished country. While 70.6% of the total population is poor, 47.5% of the poor are rural, and poverty is linked to low agricultural productivity⁴.

2. **Agriculture is Burkina Faso's predominant land use, and agricultural land cover has expanded 91% since 2001, when it accounted for just 116.9 million ha⁵.** Much of this expansion has been onto unsuitable land; just 13.3% of the national land area is suitable for cultivation⁶. Farming on unsuitable land, climate change, and desertification mean that 34% of the land area is already degraded⁷. Significant increases in productivity and intensification would enable both economic prosperity and regeneration of degraded forests. Yet its geography is challenging for agricultural production; it is an arid, land-locked country with no major rivers.

3. **Agriculture is the main source of livelihoods, employment, and exporting in Burkina Faso.** Agriculture employs 80% of the country's workforce and contributes nearly 28.6% of national GDP⁸. Smallholder producers operating on 5ha or less account for about 80% of total agricultural production. Further, women account for over 50% of the agricultural workforce and produce more than 66% of the food consumed in-country. Along with gold exports, the Burkinabe economy is fueled by exports of cotton, sesame, peanut, and soy, and cash crops such as maize, sorghum, millet, cowpeas, and sugarcane⁹. The country's primary staple crops, including maize, millet, sorghum, and rice are crucial to food security¹⁰.

4. **Climate change is apparent and widespread in Burkina Faso. Declining rainfall and increasing temperatures are now a trend since the 1970s, affecting agriculture and natural resources (water and forest) substantially.** Other climate-related hazards such as droughts, floods, heat waves, locusts, and dust storms are common. Climatic variations are widespread in the Sahel, especially variability in rainfall and temperature¹¹.

¹ FOA, "Africa Sustainable Livestock 2050." <http://www.fao.org/in-action/asl2050/countries/bfa/en/>

² World Bank, "Burkina Faso Overview" <https://www.worldbank.org/en/country/burkinafaso/overview>

³ World Bank, "Burkina Faso." 1. <https://data.worldbank.org/country/burkina-faso>

⁴ Nguyen and Dizon 2017.

⁵ Knauer, K., Gessner, U., Fensholt, R., Forkuor, G., & Kuenzer, C. (2017). Monitoring agricultural expansion in Burkina Faso over 14 years with 30 m resolution time series: The role of population growth and implications for the environment. *Remote Sensing*, 9(2), 132.

⁶ William, *Country Assessment Studies on Climate Change, Agricultural Trade and Food Security in ECOWAS BURKINA FASO REPORT*.

⁷ World Bank Climate Change Country Profile, cited in *Creating Markets in Burkina Faso*, World Bank, 2019.

⁸ UC Davis "Burkina Faso" <https://gfc.ucdavis.edu/profiles/rst/bfa.html>

⁹ UC Davis "Burkina Faso" <https://gfc.ucdavis.edu/profiles/rst/bfa.html>

¹⁰ FAO (2010). *Monitoring African Food and Agricultural Policies Project (MAFAP) Commodity Selection in the Five pilot countries*. Document attached

¹¹ World Bank, "Vulnerability, Risk Reduction, and Adaptation to Climate Change: Climate Risk and Adaptation Country Profile Burkina Faso"

5. **Burkina Faso's agriculture sector is very vulnerable to negative impacts from climate change.** High temperatures and reduced rainfall will catalyze desertification, ecosystem deterioration, and concomitant losses of arable land. Higher temperatures and intense rainfall could instigate outbreaks of crop and livestock pests and diseases. Postharvest losses may increase due to erratic and unpredictable weather conditions; low agricultural diversification intensifies the risk of crop failure. Climate change will intensify vulnerability in the north, since the region is particularly hot and dry, has lower quality agricultural lands, and is most susceptible to drought, soil erosion, and land degradation.

6. **Climate change has already decreased yields and increased the demand for water resources.** Climate shocks and the lack of sustainability of land use threatens the food security of the country¹². Extensive droughts are likely to cause crop failure and death of livestock especially because Burkina Faso operates under a rain-fed production system¹³. Burkina Faso has adequate water resources¹⁴ and approximately 233,500ha of irrigable land, but only about 34,500ha were under irrigation in 2014. With variability in rainfall and temperature, water resources are affected. Increased temperature and reduced rainfall mean wells and other water sources dry up. Projections indicate future water scarcity. Floods are less likely to cause damage compared to droughts, however a successive combination of both reduces the capability of food storage, resulting in chronic food insecurity.

7. **Climate smart agricultural investment plans (CSAIP) are a way for countries to plan for the future, and Burkina Faso is developing this national CSAIP as part of the Adaptation of African Agriculture (AAA) Initiative.** AAA is a coordinated multi-country initiative that was launched at the United National Climate Change Conference 22nd Conference of Parties in Marrakech, Morocco (2016). The development of this CSAIP was led by the Burkina Faso Ministry of Agriculture with strong input from a range of stakeholders, and the West African Science Service Center on Climate Change and Adapted Land Use (WASCAL) working as a key technical partner under the technical leadership of International Center for Tropical Agriculture (CIAT), World Agroforestry, and the CGIAR Research Program on Climate Change, Agriculture and Food Security (CCAFS). Funding and technical support were provided by the World Bank. Strong government partnership and stakeholder engagement are the foundation of this CSAIP.

8. **Burkina Faso's national CSAIP prioritized a set of nine investments and actions needed to boost crop resilience and enhance yields for nearly 1.7 million beneficiaries and their families, helping them adapt to climate change and reduce climate change causing greenhouse gas emissions as a co-benefit.** The CSA investments were identified based on a situation analysis of Burkina Faso's plans and policies, the current context of agriculture, and analysis and scenario development of climate change impacts on different crops and livestock. Burkinabe stakeholders analysed potential investments in a workshop and prioritized those presented in this CSAIP. Stakeholder experts represented a variety of organizations, including government ministries, institutions, research organizations, farmer groups, and international development organizations. An expert stakeholder workshop in Ouagadougou used an iterative, qualitative and quantitative prioritization process to rate and review 38 proposed CSA priorities. They also grouped the long list of investments by agro-ecological zones. The long list of potential investments was reduced from 38 to a shortlist of nine priority CSA investments.

9. **This CSAIP also includes elements of program design and implementation, with economic analysis, assessments of barriers and opportunities of individual priority investments, financing**

¹² USAID

¹³ FAO, "Food Security and Humanitarian Implications in West Africa and the Sahel"

¹⁴ William, *Country Assessment Studies on Climate Change, Agricultural Trade and Food Security in ECOWAS BURKINA FASO REPORT*.

avenues, and policy analysis. The plan highlights both the high-level economic benefits to the country, as well as the financial viability of investments, related projects within the country, findings from research on similar crops, and other implementation issues to be addressed. The process used to develop this plan also supports engagement and capacity strengthening.

10. Climate modeling done for this CSAIP shows that climate change could exacerbate biophysical damages for key food security and commercial crops. Climate impacts do not affect all commodity groups uniformly, but they do affect most commodity groups. Most cereals, especially maize, exhibit high vulnerability to climate change no matter what scenario is picked, from losses of between 9% to 12% in 2030 to over 18% to 22% in 2050 compared to a no climate change baseline, while other cereals do better. Other crops, especially fruits, vegetables, soybeans, groundnuts, oils seeds and sugar cane are likely to perform badly over the long term. Cotton, one of Burkina Faso's most important exports, shows declines of between about 4% to 7% under all scenarios.

11. The CSAIP climate modeling shows that some crops are climate resilient, so CSA emphasis should be on promoting practices that emphasize and maintain their resilience. Sweet potato and yams are nutritionally important and show resilience to a warmer future. Cowpeas and potatoes also show relative levels of resilience to climate change impacts under all representative concentration pathways (RCPs) for both time periods. As noted previously, other cereals and rice also appear to be relatively resilient.

12. The prioritized investments include five national activities, and four others that are regionally distributed throughout Burkina Faso. The National investments are intended to provide services, infrastructure and capacity across the country to support the outcomes of the CSAIP. The six regional investments are focussed on specific value-chains and regional issues to be addressed through targeted regional investments

13. The five national investments that build capacity, financing, resilience, support mitigation and offer human health benefits are:

- **Sustainable On-Farm Biogas Production:** increase access to and knowledge of sustainable domestic energy sources in order to conserve wood resources, reduce greenhouse gas emissions, and reduce poverty, food insecurity, and health threats to rural communities.
- **Sustainable Intensification of Livestock Production:** foster climate resilience in Burkinabe livestock systems for improved food and nutritional security and economic outcomes.
- **Financial and Insurance Products:** foster smallholder ability to invest in CSA innovations through good access to robust financial services, including credit, loan, insurance and risk instruments, savings and payment services.
- **Forest, Agroforest, And Garden Production for Climate-Smart Diversification:** foster climate-resilient livelihoods and food sources for women and youth smallholders through home gardens, agroforestry, and non-timber forest product harvesting and value-addition
- **Building Capacity in Climate Smart Agriculture:** fully integrate national climate-smart priorities into Burkina Faso's agricultural research and extension programs, and build robust extension mechanisms for delivering timely, practical climate-smart information to farmers and other stakeholders through highly accessible channels.

14. Four climate-smart investments were prioritized to support adaptation of agricultural production systems by introducing a variety of climate-smart practices into the different investments.

- **Sustainable Management of Water Resources and Irrigation:** Fully leverage Burkina Faso's water resources in sustainable ways to improve productivity, food and nutritional security, climate resiliency, and ecological health in the Nord Soudanian region for 100,000 farmers and their families.
- **Developing Climate-Smart Organic Value Chains:** Develop a burgeoning niche market to create viable careers and shorten supply chains in peri-urban areas in the Nord and Sud Soudanian regions for 60,000 farmers and their families.
- **Developing Resilient Oil-Protein Value Chains:** strengthen the quality, yield efficiency, and value chains of oil-protein crops to sustainably meet domestic demands and support international market development in the Nord Sahelian, Nord & Sud Soudanian regions for 240,000 farmers and their families.
- **Integrated Soil Management for Agricultural Productivity & Environmental:** Provide producers and extension agents with location-tailored information on soil characteristics and best management practice recommendations, as well as the tools, products, partnerships, and policy environment to implement those recommendations for 200,000 farmers and their families in the North Sahel and North and South Sudan regions.

15. To understand the CSA potential of each investment it is important to understand how vulnerable the sector is to climate impacts, how the priority investment is important, and what the future scenario looks like for that commodity or cropping system with and without the investments, and what the investments would accomplish.

Climate Change	Project Importance	Scenario WITHOUT Investment	Scenario WITH investment	Investment objectives
Diversification through FORESTS & GARDENS				
Medium resiliency	Livelihoods & food sources for women & youth thru garden, agroforestry, & non-timber forest products (NTFPs)	Low diversification increases smallholder vulnerability to economic & food insecurity under climate change; continued deforestation & environmental degradation; untapped international market opportunities	Smallholder food & economic resilience, particularly for women & youth; forest conservation & enhanced ecosystem services; national economic growth through value chain development of high-value export products.	• Food security • Resilience • Value chains • Sustainability • Mitigation • Livelihoods • Inclusivity
Sustainable management of WATER RESOURCES				
Medium resiliency	Protect & sustainably leverage water resources to improve climate resiliency.	Agriculture remains exposed to increasingly unpredictable precipitation. Crop & animal loss, food & economic insecurity. Conflict over scarce resources. Watershed degradation.	Efficient water capture allows storage & management. Irrigation stabilizes crop & animal production, food supplies, & market prices. Reduced erosion & flooding, watershed protection.	• Food security • Resilience • Value chains • Transformation • Sustainability • Environmental quality
Foster climate-smart agriculture with FINANCE SERVICES				
Foundational resilience through essential services	Enable smallholder investment in CSA through loan, insurance, & other basic services.	Smallholders have limited capacity to invest; cost of maintaining status quo increases with climate impacts; decreased food & economic security; downward poverty cycle	Credit & loan enable investment in system transformation; insurance & risk distribution instruments offer protection from loss, enabling farmers to take the risk of changing to new system	• Food security • Resilience • Value chains • Trade • Sustainability • Transformation • Mitigation
Building climate-smart CAPACITY				
Foundational resilience through essential services	Establish sustainable systems for developing & disseminating CSA innovation	Farmers make short-term decisions when facing uncertainty or high risk; perpetuates poverty cycle, degrades resources, & increases vulnerability.	Robust research, extension, & information services support a suite of site-tailored solutions that drive resilience, productivity, environmental management	• Food security • Resilience • Value chains • Sustainability • Transformation • Mitigation
Develop value chains to meet demand for ORGANIC				
Highly vulnerable	Leverage untapped domestic & international markets for diversification	Low sectoral diversity, dearth of livelihood & income opportunities, particularly for youth & women	Diversification improves economic resilience by creating livelihoods all along several organic production value chains; environmental impacts reduced; increased peri-urban food access	• Food security • Resilience • Value chains • Sustainability • Livelihoods • Economic resilience
Developing resilient value chains for OIL-PROTEIN CROPS				
Medium resiliency	Strengthen value chains to build livelihoods, reduce import dependency,	Soybean yield plummets with climate change; soil health decline; income declines; reliance on imports	Sustainable productivity, reduced reliance on imports, improved national food security, export market	• Resilience • Value chains • Sustainability • Trade

	& build export markets		development, new livelihood opportunities.	• Food security • Economic resilience
Agricultural productivity & environmental restoration through integrated SOIL MANAGEMENT				
Foundational resilience through essential services	Provide location-tailored soil information, recommendations, & tools & policy to support implementation	Agriculture & extreme weather events degrade soils; productivity decreases; cost of production rises with increased fertilizer inputs; increased greenhouse gas (GHG) emissions exacerbate climate change; environmental degradation	Soils are resilient to climate stress; sustainable productivity; food & economic security; GHG storage; strong environmental services; optimum fertilizer usage	• Food security • Resilience • Sustainability • Transformation • Mitigation
Sustainable on-farm BIOGAS				
Resilient under climate change scenarios	Leverage farm waste to produce clean energy, support health & reduce GHGs	Continued deforestation, High GHG emissions, unreliable & costly cooking fuel & fertilizer purchases, Indoor air pollution & related diseases affecting especially, women & children	Reduce deforestation & GHG emissions; Clean cooking fuel & fertilizer produced on-farm for free; improved crop yields; Improved health outcomes.	• Sustainability • Mitigation • Soil fertility • Health
Climate-resilient LIVESTOCK				
Vulnerable to climate change	Create climate resiliency to ensure future food security & income	Reduced productivity as heat stress & mortality increase, feed sources decline. Increased pastoralist-smallholder conflict.	Improved health & productivity; food & income security; reduced farmer-pastoralist conflict; reduced GHG emission intensity; reduced environmental degradation	• Food security • Resilience • Mitigation

16. **The financial and economic analysis shows that all nine investments increase productivity by at least 18% on average and have positive economic benefits for beneficiaries, while the whole portfolio reduces greenhouse gas emissions.** The chances of a positive net present value (NPV) are a good way of assessing resilience to risk. Looking at the priority investments without risks included, all have a better than 50% chance of a positive NPV. Four of the nine investments have a better than 50% chance in the face of uncertain climate and social risks. If all the priority investments were implemented in Burkina Faso, they would sequester 4.31 Mt CO₂ overall. Only one of the investments, Livestock Intensification, will be a source of net greenhouse gas (GHG) emissions, but it reduces emissions intensity from livestock (cows, goats and sheep).

17. **Of the nine priority investments analysed, some stand out in how strongly they support the three fundamental CSA pillars: productivity, resilience and mitigating greenhouse gas emissions.** Forests and Gardens appears in all three categories and is the strongest performer on resilience with risks and on reducing emissions. Based on their rank order, Integrated Soil Management, is the next strong investment across the CSA pillars, followed by Biogas, which is the third strongest for yield and reducing emissions, and then Capacity Building. Oil-Protein Crops and Finance and Insurance both appear in two of the three categories.

18. **The nine investments analyzed are predicted to provide significant benefits for farmers in Burkina Faso.** The cost-benefit analysis is presented both with and without risks, so the potential magnitude of foreseeable risks can be understood. Without risks included in the model, net present value (20-years) ranges from \$16.9 million with the Biogas program to more than \$196 million for Forests and Gardens. All investments are expected to improve the income and productivity of farmers. Investment in improved agricultural practices increases farm outputs, whether by introducing new breeds of livestock, drought tolerant seeds for crops or providing information for better farming decisions (e.g., knowledge, advisory and climate services). Each investment will present positive returns relative to their costs, based on the analysis. The analysis suggests there is potential for positive returns with each of the investments --but there are risks. Return on investment (ROI) averages 1.87% across the nine investments, in the *No Risk* and *Climate and Social Risks* scenarios, and 0.78 with risks included. Yet all of the ROIs are positive when Risk is excluded.

19. **It is important to view these proposed project investments through different lenses, for their potential transformative value.** The strongest performing projects for the three CSA pillars (Forests and

Gardens, Oil-Protein Crops, and Capacity Building) are those with the lowest mean BCR. Some types of investors (e.g. impact investing) may look for strong positive returns, yet these may not be appropriate for this CSAIP because investments in smallholders is near universally below this ROI level and these investment plans intend to catalyze change and improve the capacity and suitability so that others investors may be willing to enter. Thus, strict rules on levels of return necessary may not hold. Ultimately, these represent a public investment in growth in the agricultural sector, while benefits are accrued privately by farmers.

20. **An equivalent to a sensitivity analysis was performed for NPV, using two levels of carbon pricing (low and high), both with and without climate and social risks.** With high carbon prices and no risks included, all of the proposed investments have a positive NPV. Yet some projects, such as biogas, have high up-front costs, but also high mitigation potential. This analysis shows that the financial and economic analyses are highly sensitive to such assumptions and provides a strong rationale for careful and cautious analysis of both future carbon pricing, and risk calculations within agricultural sector investments.

21. **Climate modeling results show the positive affects that some of the investments can have on yield by implementing CSA practices.** Biogas production and Integrated soil management CSA interventions are also poised to give a boost to key cereal crops, one through the integration of livestock and cereal systems, the other through soil recovery and conservation practices. Maize yield growth in particular stands to benefit, which is otherwise projected to flatten in the near future under climate change. These practices are also projected to substantially offset Burkina Faso's steepening trade deficit in sorghum and millet. Soybean yield is projected to be one of the hardest hit trajectories by climate change. These CSA interventions could lay the groundwork to offset a fledgling trade deficit in these products. The results highlight the importance of the investment in Building CSA Capacity nationally.

22. **This CSAIP portfolio of investments is well aligned with national policies, strategies, and plans.** There is a new NDC Partnership Plan under development, and coordination between funded CSAIP investments and the Nationally Determined Contributions (NDC) is vital. The CSAIP is clearly aligned with Burkina Faso's current NDC, both for high level objectives and specific investment activities. For example, the Building Capacity investment supports seven of 10 current NDC objectives. The CSAIP also addresses and supports the goals of multiple national policies and strategies, including the National Rural Sector Program, National Climate Change Adaptation Plan, National Adaptation Program of Action, Forest Investment Program, National Climate Change Learning Strategy, the African Adaptation Program. The CSAIP builds resilience within national institutions themselves, thus positively affecting myriad aspects of national policy. These investments thus support Burkina Faso's current national policies, and also its future developments in agriculture and food security in the face of climate change.

23. **There are risks that investments in CSA program design and implementation may be constrained by the existing policy context.** There are three barriers identified in all investments: (i) poor land tenure regimes, (ii) a dearth of finance services for agricultural stakeholders, particularly smallholder producers, and (iii) weak extension services, including climate and soil services. Climate change risks may be aggravated by the policy environment, although there are climate adaptation efforts underway, from farmers changing planting times or varieties, to national-scale efforts to promote adaptation within larger sectors. All programs should take into consideration the prerequisite land tenure, finance services, extension support, and climate information that all smallholders need in order for programs to meet the broader goals of the NDCs. Furthermore, there are risks presented by internal security concerns, global financial shocks, and potential financing, related to the current 2020 COVID-19 global pandemic, and the impacts within Burkina Faso from COVID-19.

24. **A supportive implementation context is critical for the implementation success of these investments and should build on existing institutional capacities and on-going CSA-related projects.** Momentum from existing programs and initiatives can be used to leverage continued growth and development in Burkina Faso's agricultural sector. Many programs and initiatives funded by both the Burkinabe government and international donors, have driven important agriculture sector growth. Eight of the nine proposed investments saw a strong, Burkinabe knowledge base was the greatest asset supporting implementation, followed by policy alignment. Although policies are not always consistently implemented, the all-important first step of establishing CSAIP regulation is already done. Availability of needed inputs was identified as important for seven investments, and highly climate resilient and aligned interests were each important in six investments. will also support many of the proposed investments.

25. **Climate-smart innovation in Burkina Faso requires support from many sectors, including Burkina Faso's government, non-profits, the private sector, and international organizations.** Yet partnerships, and the number of potential collaborators, can bring both strong benefits and/or complexity, depending on what the arrangements are with specific partners. Many of the projects noted a variety of potential public sector partnerships, so clarity on institutional arrangements, and what government agencies have priority will be important during further project design phases. Many of the proposed investments also identified a large number of international and national collaborators, ranging from CGIAR international research institutions to Burkinabe non-governmental organizations (NGOs) to a variety of aid organizations.

26. **Burkina Faso will need to identify the financial resources to implement the suite of investments proposed in this CSAIP.** All of the investments in this portfolio help support Burkina Faso's objectives to improve food security and support rural livelihoods. However, the interest and importance of climate change, agriculture, and environment may diminish as Burkina Faso deals with a humanitarian crisis and increasing insecurity. Financing costs and needs are well within the scope of reasonable projects that are financed by a range of institutions and organizations. All of the projects identified at least once source of government funding. Blended finance from the Government and the Environmental Intervention Fund (FIE) was proposed by three projects. There was also a strong role envisioned for the private sector, with many of the proposed investments identifying potential partners for private sector financing and collaboration.

27. **The investments, and their objectives build on these recommendations and Burkina Faso's own policy priorities.** Specifically, through its Plan National de Développement Economique et Social (PNDES), Burkina sees agriculture underpinning a major effort to drive economic growth and contribute to poverty reduction. All of the investments support increasing agricultural productivity and improve sustainability. Seven of the proposed investment directly support value chain creation, jobs, and or agro-processing, with Organic Farming, Livestock, and Agroforests having the largest potential impact. Four of the proposed investments in this CSAIP have the potential to substantially transform the agricultural sector, catalyzing new approaches and actions.

Proposed Investments	Burkina Faso Agriculture & Rural Sector Objectives							
	Productivity	Food Security	Support Ag. Sector Resilience	Ag. Value Chains	Sustainable Land/Farming	Mitigation	Support Women & Youth	Ag. Sector Transformation
CAPACITY Building	✓	✓	✓	✓	✓	✓	✓	✓
FINANCE and insurance	✓	✓	✓	✓	✓	✓	✓	✓
WATER resources & irrigation	✓	✓	✓	✓	✓	✓	✓	✓
Integrated SOIL Management	✓	✓	✓		✓	✓	✓	✓
ORGANIC Farming	✓	✓	✓	✓	✓	✓		
FORESTS, Agroforests, & GARDENS	✓	✓	✓	✓	✓	✓		
LIVESTOCK Intensification	✓	✓	✓	✓	✓		✓	
OIL-PROTEIN Crops	✓		✓	✓	✓	✓		
On-farm BIOGAS	✓				✓	✓	✓	

28. **Taken together, these investment opportunities represent a well-balanced portfolio.** The investments span different sectors of the country, have different levels of risk, target divergent groups of beneficiaries, and introduce a wide range of well-demonstrated CSA technologies and practices. All of them support CSA pillars, and contribute to meeting Burkina Faso's objectives as identified in its national plans, and that are fundamental to addressing its future.

CHAPTER 1: WHY CLIMATE-SMART AGRICULTURE

1. **Burkina Faso has made significant progress in poverty reduction in the last few decades, but it faces high social and economic challenges.** While the poverty rate fell between 2003 and 2014, and GDP growth was 6% or higher during that period, Burkina Faso remains a highly impoverished country. The 2019 Human Development Index ranks Burkina as 183 of 189 countries. Poverty is especially concentrated in certain regions, and is highly linked to low agricultural productivity¹⁵. Its geography is challenging for agricultural production; it is an arid, land-locked country with no major rivers.

2. **Climate change is adversely affecting Burkina Faso, lowering agricultural production, and decreasing opportunities for future prosperity.** At the local level, climate change impacts are demonstrated by extreme and unpredictable events—including prolonged droughts, violent floods, deadly heatwaves, and erratic beginnings and ends to the rainy seasons. At the regional scale, distinct patterns become apparent in each agro-ecological region. Desertification has already claimed extensive areas, and land degradation affects nearly 34% (over 9 million hectares) of land¹⁶.

3. **A broad and robust suite of agricultural and rural development initiatives is needed to help Burkina Faso adapt to climate change by reducing vulnerability and increasing resilience, especially in its agricultural sector.** Food and nutritional security, smallholder livelihoods, and agricultural production must be resilient to extreme climate events and variability. Responding to this need, and Burkina's own National Adaptation Plan, an in-country team has identified and prioritized a portfolio of investments to support Burkina Faso's rural sector in addressing climate change through climate-smart agriculture. This document describes these investments and their context in detail.

4. **Climate-smart agriculture (CSA) addresses agricultural vulnerabilities to climate change by bolstering productivity, adaptation, and mitigation.** Increases in productivity help fill gaps in nutritional and economic security at the household level, and builds economic growth and resilience at the national level. Adaptation adjusts or transforms current practices into those that are resilient in the face of climatic vagaries. Mitigation helps reduce the risk of further climate change by reducing greenhouse gas (GHG) emissions or increasing carbon storage. Taken as a whole, these 3 pillars of CSA (Figure 1) aim to identify and implement:

methods for increasing productivity that are...resilient to extreme weather and...
reduce GHG intensity (the amount of GHG per unit of food produced).¹⁷

5. **CSA acknowledges the synergistic effects between these 3 pillars, and aims to maximize co-benefits and minimize tradeoffs between them.** When tradeoffs are necessary, the well-being of the rural poor are prioritized through a focus on boosting productivity and adaptation. Yet even when mitigation is not prioritized, CSA often brings tertiary mitigation co-benefits by increasing food production at a greater rate than GHG emissions, thus reducing GHG intensity.

6. **CSA is not a silver-bullet approach, but rather is a process of identifying and implementing highly site-specific and time-specific solutions tailored to each community's unique needs.** Thus, the

¹⁵ Nguyen and Dizon 2017.

¹⁶ CPF, 3.

¹⁷ Lipper et al., "Climate-Smart Agriculture for Food Security."

relative importance of the three pillars will vary from one situation and stakeholder group to the next. While CSA is inherently agriculture-oriented, it is multisectoral in its effects on household livelihoods, national food security, and the sustainable use of natural resources.

Figure 1: Climate-smart agriculture: The triple win of sustainability, resilience and lower emissions¹⁸



7. **Climate smart agricultural investment plans (CSAIP) are a way for countries to plan for the future, and Burkina Faso is developing this national CSAIP as part of the Adaptation of African Agriculture (AAA).** AAA is a coordinated multi-country initiative that was launched at the United National Climate Change Conference 22nd Conference of Parties in Marrakech, Morocco (2016). This CSAIP was developed with strong input from a range of stakeholders, with the West African Science Service Center on Climate Change and Adapted Land Use (WASCAL) working as a key partner. Funding and technical support were provided by the World Bank. Strong government partnership and stakeholder engagement are the foundation of this CSAIP (Annex B).

8. **Burkina Faso is invested in the success of the CSAIP.** It is also signatory to several prominent international commitments that inform this CSAIP, including the United Nations Sustainable Development Goals, the African Union Malabo Declaration, and the United Nations Framework Convention on Climate Change (UNFCCC) National Determined Contributions (NDCs). Burkina Faso is also a member of the Global Alliance for Resilience in the Sahel and West Africa. The CSAIP supports Burkina Faso's contributions to several international agreements. The CSAIP also supports numerous national policies and initiatives, especially the National Adaptation Plan, the approved readiness program prepared by International Union for Conservation of Nature (IUCN), and a December 2019 Green Climate Fund Support to Burkina Faso for Readiness.

9. **Burkina Faso launched their UNFCCC Nationally Determined Contribution (NDC) in 2015, and agriculture is a major component of both the NDC and National Adaptation Plan.** The country has three strategies that are associated with increasing agricultural output and reducing poverty, revitalizing its natural resource capital, and supporting technologies and their adoption (including in agriculture). NDCs integrate climate adaptation, increased productivity, and reduced emissions intensity to guide efforts. The NDC is currently being revised, but it may offer closer alignment with this CSAIP. The National Adaptation Plan (NAP) identifies reducing the vulnerability of specific sectors to climate change, with priorities

¹⁸ Food and Agriculture Organization of the United Nations, *Climate-Smart Agriculture Sourcebook*.

including agriculture, livestock breeding, water, forests and natural ecosystems¹⁹. At the 2019 UN Climate Change Conference in Madrid, Spain, Burkina Faso launched their partnership plan identifies six components, including capacity building, mitigation, adaptation, progress reporting, financing and communication.

10. In alignment with Burkina Faso's NDCs, this CSAIP has identified specific investments that help reduce GHG intensity, especially important given Burkina's low emissions and population growth. Each investment concept proposes actions that reduce GHG intensity by decreasing emissions and/or sequestering carbon in biomass and soils. Specific investments, such as using organic residue for biodigesters, improving livestock nutrition and emissions, reversing land degradation and improving soil carbon storage, and supporting agroforestry and non-timber forest expansion all support mitigation – while also building resilience. These are just a few examples of the mitigation activities promoted in this CSAIP. Please see Chapter 3 for details of mitigation and and Chapter 4 and Annex A for complete details of proposed investments.

11. This climate smart agricultural investment plan (CSAIP) emphasizes two of the three pillars of CSA: productivity and adaptation. The global community, including Burkina Faso, is already challenged to meet food security requirements for a growing population. This challenge will be increasingly exacerbated by climate change²⁰. In July 2019, natural disasters, food insecurity, conflict and malnutrition mean that 1.3 million people required humanitarian assistance²¹, with numbers increasing. Burkina Faso already has a vulnerable population that will feel the impacts of climate change keenly; 21% of children under age 5 are chronically malnourished, 44% lives on less than US\$1.90/person/day, and 20% are considered food insecure²². Meanwhile, Burkina Faso contributes only 0.08% of global GHG emissions. This CSAIP therefore gives priority to improving and supporting the production and adaptation efforts identified in Burkina Faso's National Adaptation Plan, by supporting Burkinabe farming systems. Through this, mitigation co-benefits are observed.

12. The CSAIP provides a tangible, vetted set of priority investments for investors and donors to consider for funding. The proposed investments herein were identified and developed through an iterative, on-the-ground process supported by scientific literature, economic analyses, and climate impact modeling. The result is a suite of synergistic investments that can be developed and implemented as stand-alone projects or a comprehensive program for innovation in Burkina Faso. This investment portfolio has been prepared with an eye toward presentation to private sector actors, public institutions, international donors, and other key stakeholders in providing the much-needed funding for transforming Burkina Faso's agricultural sector.

¹⁹ Gordon, Jessica, Rohini Kohli, and Pradeep Kurukulasuriya. 2019. Principles in Practice: Integrating Adaptation into Long-Term Strategies. World Resources Institute. <https://www.wri.org/climate/expert-perspective/principles-practice-integrating-adaptation-long-term-strategies>

²⁰ Global Commission on Adaption, "Adapt Now: A Global Call for Leadership on Climate Resilience."

²¹ <https://reliefweb.int/sites/reliefweb.int/files/resources/FFP%20Fact%20Sheet%20Burkina%20Faso%2008.14.19.pdf> USAID Food Assistance Fact Sheet, August 14, 2029 and September 30, 2019

²² USAID Nutrition Profile 2018

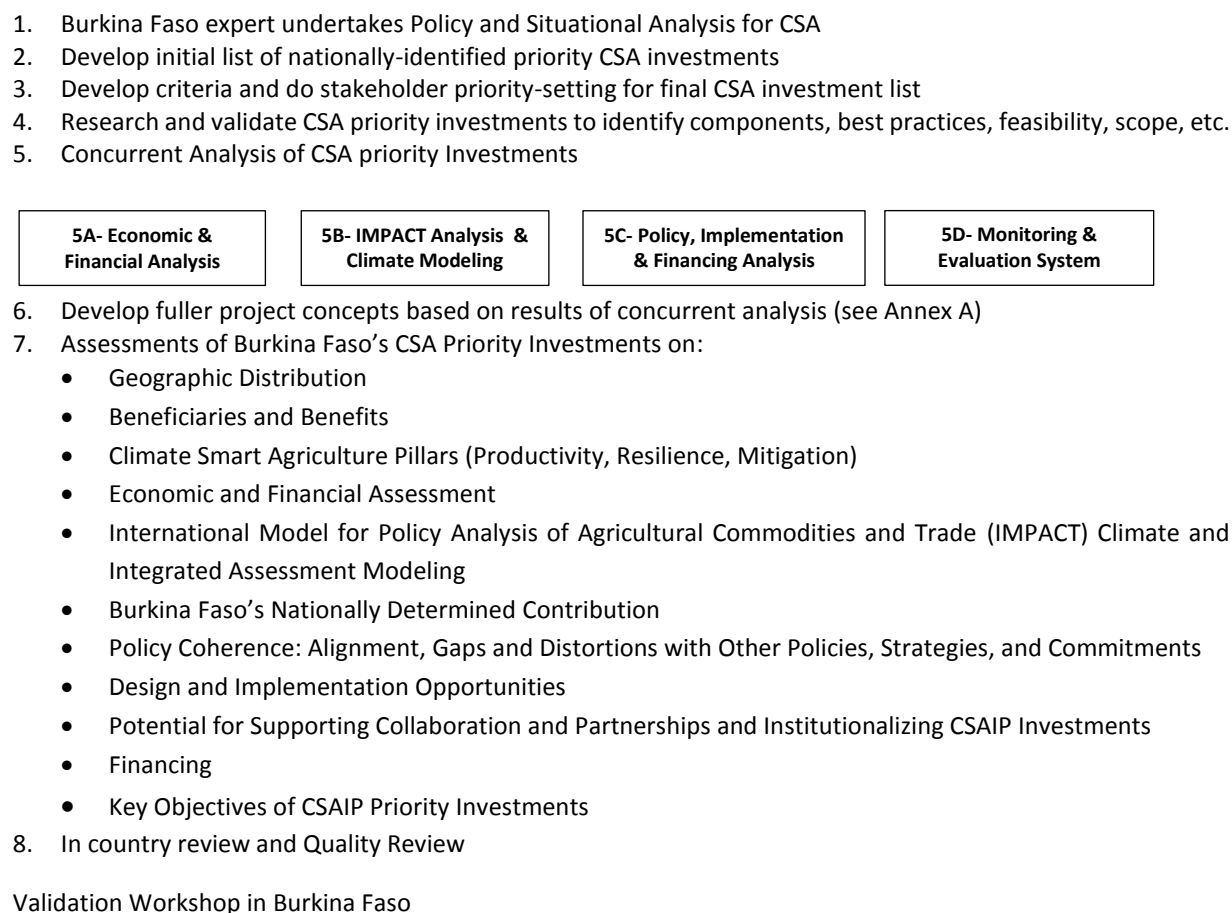
1.1 The Climate Smart Agricultural Investment Planning Framework

13. **The Burkina Faso CSAIP team has developed the investment plans described herein based on the 4 components of CSA planning and implementation.** These are²³:

- a. Situation Analysis
- b. Prioritizing Interventions
- c. Program Design
- d. Monitoring, Evaluation and Learning

These four components of CSAIP planning (Figure 2) are based on strong stakeholder engagement, topical expert contributions, and institutional and individual capacity building. This CSAIP focuses on the first two components: Situation Analysis and Prioritizing Interventions. Additionally, elements of Program Design and Monitoring, Evaluation, and Learning are discussed. In-country expertise, priorities, policies, stakeholder engagement, and capacity building were at the crux of the development of this CSAIP. The analysis used herein is built on both qualitative and quantitative assessments and methods. The process described below generally follows the CSA Prioritization Framework (see the [CSAIP Development Guide](#)), and Figure 2, below.

Figure 2: Developing Burkina Faso’s Climate Smart Agricultural Investments: Summary of Process



²³ Girvetz et al., “‘CSA-Plan’: Strategies to Put Climate-Smart Agriculture (CSA) into Practice.”

14. **A stakeholder workshop used detailed criteria to evaluate the initial list of 38 proposed investments.** Stakeholder experts represented a variety of organizations, including government ministries, institutions, research organizations, farmer groups, and international development organizations. The World Bank Burkina Faso Office led an expert workshop with the International Center for Tropical Agriculture (CIAT) and WASCAL counterparts providing technical support. At this workshop in Ouagadougou, participants used an iterative, qualitative and quantitative, prioritizing process to rate and review (see Chapter 3 and Annex C for specifics) each of the 38 proposed CSA priorities. They also grouped the initial list of investments by agro-ecological zones to represent all regions of Burkina Faso. The initial list of potential investments was reduced from 38 to 10 priority CSA investments. A second round of consultation led to a second workshop to validate and refine the results.

15. **The priority setting workshop provided a strong rationale and key information on each proposed CSA investment.** The process of reducing the initial list of priorities to a shorter list, provide detailed information on each investment. This information needs to be further refined to fully reflect the scope of each proposed investment. To do this, the CSAIP team: 1) refines the investment shortlist to insure relevant CSA best practices are followed; 2) conducts a thorough literature review on similar projects; 3) identifies current and potential yields of crops and technology packages, and other project components, such as designing value chains; and 4) reviews related projects for best practices, implementation issues, lessons learned, barriers and opportunities. Investment concept notes (see Annex A) are prepared for each priority investment.

16. **Prioritized Investments are analysed along five dimensions when concept notes are completed: (i) economic and financial; (ii) IMPACT climate modelling; (iii) policy and financing alignment; (iv) implementation issues; and (v) monitoring and evaluation.** Each of these are described in detail in a technical Annexes (See Annexes B, C, D, E), but the highlights of each are presented below.

17. **Economic and financial assessments for this CSAIP are based on analysis of how the investments will perform.** Financial assessments are derived from secondary data sources. A key input for the financial analysis is Evidence for Resilient Agriculture (ERA)²⁴, a database with information on 112 technologies used in crop, livestock and tree production and 58 indicators of performance (e.g., yield, net economic returns, soil carbon, etc). Each investment may include multiple technologies, so ERA provides field-based data for the Burkinabe agro-ecological context on outcomes for each different technology in a proposed investment. Nearly 50% of the data in ERA relates to yield benefits from use of new technologies, many of which are directly relevant to the investment plan. ERA also include data on the economic performance of technologies. These data are combined with other external data provide by national stakeholders and additional literature searches. To ensure relevance, ERA results from similar African agroecological zones are used as possible. Costs were determined in a number of ways but were largely based on the number of potential beneficiaries. (See Annex E for the economics methodology.)

18. **The results from the economic modeling across all investments are presented in Chapter 3** and include assessment of the Net Present Value (NPV); the impact of investments on the three pillars of CSA agriculture (production, resilience and mitigation); Returns on Investment and Cost-Benefit Ratio. The analyses are run in a probabilistic framework, with and without social and environmental risks using the best available information on risk and their potential impact. It allows a direct targeting of investments for the climate and pest risks threatening productivity and growth in the future. This can be contrasted to

²⁴ <http://era.ccafs.cgiar.org>

many examples of *ex-ante* assessment where the analysis presents an overly optimistic picture of likely success by not integrating specific risks to the cost-benefit analysis.

19. **Analysis of climate models, trade, and other factors were done for the nine investments in this CSAIP using the International Model for Policy Analysis of Agricultural Commodities and Trade (IMPACT).** Climate modeling is a key component to characterize the future production environment of different commodities. Using multiple potential climate scenarios facilitates understanding the impact of climate change in Burkina Faso, how this affects agricultural production, and the corresponding impact on international trade in agricultural commodities. Climate change will alter what crops are suitable in different growing regions, reducing suitability across large areas (e.g., entire countries) but also creating pockets of increased suitability for a small number of crops. At a global scale, these shifts will determine what countries can grow what crops, in turn affecting international trade. Demographic and economic growth trajectories and greenhouse gas mitigation policies will thus drive changes in demand and consumption. IMPACT models the complex interplay of all of these demographic, economic, mitigation, demand and consumption factors. IMPACT modeling is therefore more than climate modeling, providing integrated assessment modeling of multiple socio-economic and biophysical factors.

20. **IMPACT integrated assessment modeling for Burkina Faso assesses future performance of different crops and the potential for CSA practices to improve this performance, under different conditions (e.g. trade).** Specifically, IMPACT has three key components: 1) climate impact of all of Burkina Faso's basic commodities for 30 years into the future; 2) Burkina Faso's future comparative advantage in these commodities versus other countries, to understand trade advantages; 3) CSA impact of proposed investments on future yields. Assumptions regarding demographic and economic growth are reflected by incorporating the Shared Socioeconomic Pathway (SSP) for the Middle of the Road/Business as Usual pathway, where historical patterns of development, inequality, and environmental degradation continue (see Annex D). Assumptions regarding the severity of increased greenhouse gas concentration are reflected in the choice of three Representative Concentration Pathways (RCP) (4.5, 6.0 and 8.5) from least to most severe with results presented for two points in time, 2030 and 2050. IMPACT provides insights into what comparative advantage Burkina Faso will have for different commodities, given demographic, economic, demand and trade factors – into the future. It also provides insights on the difference that investments with new technologies introduced now, can have in 10 years, and in 30 years, relative to different potential climate impacts.

21. **Prioritized investments were re-assessed for their alignment to Burkina Faso's NDC and other key policies and strategies.** The emphasis here is on re-assessment, because the initial list of potential investments, and then the nine prioritized investments, were initially selected for their alignment to Burkinabe national plans and policies. This reassessment was performed to more fully indicate the nuances of policy linkages for the nine priority investments. For example, policy gaps and distortions were identified for each investment, and are summarized in Chapter 3 and in Annex A, which contains full information on each investment.

22. **Design and implementation considerations were also assessed.** Relevant projects, both for CSA and non-CSA projects for similar production systems within Burkina Faso were examined. A variety of features of these projects were assessed. Some of the findings were noted by participants at the prioritizing workshop, while other findings were based on an extensive literature review of scientific papers, project documents, white papers, etc. Some of the factors briefly considered and discussed in Chapter 3 are: a) implementation experiences to understand potential lessons; b) positive contributors or risks and barriers to project success; c) potential collaborators, both in government, in NGOs (Non-

governmental organizations), and for the private sector. This information is all included in the individual investment concepts in Annex A and summarized in Chapter 3.

23. **Financing opportunities and the potential for private sector involvement were assessed** in several ways. As part of the prioritizing workshop and investment concept development process, the potential for private sector collaboration and financing, public financing opportunities, and international financing were reviewed. The World Bank's Maximizing Finance for Development Framework was used as a guide for identifying private sector financing opportunities in Burkina Faso for the prioritized investments, and can be used in further assessments to leverage private finance for these investments. Results are presented in Chapter 3 and in the individual investment concepts in Annex A.

24. **A monitoring and evaluation plan was developed for the entire portfolio (see Chapter 5) and a theory of change was developed for the entire CSAIP and each individual priority investment (see Annex A).** The M&E (Monitoring and Evaluation) Plan aims to provide CSAIP implementers—particularly the Government—with a manageable system to generate, collect and analyze standardized data to assess the success of both individual investments and/or the portfolio of investments. The goal is to inform operational and strategic decision-making. The theory of change includes four action areas for the entire CSAIP and individual investments: a) robust research and development (R&D); b) uptake of climate-smart production technologies and value-added practices; c) stakeholder engagement and partnerships; and d) system-wide capacity to implement CSA actions.

CHAPTER 2: SITUATION ANALYSIS OF LIVELIHOODS, AGRICULTURE AND CLIMATE CHANGE

2.1 Burkina Faso's Rural and Agricultural Sector in Brief

25. **Burkina Faso's economy is growing, yet much remains to be done.** The 2018 national GDP was US\$14.4 billion, or US\$731 per capita; this is below the sub-Saharan African average of US\$1,574, and far below the world average of US\$11,300²⁵. There are 19.75 million inhabitants of Burkina Faso, and the population density is relatively high--72/ km², versus the global average of 60/ km² (

26. Figure 6 and Figure 7)²⁶. While 70.6% of the total population is poor, 47.5% of those in poverty live in rural areas. Just 9.5% of rural residents had electricity in 2017²⁷.

27. **Burkina Faso is one of the fastest growing countries in the world; the annual population growth rate of 3.1% is expected to drive the population to 42 million by 2050.**²⁸ Population growth is already high and is unevenly distributed across the country. As of 2014, approximately 40.1% of the population lived below the national poverty line, and 43.7% lives on less than US\$1.90 / day.²⁹ This is a significant improvement from 2003, when 51.1% of the population lived below the national poverty line³⁰. Food insecurity ranges from 23.8% to 54.2% of the population across regions³¹. Burkina Faso sits near the bottom of the Human Development Index; it ranked 183/189 countries in 2017 with a score of 0.423, versus the Sub-Saharan African average of 0.537. This represents a significantly improvement from a score of 0.286 in 2000.³² The national Corruption Perceptions Index score has also declined slightly in 2018 to 41/100, giving it a rank of 78 / 180 countries.³³

28. **Agriculture is Burkina Faso's predominant land use.** Burkina Faso's total land area is 273.6 million ha, with 121 million ha (44.22%) under agricultural production and 52.9 million ha (19.33%) under forest.^{34,35} Of the agricultural land, about 60 million ha (21.9%) is under arable production, 1 million ha is under permanent crops, and 60 million ha is under permanent meadows and pastures (Table 1, Figure 3, Figure 7)³⁶. The agricultural land cover has expanded 91% since 2001, when it accounted for just 116.9 million ha³⁷. Much of this expansion has been onto unsuitable land; just 13.3% of the national land area

²⁵ World Bank, "Burkina Faso Overview" <https://www.worldbank.org/en/country/burkinafaso/overview>

²⁶ World Bank, "Burkina Faso." 1. <https://data.worldbank.org/country/burkina-faso>

²⁷ World Bank, "Burkina Faso Indicators." <https://data.worldbank.org/country/burkina-faso>

²⁸ FOA, "Africa Sustainable Livestock 2050." <http://www.fao.org/in-action/asl2050/countries/bfa/en/>

²⁹ World Bank, "Burkina Faso Indicators." <https://data.worldbank.org/country/burkina-faso>

³⁰ World Bank, "Poverty and Equity Data portal." <http://povertydata.worldbank.org/poverty/country/BFA>

³¹ William, *Country Assessment Studies on Climate Change, Agricultural Trade and Food Security in ECOWAS BURKINA FASO REPORT*.

³² UNDP, "Burkina Faso Human Development Report." <http://hdr.undp.org/en/countries/profiles/BFA>

³³ Transparency International, "Corruption Perception Index 2018" <https://www.transparency.org/cpi2018>

³⁴ FAO Stata, "Burkina Faso" <http://www.fao.org/faostat/en/#country/233>

³⁵ Index Mundi "Agricultural Land" <https://www.indexmundi.com/facts/indicators/AG.LND.AGRI.ZS>

³⁶ FAO Stata, "Burkina Faso" <http://www.fao.org/faostat/en/#country/233>

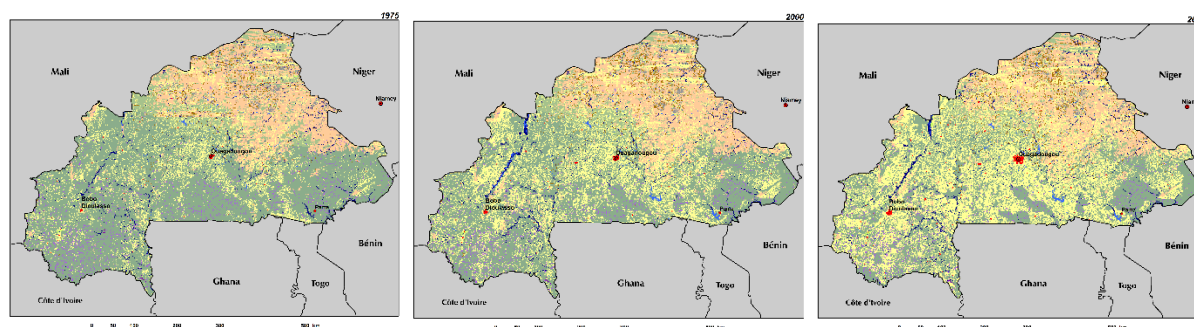
³⁷ Knauer, K., Gessner, U., Fensholt, R., Forkuor, G., & Kuenzer, C. (2017). Monitoring agricultural expansion in Burkina Faso over 14 years with 30 m resolution time series: The role of population growth and implications for the environment. *Remote Sensing*, 9(2), 132.

is suitable for cultivation³⁸. This drastic rate of agricultural expansion onto lands with low agricultural productivity has been an attempt to feed the burgeoning population³⁹. Yet farming on unsuitable land, climate change, and desertification mean that 34% of the land area is already degraded⁴⁰. Significant increases in productivity and intensification would enable both economic prosperity and regeneration of degraded forests. The country has significant carbon sequestration potential and could reduce emissions by up to 19,020,600 tCO₂e annually if deforestation and land degradation were effectively mitigated (F).⁴¹

Table 1: Land Cover Changes in Burkina Faso 1990 to 2017

Land use	Area (km ²)		% Change
	1990	2017	
Forest	68,470	52,902	-29%
Arable agriculture	95,750	121,000	21%
Irrigated agriculture	243	461	47%
Permanent agriculture	700	1,000	30%

Figure 3: Extent of Burkinabe forest and agriculture cover in 1975, 2000, and 2014⁴²



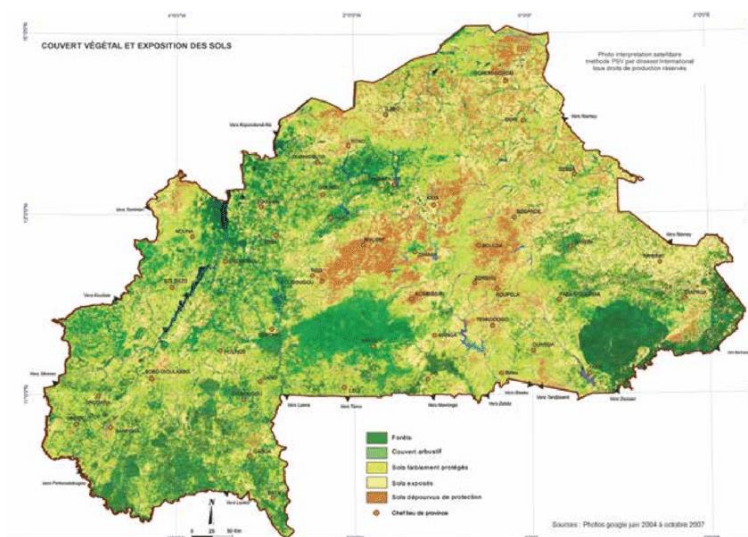
³⁸ William, *Country Assessment Studies on Climate Change, Agricultural Trade and Food Security in ECOWAS BURKINA FASO REPORT*.

³⁹ Knauer et al., "Monitoring Agricultural Expansion in Burkina Faso over 14 Years with 30 m Resolution Time Series."

⁴⁰ World Bank Climate Change Country Profile, cited in *Creating Markets in Burkina Faso*, World Bank, 2019.

⁴¹ Kambiré et al., *The Context of REDD+ and Adaptation to Climate Change in Burkina Faso : Drivers, Agents and Institutions*.

⁴² United States Department of Agriculture, "Land Use, Land Cover, and Trends in Burkina Faso."

Figure 4: Burkinabe Forest Cover and Soil Exposure 2016⁴³

29. Burkina Faso has a warm Sahelian tropical climate with significant temporal and geographic variation, with three climatic zones: the northern dry Sahel, the central North Sudan zone, and the humid South Sudan in the south (Figure 5).⁴⁴ Average temperatures are consistently high both temporally and spatially; the national monthly mean temperature sits between 25.8°C and 29.6°C year-round. Temperatures are highest in the Sahel, ranging between 15-45°C.⁴⁵ National annual rainfall ranges between 400-900 mm, with 50-70 rainy days annually. There is significant spatial and temporal variation; the Sahel receives 300-600 between July and September, followed by a 7-9 months dry season. The North Sudan receives 600-900mm between June and October, and the South Sudan receives 900-1200mm between May and October^{46,47}.

30. Agriculture is the main source of livelihoods, employment, and exporting in Burkina Faso. Agriculture employs 80% of the country's workforce and contributes nearly 28.6% of national GDP⁴⁸. Smallholder producers operating on 5ha or less account for about 80% of total agricultural production. Further, women account for over 50% of the agricultural workforce and produce more than 66% of the food consumed in-country. However, women are given markedly lower access to crucial resources, most notably access to land and land tenure rights. Youth are leaving rural areas to seek out viable career options in urban areas. Yet despite this exodus, there has been a marked increase in the rural population density (Figure 6) and agricultural expansion (Figure 7).

⁴³ Kambiré et al., *The Context of REDD+ and Adaptation to Climate Change in Burkina Faso : Drivers, Agents and Institutions*.

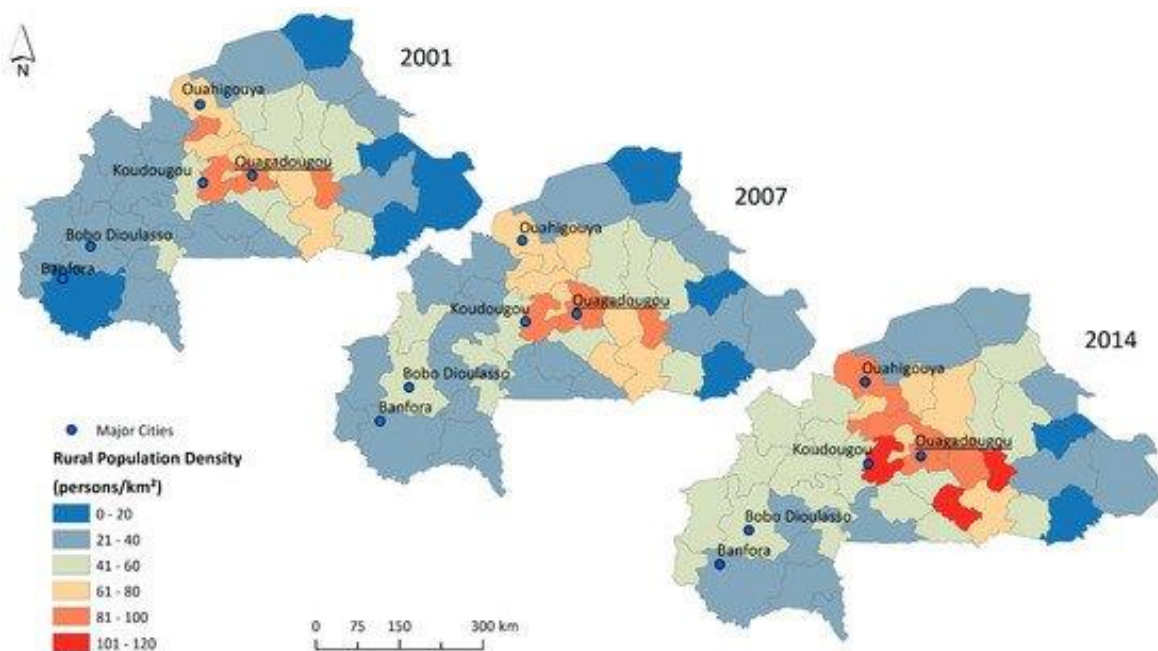
⁴⁴ UC Davis "Burkina Faso" <https://gfc.ucdavis.edu/profiles/rst/bfa.html>

⁴⁵ USAID (2017), Climate Risk in Food For Peace Geographies Burkina Faso. USAID Washington DC. Document attached

⁴⁶ Knauer, K., Gessner, U., Fensholt, R., Forkuor, G., & Kuenzer, C. (2017). Monitoring agricultural expansion in Burkina Faso over 14 years with 30 m resolution time series: The role of population growth and implications for the environment. *Remote Sensing*, 9(2), 132.

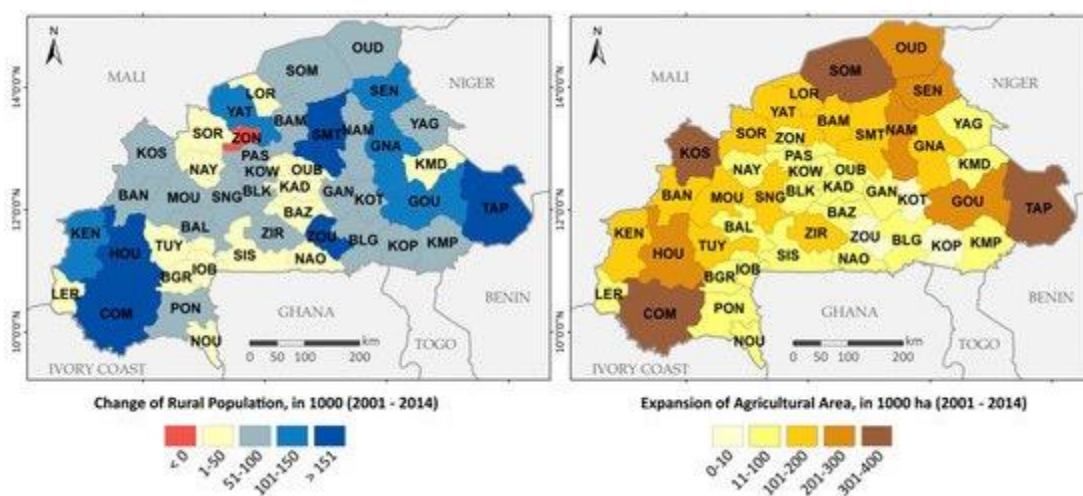
⁴⁷ UC Davis "Burkina Faso" <https://gfc.ucdavis.edu/profiles/rst/bfa.html>

⁴⁸ UC Davis "Burkina Faso" <https://gfc.ucdavis.edu/profiles/rst/bfa.html>

Figure 5: Climactic Zones of Burkina Faso⁴⁹Figure 6: Rural Population Expansion Density 2001-2014⁵⁰

⁴⁹ Kambiré et al., *The Context of REDD+ and Adaptation to Climate Change in Burkina Faso : Drivers, Agents and Institutions*.

⁵⁰ Knauer et al., "Monitoring Agricultural Expansion in Burkina Faso over 14 Years with 30 m Resolution Time Series."

Figure 7: Changes in Rural Population and Agricultural Expansion 2001-2014⁵¹

31. The agricultural economy has been growing rapidly since the 1990s and has outpaced the general economy. Burkina Faso's agricultural sector experienced 8.4% growth in 2018 alone⁵². Furthermore, the national economy is closely tied to agricultural productivity and, implicitly, weather and climate; the 2018 national economic growth of 6.8% was attributed to a rainy year and the consequent increase in cereal production⁵³. The country's primary staple crops, including maize, millet, sorghum, and rice are crucial to food security⁵⁴ (Tables 2 and 3)

32. Cereals constitute a large and increasing share of the Burkinabe diet and cultivated area (Figures 8 and 9). People have increased their daily consumption of cereals (maize, sorghum and millet), and this increased consumption and the population increase have led to a doubling of the area planted since 1973 (Figure 9). Pulses, mainly cowpea, production has increased and occupies over 10% of the cultivated land, but cowpea consumption has not kept pace with increased production.

⁵¹ Knauer et al.

⁵² World Bank, "Burkina Faso Indicators." <https://data.worldbank.org/country/burkina-faso>

⁵³ World Bank, "Burkina Faso Overview" <https://www.worldbank.org/en/country/burkinafaso/overview>

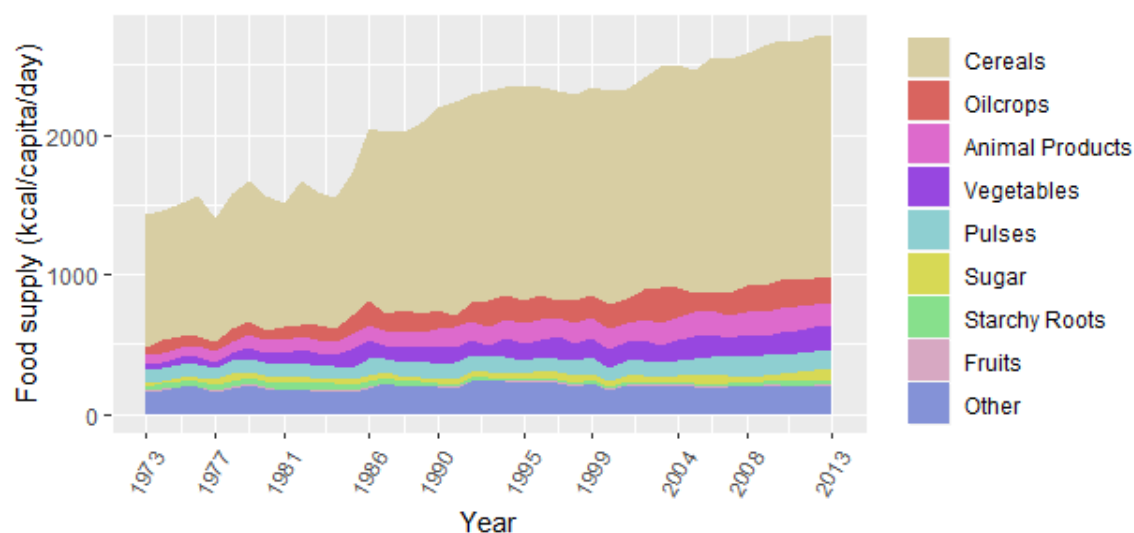
⁵⁴ FAO (2010). Monitoring African Food and Agricultural Policies Project (MAFAP) Commodity Selection in the Five pilot countries. Document attached

Table 2: Key Burkinabe food and commodity crops

Darker colors show higher values for each indicator. Grey color shows missing value. Data resources vary from (2014-2017)^{55,56}

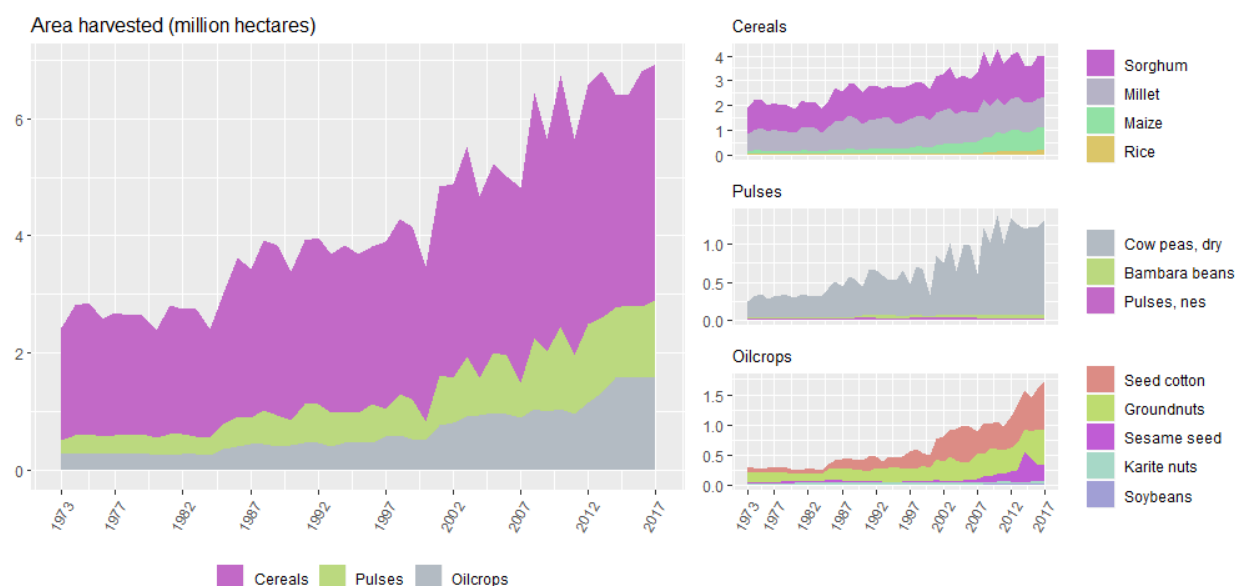
	% of Cultivated Land	Yield (tonnes/ha)	Production (1000 tonnes)	Food consumption (g/capita/day)	Kcal/capita /day
Cotton	6.98%	1.00	844,337		
Sorghum	13.78%	0.80	1,365,898	158.69	438
Millet	10.04%	0.68	828,234	155.83	404
Maize	7.90%	1.60	1,533,431	187.70	597
Cashew nuts	0.82%	1.00	99,027	2.75	9
Beans	0.006%	13.43	9,913		
Sesame	2.41%	0.56	164,787	2.78	15
Sweet potatoes	0.04%	11.72	58,749	24.40	23
Sugarcane	0.04%	100.54	487,643		
Rice	1.36%	1.97	325,566	88.04	207
Cowpeas	10.37%	0.48	603,966		
Cassava	0.03%	1.26	4,842	2.35	2
Ground Nut	4.58%	0.60	334,328	28.93	159
Soybean	0.17%	0.93	18,500	3.34	11
TOTAL	58%			1.47	1815

Figure 8: Calorie consumption per capita per day in Burkina Faso, 1973-2013 (Source: Food and Agriculture Organisation (FAO)).



⁵⁵ FAO, "Crop" <http://www.fao.org/faostat/en/#data/QC>

⁵⁶ FAO, "Food Supply." <http://www.fao.org/faostat/en/#search/Food%20supply%20kcal%2Fcapita%2Fday>

Figure 9: Harvested area in Burkina Faso, 1973-2017 (Source: FAO).

33. **Primary agricultural commodities produced include maize, sorghum, millet, rice, cowpea, sesame and cotton** (Table 3). Along with gold exports, the Burkinabe economy is fueled by exports of cotton, sesame, peanut, and soy, and cash crops such as maize, sorghum, millet, cowpeas, and sugarcane⁵⁷ (Table 3). While maize is exported, as noted above, domestic demand for it has increased. Demand for additional cereals, (rice and wheat) is increasing and driving imports, as is demand for cattle meat.

Table 3: Import, export, and production of key commodities in Burkina Faso

Darker colors show higher values for each indicator. Grey color shows missing value. Data resources from 2015-2017, produced for this analysis.

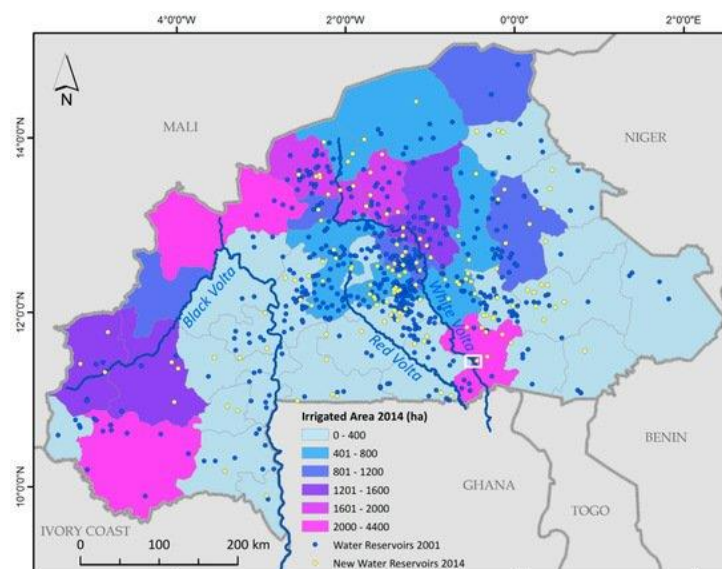
	Export (1000 tons)	Imports (1000 tons)	Production (1000 tons)
Millet	8,700	475	828,234
Sorghum	3,020	-	1,365,898
Maize	44,902	2,036	1,533,431
Rice	2,047	431,917	325,566
Sesame	110,490	1,208	164,787
Soybean	1159	43	18,500
Cotton	250,233	2,806	844,337
Cattle Meat	2	6	-
Wheat	278	246,593	-

⁵⁷ UC Davis "Burkina Faso" <https://gfc.ucdavis.edu/profiles/rst/bfa.html>

34. **Agricultural productivity is persistently low and has increased only 10% in the last decade.** For example, Burkina Faso has regional competitive advantages in the livestock sector, but current policies substantially limit opportunities⁵⁸. Some of the challenges facing the agricultural sector are low diversification, vulnerability to weather and climate impacts, and persistent discrimination. Agricultural value chains remain largely undeveloped, the policy and finance environments are non-conducive to business development. Most workers do not have the skills demanded by firms.⁵⁹

35. **Significant opportunities exist to increase national productivity and economic growth⁶⁰.** The first step is removing some of the barriers cited above. Additional opportunities can be found through intensifying smallholder production. The agricultural sector in Burkina Faso is highly reliant on a rain-fed production system. Yet expanding irrigated agriculture is an important step toward delinking climate and economy; Burkina Faso has abundant water resources⁶¹ and approximately 233,500ha of irrigable land, but only about 34,500ha were under irrigation in 2014 (Figure 10)⁶².

Figure 10: Irrigated area per province 2014 and locations of artificial water reservoirs in 2001 and 2014⁶³



⁵⁸ William, *Country Assessment Studies on Climate Change, Agricultural Trade and Food Security in ECOWAS BURKINA FASO REPORT*.

⁵⁹ Country Management Unit AFCF2, "Country Partnership Framework for Burkina Faso."

⁶⁰ Ksoll, C., Morgan, S., Bos, K. & Blair, R. (2018) Evaluation of the Burkina Faso Agricultural Development Project: Baseline Report. Mathematica Policy Research, NE. Document attached

⁶¹ William, *Country Assessment Studies on Climate Change, Agricultural Trade and Food Security in ECOWAS BURKINA FASO REPORT*.

⁶² Knauer et al., "Monitoring Agricultural Expansion in Burkina Faso over 14 Years with 30 m Resolution Time Series."

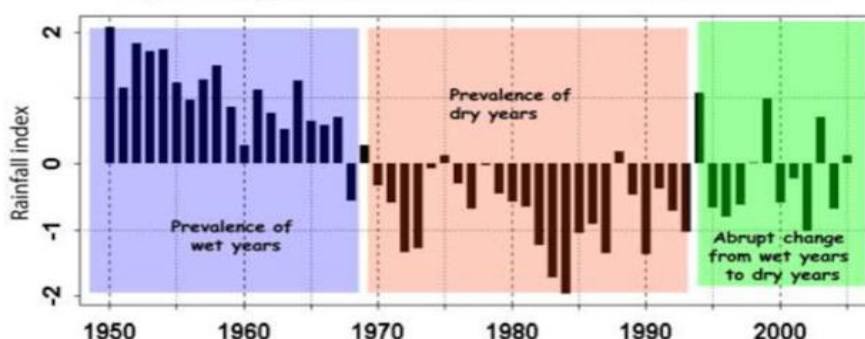
⁶³ Knauer et al.

2.2 Climate Change Impacts on Burkina Faso's Agriculture

36. **Burkina Faso has experienced extreme climatic impacts.** Declining rainfall and increasing temperatures are recurring, affecting agriculture and natural resources (water and forest) substantially. Other climate-related hazards such as droughts, floods, heat waves, locusts, and dust storms are common in the country. Climatic variations are widespread in the Sahel, especially variability in rainfall and temperature⁶⁴.

37. **Burkina Faso's agriculture is affected by erratic precipitation patterns⁶⁵ related to climate change, with overall trends showing declining rainfall since 1950.** Yet when it does rain, it rains harder (more rain), and pattern of rainfall has changed, with fewer consecutive wet days, and more consecutive dry days. Three main distinctive trends for different time periods have been observed (Figure 11); persistent wet years between 1950-1970, persistent dry years in the period 1970-1990 and alternating between wet and dry years from 1990 until recently⁶⁶.

Figure 11: Distinctive trends in Burkina Faso's rainfall (1950-2005)⁶⁷



38. **Burkina Faso is getting warmer with temperatures steadily increasing.** The average temperature range is from 27 - 30 °C., but since 1975, temperatures have increased by 0.6°C with an estimated warming rate of more than 0.15°C per decade. The period 1975-2009 experienced warming in the rainy season in the months of June to September.⁶⁸ Similarly, high evapotranspiration is observed across the country with an average of 100 mm per month, and a maximum of 200 mm in February to March⁶⁹. This magnifies the drought scenario in the country as a whole⁷⁰ (Figure 12).

⁶⁴ World Bank, "Vulnerability, Risk Reduction, and Adaptation to Climate Change: Climate Risk and Adaptation Country Profile Burkina Faso"

⁶⁵ Longueville et al., "Long-term analysis of rainfall and temperature data in Burkina Faso"

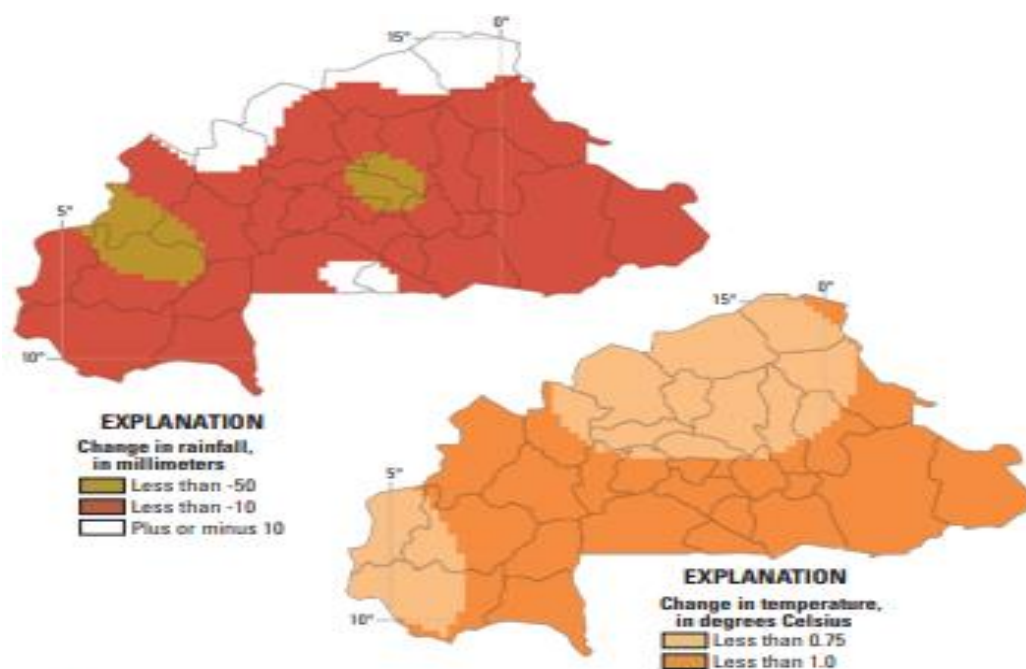
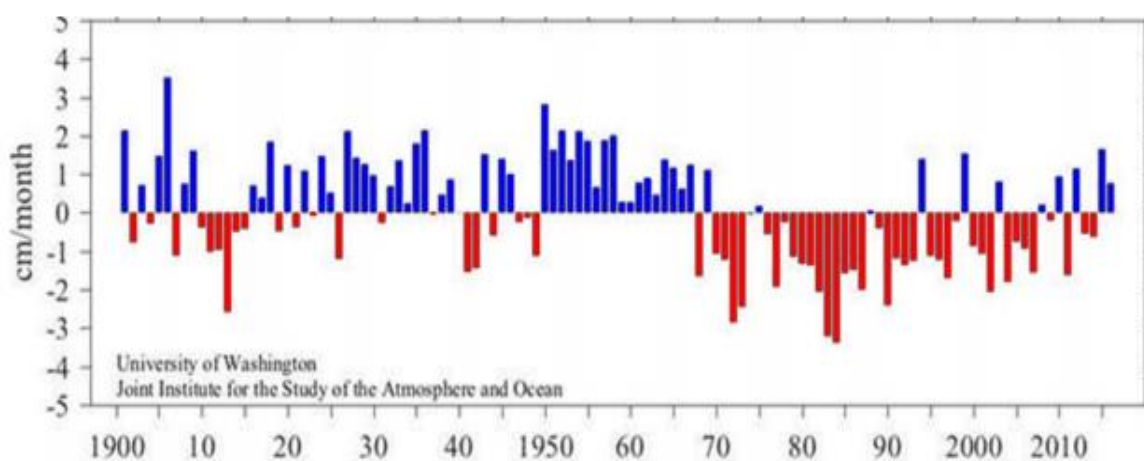
⁶⁶ World Bank, "Burkina Faso climate smart agriculture profile"

⁶⁷ World Bank

⁶⁸ USAID, "A Climate Trend Analysis of Burkina Faso"

⁶⁹ Crawford et al., "Review of current and planned adaptation action in Burkina Faso"

⁷⁰ Nana, "Impact of Climate Change on Cereal Production in Burkina Faso"

Figure 12: Changes in rainfall and temperature in Burkina Faso (1960-2009)⁷¹Figure 13: Annual rainfall variability compared to average in the Sahel (1900–2016)⁷²

39. **The Northern region of the country has been greatly affected by the temperature and rainfall variations (since 1950).** Burkina Faso has had both inter-temporal and spatial variation in rainfall which exhibits decreasing tendencies. This has had notable impacts in all climatic zones of the country. Consequently, there has been an upward trend in temperature throughout the country. However, the highest temperatures are in the Northern region which lie in the desert zones⁷³, with dry regions extending

⁷¹ USAID

⁷² USAID, "Climate risks in food for peace geographies Burkina Faso"

⁷³ Crawford et al.

to the South over the 20th century⁷⁴. Extreme events like floods, drought and dust winds have become recurrent in various localities of the country. Specifically, increased flash flooding occurs in the Central and Northern parts of the country during the rainy season⁷⁵ (Figure 13).

40. **Climate risks, such as floods and droughts, are increasing.** Given the steady temperature increase and rainfall decrease, droughts are more common. Burkina Faso encountered three major droughts in 1990-91, 1995-96 and 1997-98. Although floods are less frequent than droughts, the country has experienced 11 major floods between 1991 and 2009, affecting 380,000 people and 93 reported deaths⁷⁶.

41. **Climatic projections for Burkina Faso show increasing temperatures and decreasing rainfall.** Future projections show a probable increase of 1.4-1.6°C rise in temperature by the year 2050⁷⁷. In the north, warming is more likely during the wet season than in the dry season⁷⁸. The number of both dry and wet years is expected to increase, and semi-arid areas will become more arid. There are contrasting projections for rainfall patterns, but intensity of events may increase, with less rainfall overall, although there is uncertainty⁷⁹.

42. **Climate change has the greatest impact on the agricultural sector since subsistence agriculture, and forestry supports an estimated 80% of the population**⁸⁰. With extreme rainfall, there is a risk of flooding that causes soil leaching, erosion and run-off. Yet little or no rainfall causes prolonged drought and water scarcity. Higher temperatures increase evapotranspiration, stunt crop growth and increase prevalence of both pests and diseases. Planting calendars are distorted, and farmers may lose livestock and crops, and/or have lower yields. Agricultural production in North and central Burkina is likely to be affected due to drought and flood exposure of the regions⁸¹.

43. **Women are the most vulnerable group in the agricultural sector.** Women are more vulnerable to climate change compared to men since they differ in asset ownership to tackle climate change impacts⁸². For example, during droughts, men can seek greener pastures elsewhere, while women are less able to migrate. Women also have less education and lower human capital, and men are prioritized when it comes to training and extension services. Financial access is also a challenge as men control assets needed to secure credit.

44. **Water resources have been adversely affected by climate change.** A larger proportion of farmers rely on rain and underground water to sustain their agricultural production⁸³. With variability in rainfall and temperature, water resources are affected. Increased temperature and reduced rainfall mean wells and other water sources dry up. Yet excess rainfall increases flooding that destroys infrastructure, such as water pipes, and siltation and sedimentation of rivers causes water pollution. Ultimately, there are projections indicating there could be water scarcity.

⁷⁴ USAID, "Climate vulnerabilities and development in Burkina Faso and Niger"

⁷⁵ USAID

⁷⁶ Crawford et al.

⁷⁷ UNDP, "Climate change adaptation Burkina Faso"

⁷⁸ World Bank

⁷⁹ World Bank

⁸⁰ World Bank, "Climate knowledge portal Burkina Faso"

⁸¹ Crawford et al.

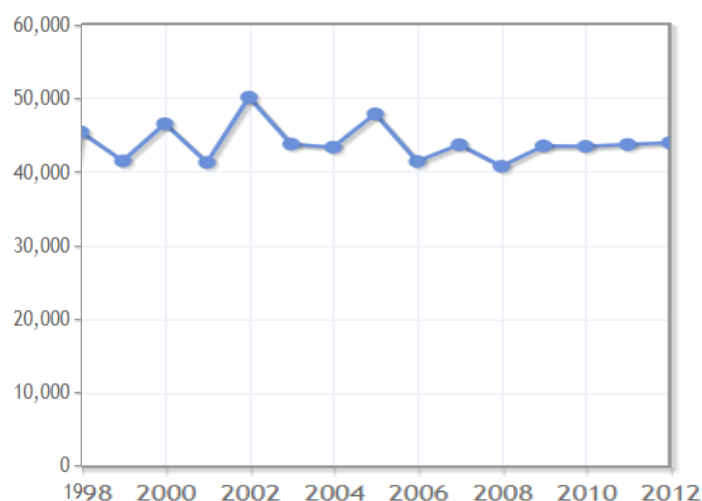
⁸² González et al., "Climate Change And Women Farmers In Burkina Faso: Impact and adaptation policies and practices"

⁸³ World Bank

45. **Climate change is a threat to the peace and political stability of Burkina Faso.** The implications of climate change on peace and security is not only a global concern but also a local concern in West Africa in general⁸⁴. Climate change extensively affects water availability, food security, population distribution, disease prevalence and coastal boundaries. Given the variability and decline in annual precipitation, there already are disputes over water resources in Burkina Faso. With a rapidly growing population in Burkina Faso, there is a likelihood of boundary conflicts and tensions over the Volta basin in the coming years, which Burkina Faso shares with Ghana. Moreover, external factors such as poor governance, recession and ethnic tensions could further undermine country's economic and political stability⁸⁵. Additionally, migrations of people and expansion of agricultural communities into pastoral lands could trigger conflicts.

46. **Burkina Faso contributes only 0.07% to global Greenhouse Gas Emission (GHGs).** Burkina Faso produces 40,000-50,000 kt of CO₂ equivalent (Figure 14). By 2014, Burkina Faso contributed to less a cumulative total of 32.60 (MtCO₂e)-1 to global GHGs⁸⁶. The agricultural sector has been the largest source of emissions⁸⁷ (Figure 15).

Figure 14: Total greenhouse gas emissions (Kt of CO₂ equivalent, 1998-2012)⁸⁸



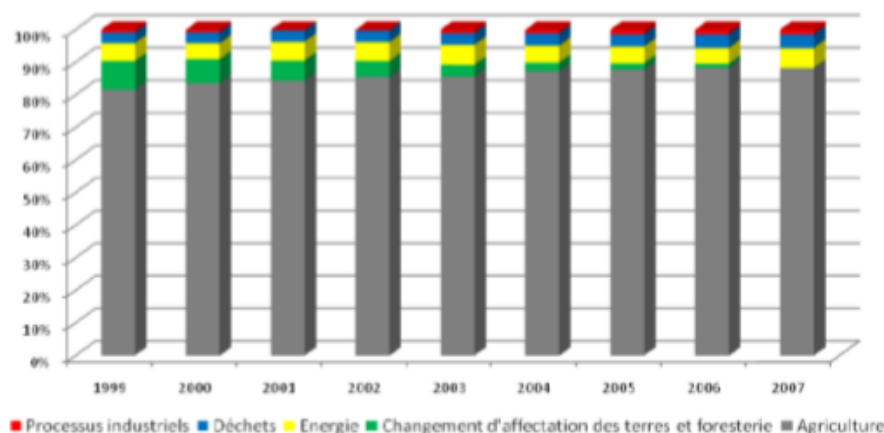
⁸⁴ UNDP, "Human Development report"

⁸⁵ Brown and Crawford, "Climate change: A new threat to stability in West Africa? Evidence from Ghana and Burkina Faso"

⁸⁶ USAID, "Greenhouse Gas Emissions in the West Africa Region"

⁸⁷ Ministry of Environment and Sustainable Development, "Second national communication of Burkina Faso on climate change"

⁸⁸ <https://data.worldbank.org/indicator/EN.ATM.GHGT.KT.CE?end=2012&locations=BF&start=1998&view=chart>

Figure 15: Sectoral contribution to GHGs emissions in Burkina Faso (1999-2007)⁸⁹

2.3 Climate Change Impacts on Burkina Faso's Food Security and Agricultural Economy

47. **Burkina Faso is highly vulnerable to climate change impacts and has a low adaptive capacity.** Burkina Faso is ranked 39th in a list of countries with low readiness to climate change⁹⁰. It recognizes the need for climate change adaptation in national strategies, policies and programs. Yet low adaptive capacity and high vulnerability are due to inadequate information and technology, lack of synergies between relevant stakeholders and a weak education system.

48. **Climate change has already negatively affected food security, bringing climatic shocks and increasing demand for natural resources.** These shocks, the lack of sustainability of land use threatens the food security of the country⁹¹. Extensive droughts are likely to cause crop failure and death of livestock especially because Burkina Faso operates under a rain-fed production system⁹². Floods are less likely to cause damage compared to droughts, however a successive combination of both reduces the capability of food storage thereby resulting to chronic food insecurity.

49. **Most predictions of climate change impacts on biophysical suitability hold management and technology constant at current levels.** In reality, ongoing agricultural technology improvement in breeding and management will likely be able help offset some of these reductions. Moreover, farmers will certainly exercise adaptive agency—intentionally switching to an improved variety or an alternative crop or changing levels of inputs or farming methods—actively responding to shifting economic incentives induced by climate change. These techniques may bolster the resiliency of farmers, and the agricultural sector as a whole, to climate change.

50. **An analysis of the nine investments in this CSAIP was performed using the International Model for Policy Analysis of Agricultural Commodities and Trade (IMPACT).** IMPACT models the complex

⁸⁹Ministry of Environment and Sustainable Development

⁹⁰GAIN, "Notre Dame Global Adaptation Index Burkina Faso"

⁹¹ USAID

⁹² FAO, "Food Security and Humanitarian Implications in West Africa and the Sahel"

interplay of demographic, economic, mitigation, demand and consumption factors to understand what crops are likely to be viable and competitive in the future. Not all crops and commodities are affected the same way by climate changes – for example, there are strong differences both within groups of crops (e.g. cereals) – with maize, sorghum, rice and other cereals having different responses—as well as across different groups of crops, e.g. cereals versus fruits and vegetables or roots and tubers). Different crops show different levels of impact and resilience, which is reflected in different yield estimates.

51. **IMPACT modeling of climate change shows it will negatively affect the yield of all staple crop groups, in both the medium and long term, under a variety of socio-economic and representative carbon concentration scenarios.** Projected yields are shown for low, medium and high scenarios and for two time periods in **Other crops, especially fruits, vegetables, soybeans, groundnuts, oils seeds and sugar cane are likely to perform badly over the long term.** For many of these crops, it doesn't matter which scenario is viewed, there are similar declines in yield across all three RCPs at each time point. This is the case for temperate fruits, (declining around 5% in the short term and 10% by 2050), vegetables (declining around 5% in the short-term and 8% by 2050) and sugar cane, declining between 4-5 % in the short term and 9-12% in the long term.

52. **Cotton, one of Burkina Faso's most important exports, shows declines of between about 4% to 7% under all scenarios.** Given yield gaps in this sector, and the potential to improve resilience, these numbers are not so bad. They suggest that with appropriate interventions, there is the potential to increase the resilience of Burkina Faso's most important export crop.

53. **Some crops are climate resilient and offer potential for expansion.** Sweet potato and yams are nutritionally important and show resilience to a warmer future. Cowpeas and potatoes also show relative levels of resilience to climate change impacts under all RCPs for both time periods. As noted previously, other cereals and rice also appear to be relatively resilient.

54. **Cultivated area is projected to be higher for more crops under climate change than under a no-climate change reference scenario,** although many crops show less than a 1% change either way. Increased area under production is not necessarily a good thing, since expansion on to land with poor agricultural suitability drives land degradation (Table 5). CSA practices that improve intensification, in turn reducing land conversion, can therefore directly contribute to mitigation. Potatoes, groundnut, oilseeds, cowpeas, rice, millet and sugarcane all show an increase in harvested area expansion, in both the short and long term. For most time periods, pulses, vegetables, temperate fruits and other cereals show a decline in area.

55. Table 4. Nearly all crops, under all scenarios, in both time periods fare worse under climate change. However, because there is such low productivity with high yield gaps, the trends are not as dire as they may first appear.

56. **Most cereals, especially maize, exhibit high vulnerability to climate change no matter what scenario is picked,** from losses of between 9% to 12% in 2030 to over 18% to 22% in 2050 compared to a no climate change baseline. The good news is that millet and sorghum show relatively smaller declines in the short-term, but both only show losses of around 2% - 5% by 2050, while other cereals do better. Rice also declines, but less so than other cereals, suggesting there is room to grow resilient rice. Overall however, climate change impacts on cereals are of particular relevance in Burkina Faso, where they

constitute a large share of daily caloric intake and cultivated area (maize, sorghum, and millet are especially dominant).

57. **Other crops, especially fruits, vegetables, soybeans, groundnuts, oils seeds and sugar cane are likely to perform badly over the long term.** For many of these crops, it doesn't matter which scenario is viewed, there are similar declines in yield across all three RCPs at each time point. This is the case for temperate fruits, (declining around 5% in the short term and 10% by 2050), vegetables (declining around 5% in the short-term and 8% by 2050) and sugar cane, declining between 4-5 % in the short term and 9-12% in the long term.

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Table 4: Percentage difference in Burkina Faso rainfed crop yields over a no-climate change reference scenario for 2030 and 2050, under different representative carbon concentration scenarios (RCPs), with business-as-usual demographic and economic growth trajectories (SSP2).

crops	2018 Baseline Value(MT/ha)	RCP4.5_SSP2		RCP6.0_SSP2		RCP8.5_SSP2	
		2030	2050	2030	2050	2030	2050
CER-Maize	1.9391	-9.42	-18.12	-8.53	-16.34	-11.50	-21.69
CER-Millet	1.0411	-0.47	-0.97	-1.97	-3.88	-2.56	-5.03
CER-Other Cereals	1.0409	-0.98	-1.53	-1.08	-1.63	-1.88	-3.08
CER-Rice	1.5953	-1.87	-3.77	-1.41	-2.87	-0.97	-1.99
CER-Sorghum	1.2922	-3.77	-7.46	-3.70	-7.28	-4.75	-9.32
COT-Cotton	0.4965	-2.04	-3.87	-3.66	-6.78	-3.64	-6.50
COT-Other	1.1522	-0.18	-0.29	-1.05	-1.87	-0.78	-1.17
F&V-Temperate Fruit	6.7094	-5.66	-10.99	-4.93	-9.63	-5.66	-10.85
F&V-Tropical Fruit	6.9021	-1.25	-2.18	-3.37	-6.08	-3.57	-6.13
F&V-Vegetables	9.9878	-5.02	-9.73	-4.23	-8.24	-4.64	-8.87
OLS-Groundnut	0.8500	-1.70	-3.01	-4.10	-7.34	-4.47	-7.62
OLS-Other Oilseeds	1.3220	-1.70	-3.04	-3.73	-6.80	-4.07	-6.99
OLS-Soybean	1.2798	-3.65	-6.97	-4.97	-9.16	-5.46	-9.64
PUL-Cowpeas	0.5919	0.16	0.41	-0.79	-1.31	-0.46	-0.57
PUL-Other Pulses	0.8835	-1.75	-3.40	-2.71	-5.11	-2.61	-4.59
R&T-Cassava	4.3989	-2.28	-4.53	-1.95	-3.89	-2.00	-3.92
R&T-Potato	5.5495	0.28	-0.58	-1.91	-4.70	-0.12	-1.82
R&T-Sweet Potato	11.7151	-0.56	-0.75	-0.91	-1.41	-0.52	-0.33
R&T-Yams	10.1577	0.17	0.46	-0.45	-0.64	0.12	0.64
SGC-Sugarcane	104.8499	-4.83	-9.53	-4.34	-8.55	-6.04	-11.79

CER- cereals; COT-cotton; F&V- fruits and vegetables; OLS-oilseeds; PUL- pulses; R&T – roots and tubers; SGC- Sugarcane

Table 5: Percentage difference in Burkina Faso rainfed crop area over a no-climate change reference scenario for 2030 and 2050, under low, medium and high and carbon emission scenarios (different representative carbon concentration scenarios (RCPs)), with business-as-usual demographic and economic growth trajectories (SSP2).

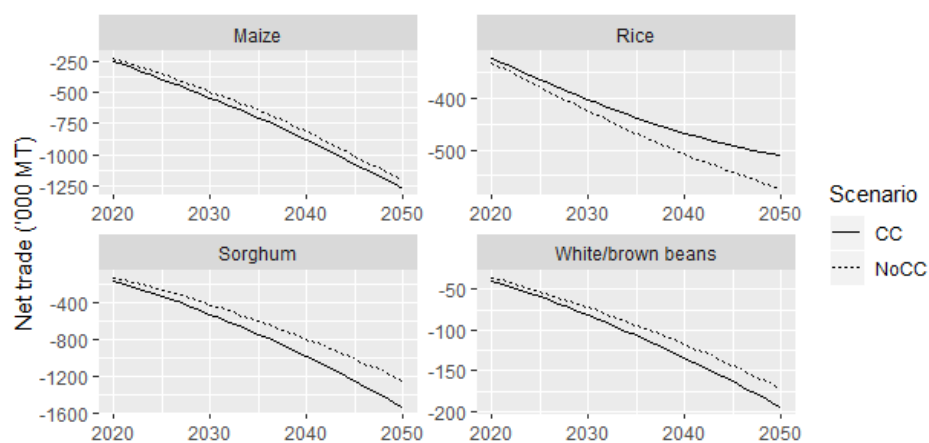
Crops CER- cereals; COT-cotton; F&V- fruits and vegetables; OLS-oilseeds; PUL- pulses; R&T – roots and tubers; SGC- Sugarcane	Percentage Change in Area Harvested from No Climate Change Scenario					
	Low Emissions		Medium Emissions		High Emissions	
	2030	2050	2030	2050	2030	2050
CER-Maize	-0.43	-1.48	-0.34	-1.15	0.35	-0.25
CER-Millet	0.83	1.43	0.45	0.68	1.13	2.06
CER-Other Cereals	-3.27	-6.19	-3.07	-5.86	-3.59	-6.58
CER-Rice	1.19	2.41	1.54	2.97	2.16	4.40
CER-Sorghum	-0.55	-1.22	-0.46	-0.95	-0.28	-0.62
COT-Cotton	-0.07	-0.05	-0.67	-1.24	-0.67	-1.19
COT-Other	0.04	-0.12	-0.04	-0.23	-0.18	-0.54
F&V-Temperate Fruit	-2.10	-4.09	-1.75	-3.48	-2.09	-4.12

F&V-Tropical Fruit	0.51	1.10	-0.12	-0.22	-0.27	-0.40
F&V-Vegetables	-2.86	-5.29	-2.09	-3.89	-2.66	-4.85
OLS-Groundnut	2.51	5.02	1.65	3.23	1.72	3.37
OLS-Other Oilseeds	1.31	1.96	1.87	2.51	1.79	1.86
OLS-Soybean	-0.08	-0.21	-0.32	-0.64	-0.46	-0.94
PUL-Cowpeas	1.10	2.22	0.95	1.90	0.86	1.76
PUL-Other Pulses	-5.08	-10.04	-5.24	-10.40	-6.22	-11.81
R&T-Cassava	0.39	0.91	0.25	0.56	-0.06	0.09
R&T-Potato	3.28	4.50	1.71	1.19	4.47	5.27
R&T-Sweet Potato	-0.15	0.09	0.03	0.32	-0.17	0.12
R&T-Yams	0.61	1.35	0.52	1.12	0.51	1.18
SGC-Sugarcane	2.39	3.99	2.16	3.60	3.00	4.69

Climate change will change Burkina Faso's comparative trade advantage for different crops. **Climate change suitability and yield across countries, and also affects the complex international interplay of socio-economic factors. Maize is expected to decline sharply, and so too will maize exports. Burkina Faso is dependent upon imports to meet its internal cereal demand—especially for maize and sorghum—and this dependence is aggravated by climate change (**

61. Figure 16). Rice does a bit better than expected, as it is more resilient to climate change impacts than other cereals. While the net trade declines, it is possible that actions could be taken to improve modelled trajectories. Sorghum and white and brown beans will also decline. The steepening trade deficit in maize shows little difference under climate change.

Figure 16: Net trade projections out to 2050. SSP2 RCP 8.5



CHAPTER 3: ASSESSING PRIORITIZED INVESTMENTS FOR CLIMATE SMART AGRICULTURE PORTFOLIO IN BURKINA FASO

62. **The initial list of 38 potential investments was reduced to nine priority investments for Burkina at a priority-setting workshop.** Participants used an iterative, qualitative and quantitative prioritizing process to review each of the 38 proposed CSA priorities. Specifically, eight clusters of criteria were used to reduce the 38 identified investments to the nine final investment priorities (see Annex C). These criteria used to evaluate and identify the nine final priorities are:

- a) **On-farm Value:** economic; nutritional; food security.
- b) **CSA Smartness:** productivity; adaption/resilience; mitigation.
- c) **Investment Objective:** growth in a new sector; resilience in a crop/sector that is already important.
- d) **Boosting Agriculture:** Agriculture value diversification; Infrastructure and connectivity.
- e) **Climate risks, climate mitigation, and productivity:** Addresses key climate risks; Increases agricultural productivity; Provides adaptation and builds resilience to climate risks; Reduces greenhouse gas emissions (absolute emissions or emissions intensity).
- f) **Finance and private sector engagement:** Improves access and affordability of finance for CSA; Improves competitiveness of the agriculture sector through infrastructure, incentives and other means; Provides private sector investment and business opportunities (including PPPs); Long term sustainability of investment and adoption of interventions.
- g) **Policies and institutions:** Aligns with national policies (for CSA); Improves institutional capacities (economic, financial, natural resource management and local government); Engages and strengthens farmer networks/organizations; Improves extension (public and/or private); Improves research & development to support CSA.
- h) **Social benefits and safety nets:** Provides job creation opportunities, enhances gender equity and engages the youth, Protects the poor and vulnerable through social safety nets, Improves access to clean water and sanitation.

63. **Names used in this report, especially in tables, have been shortened from the names identified at the prioritizing workshop.** Many of these names are simply too long to easily fit in a table, so for the purposes of this CSAIP, names are often shortened (as shown in Table 6), so that “Sustainable on-farm biogas production” appears as simply “On-farm biogas” or Biogas. The proposed investment called “Forest, Agroforest, and Garden Production for Climate-Smart Diversification,” has multiple components spanning forests and agroforests to improve their health and create value chains (among other things) is shortened to “Forest, Agroforest, and Garden” or more simply, “Forest and Garden”. Use of these shorter names does not detract from the fact that they represent from the full names or practices to support CSA; the shorter names are needed for the tables and simplicity.

Table 6: Complete and Shortened Names of Proposed Project Investments.

Full name	Medium length	Short
Sustainable on-farm biogas production	On-farm biogas	BIOGAS
Sustainable intensification of livestock production	Sustainable Livestock Intensification	LIVESTOCK
Finance and insurance services to foster climate-smart agriculture	Finance and insurance	FINANCE
Forest, agroforest, & garden production for climate-smart diversification	Forest, agroforest, & garden	FOREST & GARDEN
Building capacity in climate-smart agriculture	Capacity Building	CAPACITY
Sustainable management of water resources and irrigation	Water resources and irrigation	WATER RESOURCES
Developing climate-smart organic value chains	Organic farming	ORGANIC
Developing resilient oil-protein value chains	Oil-protein Crops	OIL-PROTEIN
Integrated Soil Management for Agricultural Productivity and Environmental Restoration	Integrated soil management	SOILS

64. All of the prioritized investments embody cutting edge and proven technologies and practices for climate smart agriculture (Table 7). Each of the nine priority investments was selected based on the criteria given above, and for their importance either to Burkina Faso, or to the region where they are located. Each project's main objective is shown in Table 7.

Table 7: Proposed Project Investments, Regions, Beneficiaries, and Objectives

NATIONAL SCALE INVESTMENTS
SUSTAINABLE ON-FARM BIOGAS PRODUCTION BENEFICIARIES: 65,000 farmers and their families OBJECTIVE: increase access to and knowledge of sustainable domestic energy sources in order to conserve wood resources, reduce greenhouse gas emissions, and reduce poverty, food insecurity, and health threats to rural communities.
SUSTAINABLE INTENSIFICATION OF LIVESTOCK PRODUCTION BENEFICIARIES: 150,000 farmers and their families OBJECTIVES: Foster climate resilience in Burkinabe livestock systems for improved food and nutritional security and economic outcomes
FINANCIAL AND INSURANCE PRODUCTS BENEFICIARIES: 200,000 farmers and their families OBJECTIVES: foster smallholder ability to invest in CSA innovations through good access to robust financial services, including credit, loan, insurance and risk instruments, savings and payment services.
FOREST, AGROFOREST, AND GARDEN PRODUCTION FOR CLIMATE-SMART DIVERSIFICATION BENEFICIARIES: 180,000 farmers and their families OBJECTIVES: Foster climate-resilient livelihoods and food sources for women and youth smallholders through home gardens, agroforestry, and non-timber forest product harvesting and value-addition.
BUILDING CAPACITY IN CLIMATE SMART AGRICULTURE BENEFICIARIES: 500,000 farmers and their families OBJECTIVES: Fully integrate national climate-smart priorities into Burkina Faso's agricultural research and extension programs, and build robust extension mechanisms for

delivering timely, practical climate-smart information to farmers and other stakeholders through highly accessible channels.
REGIONAL SCALE INVESTMENTS
SUSTAINABLE MANAGEMENT OF WATER RESOURCES AND IRRIGATION ZONES: Nord Soudanien BENEFICIARIES: 100,000 farmers and their families OBJECTIVE: Fully leverage Burkina Faso's water resources in sustainable ways to improve productivity, food and nutritional security, climate resiliency, and ecological health.
DEVELOPING CLIMATE-SMART ORGANIC VALUE CHAINS ZONES: Nord Soudanien and Sud Soudanien BENEFICIARIES: 60,000 farmers and their families OBJECTIVE: Improve organic production and develop markets to create viable careers and shorten supply chains in peri-urban areas.
DEVELOPING RESILIENT OIL-PROTEIN VALUE CHAINS ZONES: Nord Sahelien, Nord & Sud Soudanien BENEFICIARIES: 240,000 farmers and their families OBJECTIVES: Strengthen the quality, yield efficiency, and value chains of oil-protein crops to sustainably meet domestic demands and support international market development
INTEGRATED SOIL MANAGEMENT FOR AGRICULTURAL PRODUCTIVITY & ENVIRONMENTAL RESTORATION ZONES: North Sahel, North Sudan, South Sudan BENEFICIARIES: 200,000 farmers and their families. OBJECTIVE: provide producers and extension agents with location-tailored information on soil characteristics and best management practice recommendations, as well as the tools, products, partnerships, and policy environment to implement those recommendations.

3.1 Assessing Geographic Distribution of Burkina Faso's Priority Investments

65. The prioritized investments include five national plans, and four others that are distributed throughout Burkina Faso (Table 8). Five investments are national in scope, providing a set of investments that build capacity, resilience, support mitigation and offer human health benefits. The capacity project in particular benefits from economies of scale. Additionally, five projects focus on the North Sudan zone, including the organic agricultural project, which is focused on urban and peri-urban areas. Four additional projects also focus on the South Sudan zone. The North Sahel features two additional projects that will also be conducted in the Soudanian zones. No investments were focused on the Southern Sahel zone, with the exception of national investments, which could be targeted there. Stakeholders identified these agroecological zones, but they may also be called regions within this document, acknowledging that administrative areas that correlate need to be defined in future project stages.

Table 8: Prioritized CSA Investments by Agroecological Zone in Burkina Faso

North Sahel	South Sahel	North Sudan	South Sudan
		WATER RESOURCES	
		ORGANIC	
OIL-PROTEIN		OIL-PROTEIN	
SOILS		SOILS	
BIOGAS			
LIVESTOCK			
FINANCE			
FOREST & GARDEN			
CAPACITY			

3.2 Assessing Beneficiaries and costs of Burkina Faso's Priority Investments

66. **If all of the recommended investments for Burkina Faso were implemented, they would reach nearly 1.7 million people (Table 9).** Taken together, the four national-scale projects would reach 895,000 small farmers across Burkina Faso. The on-farm biogas project is projected to reach a relatively low number of beneficiaries, while the investment in Capacity Building would reach the greatest number of beneficiaries (500,000), but at a lower intensity. In contrast, the Oil Protein Crop investment is a large project that would reach 240,000 farmers, intended to build resilience and exports of an important crop that is largely managed by women. Integrated soil management would transform farming for 200,000 smallholders, and the Finance and Insurance investment would reach the same number. The smallest project is for Organic Farming, reaching 60,000 small farmers (See Annex E for further methodology).

67. **To reach these beneficiaries, the total cost of all investments is USD 472.5 million (Table 9).** The individual project average USD 52.5 million ranging in size from USD 37.5 million (Forest, agroforest & garden) to USD 65 million (sustainable livestock intensification). The cost per beneficiary averages USD 252 ranging from USD 110 (capacity building) to USD 1000 (organic farming).

Table 9: Beneficiaries and costs in Each Project.

(N) denotes national-scale projects.

Proposed Investments	Beneficiaries	Cost (Million USD)	Cost per Beneficiary
Organic farming	60,000	60.0	1000
On-farm biogas (N)	65,000	50.0	769
Water resources and irrigation	100,000	55.0	550
Sustainable livestock intensification (N)	150,000	65.0	433
Forest, agroforest, & garden (N)	180,000	37.5	208
Finance and insurance	200,000	40.0	200
Integrated soil management	200,000	55.0	275
Oil-protein Crops	240,000	55.0	229
Capacity building (N)	500,000	55.0	110
Total	1,695,000	472.5	252

68. **While all of the projects are likely to mostly reach poor Burkinabe farmers, some of the projects are more likely to primarily benefit women and youth.** Two of the smaller projects, Organic Farming and Biogas, will have strong benefits for women and youth. Fruits and gardens are mainly tended by women. Water Resources and Irrigation and the biogas investment reduce the time and risk of having women and children collect and haul water and fuel, and the poor health outcomes of indoor air pollution of burning

unclean fuel⁹³. Livestock, Forest and Garden, and Oil-protein crops are likewise primarily the responsibility of women within Burkina Faso.

3.3 Assessing Climate Smart Agriculture Pillars (Productivity, Resilience, Mitigation) of Burkina Faso's Priority Investments

Climate smartness is comprised of improving productivity, resilience and mitigation. Results from assessing these core components of CSA for the nine priority investments are presented below.

Productivity

69. **All of the CSA priority investments increase productivity by at least 18%, looking at the change in yield for beneficiaries with and without the project (Table 10).** Water Resources and Irrigation is the priority investment that shows the highest yield increase per beneficiary at 56%. Organic Farming is close behind with a 54% yield increase. Two other projects also perform over 40%: Biogas and Forest and Garden. Three projects, Oil Protein Crops, Livestock, and Soils have strong yield increases between 29% and 39%. The two projects that provide less direct benefits, Finance and Insurance and Capacity Building, have lower relative yield improvements, but still come in at 18% to 19%. Looking at the Standard Deviations (SD) of the projects, three of them have relatively narrow deviations: Capacity Building, Finance and Insurance, and Oil Protein Crops.

Table 10: Change in Yield (% , SD)

	Change in Yield (% , SD)
Water Resources and Irrigation	56% (41%)
Organic Farming	54% (45%)
On-Farm Biogas	45% (28%)
Forests, Agroforests, and Gardens	40% (21%)
Oil-Protein Crops	39% (9%)
Sustainable Livestock Intensification	38% (28%)
Integrated Soil Management	29% (34%)
Finance and Insurance	19% (10%)
Capacity Building	18% (8%)

Note: Change in Yield (% , SD): Average percentage change between beneficiaries with vs without project

Resilience and Risk

70. **One way of showing the resilience of projects is by understanding the risks, and then using the probability of a positive NPV (Net Present Value) as an indicator of resilience.** This investment plan and

⁹³ CITATION

the economic analysis provide for a direct targeting of investments for the climate and pest risks threatening productivity and growth in the future. As described in Annex E, the midpoint of the discount rate was 13%, but was based on a distribution ranging from 4% on the low side to unbounded, with 90% of values less than 19.5%. This guards against an overly optimistic analysis when risks are excluded from economic analysis. As shown in Table 11, four of the nine investments have a better than 50% chance in the face of uncertain climate and social risks. This suggests that these particular investments are strongly robust for tomorrow's environmental conditions. Two additional investments have a better than 40% change of a positive NPV with risks. The three riskiest projects are Biogas, Organic Farming, and Water Resources and Irrigation. Notably, those three projects also provide the highest benefits (see Production above) to the smallest numbers of people (see Beneficiary section above).

71. **If risks are ignored, the chances of a positive NPV increase in all investments.** Looking at the priority investments without risks included, all have a better than 50% chance of a positive NPV. Excluding risks, Forest and Garden and Oil Crops show a better than 90% change of a positive NPV. For the projects with a higher chance of a positive NPV, there is only a small difference in results with and without risk. Some of the projects also show a higher sensitivity to risk. Water Resources, Oil-Protein Crops, and Organic Farming show a higher sensitivity and lower resilience to potential risks (e.g. over 25% difference). For this analysis of risk and resilience, over 100 model runs were done. While the modeling is robust, the reality is that risks make the likelihood of success more uncertain. For this analysis of risk and resilience, over 100 model runs were done. While the modeling is robust, the reality is that risks make the likelihood of success more uncertain. Overall, the chance of a positive NPV are good for most of the investments. This investment plan took a different approach by using the best available information on the specific risks that farmers in Burkina Faso are facing, and therefore simulating how investments may perform under the conditions of tomorrow.

Table 11: Chance of a Positive NPV With and Without Climate and Social Risks

	WITH	WITHOUT
	RISKS	RISKS
	Chance Positive NPV (%) ²	
Forests, Agroforests, and Gardens	89%	92%
Integrated Soil Management	71%	83%
Capacity Building	62%	80%
Oil-Protein Crops	60%	90%
Finance and insurance	45%	63%
Sustainable Livestock Intensification	43%	65%
On-Farm Biogas	32%	51%
Organic Farming	29%	54%
Water Resources and Irrigation	21%	54%

² Average of 100 model runs

Emissions

72. **If all the priority investments were implemented in Burkina Faso, they would sequester 4.31 Mt CO₂ overall.** Only one of the investments, Livestock Intensification, will be a source of net greenhouse gas

(GHG) emissions (Table 12). Overall, this investment reduces emissions intensity from livestock (cows, goats and sheep). It also can be justified because it supports increased consumption of animal source foods, a critical nutrition intervention to combat childhood stunting and wasting. The investments with the greatest emissions reduction potential are Integrated Soil Management and Forest and Garden. Both of these increase soil carbon storage, and reduce practices that generate emissions. Other investments may lead to both the extensification and intensification of land with concomitant increases in absolute levels of emissions. Creating and supporting the appropriate structures (e.g. through the Capacity Building investment) to nurture pathways of sustainable intensification so that productivity grows without creating additional environmental degradation will be critical.

Table 12: Emissions from priority investments (mtCO₂)

Sustainable Livestock Intensification	-0.69
Water Resources and Irrigation	0.03
Organic Farming	0.04
Oil-Protein Crops	0.11
Capacity Building	0.21
Finance and insurance	0.25
On-Farm Biogas	0.53
Integrated Soil Management	1.25
Forests, Agroforests, and Gardens	2.58
TOTAL CARBON SEQUESTERED	4.31

73. **Of the nine priority investments, some stand out in how strongly they support the fundamental CSA pillars.** A summary of the top five investments for each of the three CSA pillar categories is shown in Table 13. Of the nine investments, only Forest and Garden appears in all three categories, and is the strongest performer on resilience with risks and on reducing emissions. Based on their rank order, Integrated Soil Management, with the second highest rank in two categories, is the next strong investment across the CSA pillars. This is followed by Biogas, which is the third strongest for yield and reducing emissions, and then Capacity Building. Oil-Protein Crops and Finance and Insurance are with the lowest ranks, but both appearing in two categories.

Table 13: Climate Pillars: Ranking the top five investments in each CSA Smartness category.

YIELD	RESILIENCE WITH RISKS	REDUCING EMISSIONS
Water Resources and Irrigation	Forests, Agroforests, and Gardens	Forests, Agroforests, and Gardens
Organic Farming	Integrated Soil Management	Integrated Soil Management
On-Farm Biogas	Capacity Building	On-Farm Biogas
Forests, Agroforests, and Gardens	Oil-Protein Crops	Finance and insurance
Oil-Protein Crops	Finance and insurance	Capacity Building

3.4 Economic and Financial Assessment of Burkina Faso's Priority Investments

74. **The nine investments are predicted to provide significant benefits for farmers in Burkina Faso.** As shown in Table 14, the cost-benefit analysis is presented both with and without risks, so the potential magnitude of foreseeable risks can be understood. Without risks included in the model, net present value (20-years) ranges from \$16.9 million with the Biogas program to more than \$196 million for Forest and Garden.

75. **The projects with lower NPV generally have high up-front investment costs per beneficiary, but they are also the same projects with the highest yield per beneficiary⁹⁴.** Costs per beneficiary also support the entire project, and are pro-rated to beneficiaries. Some projects may have more complex and higher overall costs. Others may have lower administrative costs, but high start-up costs. The relatively low NPV of the Biogas investment can likely be attributed to the high initial investment cost for the equipment. Biodigesters often cost more than \$500 per household and that, plus the program implementation costs create poor conditions for very productive investments. It may be however that NPV is the wrong indicator to judge a green energy product. There are many reasons, ranging from greenhouse gas emissions to health and sanitation, to promote on-farm biogas. These values are not captured in the project's NPV. Other projects with a lower NPV include Organic Farming, which includes solar irrigation systems and improved storage for vegetables and fruits (which is usually cool storage). Cost for solar irrigation systems is about \$1000 per system, and cool storage units are similar. Water resources and irrigation has costs of about \$1000 per smallholder system and rehabilitation of ponds/water bodies. In contrast, the Farm and Garden valuation represents the opposite condition, low investment requirements, large numbers of relevant beneficiaries, and high prices for carbon. The parkland agroforestry systems of Burkina Faso create a ripe opportunity for sustainable development. These wild harvesting systems often promote gender inclusion in the marketplace, with knock-on effects for health and nutrition of households.

76. **All investments are expected to improve the income and productivity of farmers.** Evidence shows that investment in improved agricultural practices increases the outputs of the farm. Key types of investments advocated for in Burkina Faso CSAIP include organic waste management which provides nutrient rich composts to the soil, preserving and building soil fertility, improved breeds and feeds in Sustainable Livestock Production and improved water distribution and management. Importantly, the investment plan suggests interventions that are also geared to help farmers perform under uncertain and harsh conditions, such as improved access to Financial Products and Insurance (e.g., credit and index-based insurance).

Table 14: Assessing Economic and Financial Performance of the nine priority investments within the Burkina Faso Climate-Smart Agriculture Investment Plan (with and without climate and social risks).

NPV = Net Present Value, ROI = Return on Investment, BCR = Benefit Cost Ratio with standard deviations (sd). NPV and ROI are average of 100 model runs

⁹⁴ The midpoint of the distribution used for discount rate was 13% and ranged between a low of 4% and unbounded on the high side, with 90% of values less than 19.5%. See Annex E.

CSA INVESTMENT	Estimated Project Budget (Millions USD)	Mean NPV (million USD)		Mean ROI		Mean BCR (SD)	
		WITH	WITHOUT	WITH	WITHOUT	WITH	WITHOUT
Integrated Soil Management	60.00	72.0	128.9	1.35	2.46	5.30 (7.85)	5.64 (8.00)
Organic Farming	50.00	-5.1	21.8	-0.15	0.48	2.99 (14.31)	4.99 (17.12)
On-farm biogas	55.00	-1.5	16.9	-0.09	0.30	2.72 (3.96)	3.33 (4.33)
Water resources and irrigation	65.00	-15.8	22.9	-0.35	0.35	2.63 (5.54)	5.04 (6.96)
Sustainable Livestock	37.50	16.4	76.1	0.67	2.57	1.97 (4.90)	2.88 (5.70)
Finance and insurance	40.00	15.9	52.2	0.43	1.51	1.54 (2.36)	1.84 (2.47)
Forests, Agroforests, & Gardens	55.00	168.2	196.1	3.47	4.06	0.77 (7.26)	1.78 (7.74)
Oil-Protein Crops	55.00	32.5	106.1	0.62	2.19	0.36 (2.39)	1.15 (2.76)
Capacity Building	55.00	52.5	140.4	1.06	2.95	0.21 (0.92)	0.62 (1.24)

77. **Overall, differences in the estimated productivity are due to inherent variation in the costs of interventions, the number of beneficiaries reached, and the relative speed at which interventions reach scale.** When considering the realities of inter-annual weather fluctuations, pest outbreaks, and social and political drivers of adoption, the estimated net present value of the investments drops. This is perhaps most noticeable for Organic Farming. This reflects the extent of negative impacts climate change is expected to have on crop and livestock production in Burkina Faso and provides further justification of the need for CSA practices. Importantly, the investment plan suggests interventions that are also geared to help farmers perform under uncertain and harsh conditions, such as improved access to Financial Products and Insurance (e.g., Credit and Index-based Insurance).

78. **The extent to which these on-farm improvements translate to overall investment productivity varied as a result of the variable costs of interventions and the pace of adoption.** A few investments stuck out as high performing, such as Capacity Building. This investment had a relatively low level of individual change but when aggregated across the large domain of applicability has potential to transform the landscape. With its mixed face-to-face training with smart use of information and communication technologies, the investment will be able to reach many beneficiaries quickly and thus has potential to create substantial change despite its lower per beneficiary result. Common types of interventions that are the result of receiving information, such as on-time planting, have a relatively low impact on average yields (i.e., 4-8%) compared to planting at a suboptimal time. Of course, when poor planting decisions occur, they create crop failures in households not participating in the program, so relative changes in production can be more substantial, and are captured in the tails of distributions used in the models. Or, soil management improves soil fertility, but typically this happens over longer time horizons, and depends on specific management interventions. So an important improvement over a long time may not show dramatic results, but such investments can be critical when looking at the health of agricultural systems.

79. **The analysis suggests there is potential for positive returns with each of the investments --but there are risks.** ROI averages 1.87% across the nine investments, in the *No Risk* and *Climate and Social Risks* scenarios, and 0.78 with risks included. Yet all of the ROIs are positive when Risk is excluded. While some types of investors (e.g. impact investing) may look for strong positive returns, these may not be appropriate for this CSAIP because: investments in smallholders is near universally below this ROI level and these investments intend to catalyze change and improve the capacity and suitability so that others investors may be willing to enter. For the On-Farm Biogas, the low return rate may simply be due to the

inability to value the ancillary ecosystems services delivered. As expected, including climate and social risks decrease the likely ROI.

80. **Benefit-cost ratios for practically all of this CSAIP portfolio investments are within the range of the investments identified by the Global Commission on Adaptation⁹⁵, suggesting these programs are consistent with and equally good investments as those identified in other initiatives.** Ironically, some of the strongest performing projects for the CSA pillars (Table 14) (Forest and Garden, Oil-Protein Crops, and Capacity Building) are those with the lowest mean BCR. This suggests that it is important to view these proposed project investments through different lenses, for their potential transformative value. Ultimately, they represent a public investment in growth in the agricultural sector, while benefits are accrued privately by farmers. Thus, strict rules on levels of return necessary may not hold.

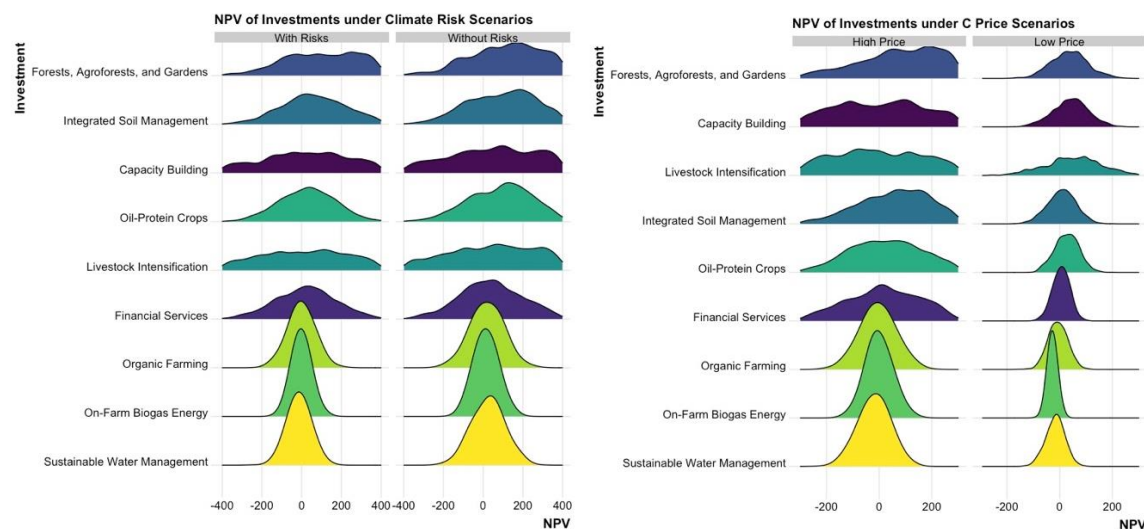
81. **An equivalent to a sensitivity analysis was performed for NPV, using two levels of carbon pricing (low and high), both with and without climate and social risks (Table 15 and Figure 17).** With high carbon prices and no risks included, all of the proposed investments have a positive NPV. The mean NPV shows an especially high sensitivity to high carbon prices for the three investments (Forest and garden, integrated soils, and on-farm biogas) that strongly support mitigation. Yet for some of the projects (e.g. on-farm biogas) even high carbon prices are insufficient to offset potential climate and social risks, even though it makes them more attractive financially, and goes a long way to offsetting the high cost/person. It also highlights the tradeoffs; the biogas project has high up-front costs, but also high mitigation potential. This analysis shows that the financial and economic analyses are highly sensitive to some of the model assumptions and provides a strong rationale for careful and cautious analysis of both future carbon pricing, and risk calculations within agricultural sector investments.

Table 15 : Comparison of mean Investment NPV (million USD) under Carbon Price & Risk Scenarios

Investment	No Risk - Low C Price	No Risk - High C Price	With Risks - Low C Price	With Risks - High C Price
Capacity Building	129.9	140.4	42	52.5
Sustainable Livestock Intensification	110.3	76.1	50.7	16.4
Oil-Protein Crops	100.8	106.1	27.2	32.5
Forests, Agroforests, and Gardens	66.9	196.1	39	168.2
Integrated Soil Management	66.3	128.9	9.4	72
Finance and insurance	39.6	52.2	3.2	15.9
Water resources and irrigation	21.5	22.9	-17.2	-15.8
Organic Farming	19.8	21.8	-7.2	-5.1
On-farm biogas	-9.6	16.9	-28	-1.5

⁹⁵ Global Commission on Adaptation, WRI, 2019

Figure 17: NPV of Prioritized Burkina Faso Investments with and Without Risks and for Low and High Carbon Prices



3.5. Climate Modeling Assessment of CSAIP Priority Investments

82. **The CSA investment plans are built upon the strengths of commodities exhibiting resilience under climate change, while simultaneously offsetting potential damages to commodities exhibiting vulnerability.** We assessed the potential impact of CSAIPs on yield, food security, and trade trajectories in Burkina Faso out to 2050 under climate change, given different assumptions regarding demographic, economic, and emissions over this period. This analysis identified the crops that exhibit resilience and vulnerability under climate change and focus on how the CSAIPs can leverage/offset this resilience/vulnerability. For this analysis, the team considered yield shocks, expected time horizons and adoption rates associated with each of the CSAIPs. This input came from experts familiar with these technologies and practices. The IMPACT scenarios below show “with CSAIP” and “without CSAIP” future climate change scenarios out to 2050, for the Business as Usual Shared Socioeconomic Pathways (SSP 2), and different assumptions about future greenhouse gas concentration scenarios (Representative Concentration Pathways).

83. **The Sustainable on-farm biogas production and Integrated soil management CSA interventions are also poised to give a boost to key cereal crops,** one through the integration of livestock and cereal systems, the other through soil recovery and conservation practices. Maize yield growth in particular stands to benefit, which is otherwise projected to flatten in the near future under climate change. These practices are also projected to substantially offset Burkina Faso’s steepening trade deficit in sorghum and millet (**Error! Reference source not found.**).

84. **The Oil protein crop CSA investment is projected to offset declining soybean yields and to enhance white and brown bean yields.** Soybean yield is projected to be one of the hardest hit trajectories by climate change. These CSA interventions could lay the groundwork to offset a fledgling trade deficit in these products (Figure 19).

Figure 18: Potential impact of the On-farm Biogas and the Integrated Soil Management Investments given their potential to improve maize, millet, and sorghum yields and balance of trade.

Trajectories modeled using IMPACT under a business-as-usual shared socioeconomic pathway (SSP 2) and a pessimistic representative carbon concentration scenario (RCP 8.5).

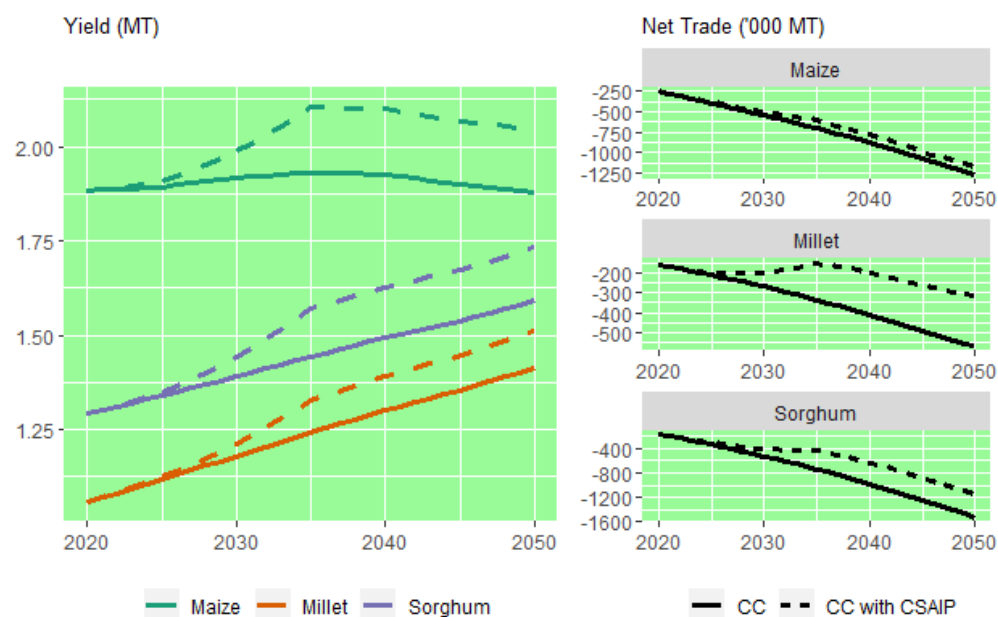


Figure 19: Potential impact of the development of oil protein crops CSA investment on soybean, white and brown bean yields and balance of trade.

Trajectories modeled using IMPACT under a business-as-usual shared socioeconomic pathway (SSP 2) and a pessimistic representative carbon concentration scenario (RCP 8.5).



3.6 Assessing CSAIP Investment Alignment with Burkina Faso's Nationally Determined Contribution

85. **Due to the high dependency on the Agriculture, Forest and Land-Use sector and its vulnerability to climate change, the Burkinabe government has emphasized mitigation and adaption actions in this sector.** The Burkinabe government has put forth Nationally Determined Contributions to the Paris Climate Agreement. Burkinabe's commitment to a climate-smart future is also visible in the goals of the Fond d'Intervention pour l'Environnement (FIE)⁹⁶ whose mission is to contribute to the environmental objectives of Burkina Faso. Those initiatives are supported by the international community, including the United Nations, Economic Community of West African States (ECOWAS), and donors such as the West African Development Bank, the African Development Bank (AfDB), the World Bank (WB), and the European Union (EU)⁹⁷.

86. **This CSAIP is clearly aligned with Burkina Faso's Nationally Determined Contributions, both for high level objectives and specific investment activities.** The World Bank is supporting Burkina Faso's development of this CSAIP in alignment with and under the auspices of the NDC. A new NDC Partnership Plan is under Development, and coordination between funded CSAIP investments and the NDC is vital. It will be crucial to ensure that this alignment and support continues. The NDC includes two types of objectives: mitigation and adaption. While the contribution of Burkina Faso to global greenhouse gas emissions is very low, the country has committed to international initiative such as Reducing Emissions from Deforestation and Forest Degradation Program (REDD+), Forest Investment Program (PIF), Nationally Appropriate Mitigation Actions (NAMA). The adaptation objective is more ambitious as it aimed to use a Project/Activity/Action. AFOLU is the sector producing the most GHG, however with more than 80% of the population working in that sector, it is also the sector contributing the most to the Burkinabe economy and the most vulnerable to climate change. The adaption objective stresses first improving environmental services through actions supporting food security, soil and water conservation, sustainable agriculture etc., which then lead to a long-term reduction in GHG intensity through projects. Those projects fall in the 4 sectors: AFOLU, Habitat and urbanism, Health, extreme climatic events management. The Burkinabe NDC integrates all 3 pillars of CSA: productivity, adaptation, and mitigation. Given unavoidable, continuing, erratic and variable climate change impacts, greater additional focus on adaptation is warranted to ensure continued productivity.

The proposed investments make robust contributions to the adaptation objective in the AFOLU sector at the national scale (Table 16). For example, of ten components in the NDC, the Building Capacity investment supports seven objectives. Developing the capacity for CSA implementation is key to better disseminate technologies and innovations to the farm level. Introducing CSA into the national extension system supports reduced emissions, climate resiliency, and equitable development via sustainable intensification and natural resource management. The Finance and Insurance investment supports five objectives, and helps to adaptation objectives by developing climate insurance, helping farmers adapt to new crops and technologies, and improving long-term productivity and resilience.

⁹⁶ FIE, "Fonds d'Intervention pour l'Environnement."

⁹⁷ Governemnt of Burkina Faso, "Contribution prevue determinee au niveau national (CPDN) au Burkina Faso."

Table 16: Alignment of CSA Investment Plans with NDC Partnership Investments

Only those with at least one area of strong alignment or two areas of partial alignment are shown here. See Annex C for additional information.

	Strong alignment	Partial alignment	Little or no alignment							
BURKINA FASO CSA PRIORITY INVESTMENTS										
NDC Components	Building Capacity	Finance	On-farm Biogas	Livestock	Forests Gardens	Organic Farms	Oil-protein	Soils	Water Mngmt	Dams
Improved farming technologies/ CA										
Conserve water & irrigation mgmt										
Soils: manage, conserve fertility										
Governance reform										
Post-harvest handle & process										
Renewable energy support										
Livestock & aqua-culture										
Info mgmt/ advisory services										
Restoring forests/ agroforestry										
Financial services support										

3.7 Policy Coherence: Alignment, Gaps and Distortions with Other Policies, Strategies, and Commitments

87. **The CSAIP addresses and supports the goals of multiple national policies and strategies.** This includes the National rural sector program, national climate change adaptation plan, national adaptation program of action, the forest investment program, the national climate change learning strategy, the African adaptation program (**Error! Reference source not found.** and Annex C). The CSAIP also strongly supports several main components of national forest policy, national action on integrated water management, national plan for capacity building on risk reduction and response to emergencies in Burkina Faso and other policies key to maintaining the agricultural sector's growth and contribution to the national economy. Furthermore, by promoting and expanding the adaptation approaches previously discussed, the CSAIP builds resilience within national institutions themselves, thus positively impacting myriad aspects of national policy. These investments thus support Burkina Faso's current national policies, and also its future developments in agriculture and food security in the face of climate change.

88. **Implementation of the CSAIP will make significant strides toward meeting several of Burkina Faso's international commitments.** In addition to the NDCs, Burkina Faso is signatory to other international commitments, including:

- Sustainable Development Goals (SDGs)
- Comprehensive Africa Agriculture Development Program (CAADP)

- Malabo Declaration on the Transformation of Agriculture
- African Union 2063 Agenda
- African Forest Landscape Restoration Initiative (AFR100)
- Land Degradation Neutrality Target Setting program of the United Nations Convention to Combat Desertification (UNCCD)

89. **The CSAIP will move Burkina Faso closer to meeting all of these international commitments.** Every one of the proposed investments address the poverty reduction and environmental conservation aims of the 2030 Sustainable Development Goals and the African Union 2063 agenda for inclusive and sustainable development⁹⁸. Burkina Faso is on track with the re-commitment to CAADP Process (score 5.63/10, minimum 3.33)⁹⁹, nevertheless, the country only has 7.7% of agriculture land under sustainable land management practices. Implementing the investments in this CSAIP will help Burkina Faso get back on track to meeting its Malabo Declaration commitment to enhancing investment finance in agriculture, specifically access to finance and to ending hunger by 2025 by doubling agricultural productivity and reduction of post-harvest losses¹⁰⁰. Finally, these investments will significantly augment Burkina Faso's contribution to AFR100's goal of restoring 100 million hectares of deforested and degraded landscapes across Africa by 2030 with 5 million hectares in Burkina Faso.

Policy Gaps and Distortions

90. **Investments in CSA program design and implementation may be constrained by the existing policy context.** As shown in Table 17, there are some barriers that are more common (top of the table), and investments that are more likely to be threatened by multiple policy gaps or barriers (more affected investments on the left side). There are three barriers present in all projects: (i) poor land tenure regimes, (ii) a dearth of finance services for agricultural stakeholders, particularly smallholder producers, and ii) weak extension services, including climate and soil services. Additional important barriers include: (i) institutional capacity (ii) market distortions, (iii) and low political will and a lack of inter-institutional coordination. Distortions or gaps in digital policy, conservation, standards and certifications, funding, and research and development may also affect the implementation of some investment programs.

91. **Investments in CSA are disincentivized by current tenure regimes, input costs, and land-use planning practices.** Tenure insecurity and poor marketing systems are major limiting factors to national agricultural development. The cost of inputs to improve productivity on existing lands often exceeds the cost of agricultural expansion, leading to forest clearing. Minimal resources and low capacity make rural land use planning and forestry conservation laws are difficult to enforce.

⁹⁸ African Union, "Agenda 2063: The Africa We Want."

⁹⁹ AUDA-NEPAD, "Tracking Progress: Burkina Faso."

¹⁰⁰ AUDA-NEPAD.

Table 17: Policy Gaps and Distortions

Policy Gaps	Proposed Investments									
	Finance	Capacity	Organic	Oil-Protein	Soils	Live-stock	Forest Garden	Biogas	Water	Dams
Land Tenure										
Finance Services										
Extension										
Institutional Capacity										
Markets										
Political Will /Coordination										
R&D										
Digital										
Program										
Standards/										
Conservation Management										

92. **Climate change risks may be aggravated by the policy environment, and vice versa.** Some of these risks are incorporated into the economic models (see Annex E). While climate stresses cannot be ignored, there are climate adaptation efforts underway, from farmers changing planting times or varieties, to national-scale efforts to promote adaptation within larger sectors. Urban and rural impacts and risks will be quite different and will have different pathways in terms of who is affected, how, and to what extent. The IMPACT modeling reflects this for global trade competitiveness given changing advantages in the agricultural sectors of different countries.

93. **Several pro-CSA policies are not fully implemented due to weak collaboration between sectors and governmental ministries.** Multiple Burkinabe policies outline CSA goals without specifying precisely how they will be achieved or by whom. As a result, there is poor coherence between sectoral policy documents and approaches. Competition and disagreement between sectors and ministries exist rather than alliances and collaboration toward common goals. This has been recognized as a major limiting factor to development by the Ministry of Food and Agriculture, among other key stakeholders. Defining of roles, synergies, compromises, and trade-offs between stakeholders, and in particular state institutions, will be a vital step toward implementing and improving the effectiveness of existing pro-CSA policies, particularly given limited resources and budgets. Better coordination across government entities or a single unit to coordinate between the government and private sector could increase the very limited private sector engagement in Burkinabe CSA programs at the present time.

94. **Most national policies give attention to climate change issues, and there is opportunity to better align these policies with national objectives.** Legislation going as far back as the 1995 National Forest Policy acknowledge climate change. However, many stop short of offering concrete steps toward adaptation, mitigation, or climate-smart agriculture. For example, the Strategy for Accelerated Growth

and Sustainable Development (2011-2015) and National Economic and Social Development Plan (2016-2020) mention climate change only in passing, and do not address any specific adaptation or mitigation measures. The National Climate Change Learning Strategy (2016), the African Adaptation Program (2012), and the Reducing Emissions from Deforestation and Forest Degradation Strategy (2010) do not address climate-smart agriculture. Identifying the specific cohesive (and conflicting) elements of these policies with climate-smart initiatives can help policymakers devise coordinated approaches to holistic climate-smart development goals.

95. **Distortions of some pro-CSA policies and subsidies are creating potential further barriers to CSA implementation.** For example, the Burkinabe government has prioritized the soybean industry to help fill domestic demand gaps and reduce reliance on imports. However, the monetization of this and other oil-protein crops are causing farmers to abandon CSA practices such as intercropping and fallowing, and is pushing women out of production of oil-protein crops. Both of these shifts increase smallholder and national vulnerability to climate impacts. Careful design and implementation of policies, and a legislative understanding that revisions are expected as policies come into force, can resolve these unintended consequences. Maximizing policy coherence is crucial to achieving all agreed national objectives.

96. **Institutional capacity increases will be foundational to meeting Burkina Faso's national climate-smart goals.** Stakeholders, including the Ministry of Agriculture, have repeatedly highlighted the need for investments in the capacity needed to move toward integrated implementation of national planning priorities. Enhanced institutional capacity would prepare ministries for the necessary realignment of existing policy frameworks to achieve coherence across policies and establish coordinated efforts across departments, ministries, and sectors.

97. Burkinabe farmers have little or no access to the essential services necessary for successful agricultural production. This includes land climate, soil, market, and technical extension information, finance and insurance services, and land tenure. Information services support farmers' capacity to implement new CSA practice, and inform their decision making regarding the same. Finance and insurance services give farmers access to the capital and risk protection needed to implement new practices. Land tenure ensures they will be able to benefit from these investments in the future.

98. **Without climate information services, most Burkinabe smallholders rely on traditional techniques and indigenous knowledge to forecast near-term weather and season climate.** Traditional knowledge such as when birds migrate, when plants germinate, and changes in cloud cover and wind patterns have historically guided farmers' decision-making. However, the erratic weather conditions brought by climate change have made such knowledge and practices increasingly unreliable. The current national extension model is costly to implement, and has not scaled to reach the majority of farmers.

99. **Farmers are not receiving agricultural extension services based on state of the art research, technology, or science innovations.** Farmers respond to this lack of information by delaying farming decisions, hoping they will receive some useful information later. Delayed decision-making, combined with erratic weather patterns, often results in farmers missing crucial windows for field management activities. Climate and knowledge services fix this gap by providing timely, reliable agricultural forecasting

if all stakeholders are involved in both producing and disseminating climate information. Climate and knowledge must be available and designed to reach to smallholders via a variety of channels and formats. For example, the CCAFS has been a leader in participatory production of climate services that integrate farmer knowledge and experiences with meteorology.

3.8 Assessing Design and Implementation Opportunities in CSAIP Priority Investments

Opportunities: Building on Existing CSA Activities

100. **Momentum from existing programs and initiatives can be used to leverage continued growth and development in Burkina Faso's agricultural sector.** Many programs and initiatives (Table 18), funded by both the Burkinabe government and international donors, have driven important agriculture sector growth. Most projects have focused on enhancing productivity, sustainability, and rural livelihoods; some projects also include capacity development. Several initiatives focus on fostering an enabling environment for private sector actors in various commodity value chains. Smallholders' adaptive capacity remains low, particularly in northern Burkina Faso. All programs should take into consideration the prerequisite land tenure, finance services, extension support, and climate information that all smallholders need in order for programs to meet the broader goals of the NDCs.

101. **Continued climate-smart innovation in Burkina Faso requires support from many sectors, including Burkina Faso's government, non-profits, the private sector, and international organizations.** Governmental policy is already strongly aligned with several programs in this portfolio, in particular biogas, oil-protein crops, and water management. The government of Burkina Faso has also been the sole funder of several programs. Bilateral and multilateral donors have been major programs funders to date, and have recently committed to strong continued support, particularly of climate adaptation efforts. Non-profits have played a key role in program implementation. A limited number of programs have been funded by private sector actors.

102. **International alliances strongly support various projects in this portfolio.** In addition to the NDCs, this investment plan closely aligns with multiple international accords to which Burkina Faso is party, including the Dakar Declaration on Irrigation, The Comprehensive Africa Agriculture Development Programme, the Sahel Irrigation Initiative, the Economic Community of West African States, Organics International, and the African Biogas Partnership Program. The international coalitions around water issues are particularly robust. Given the current and predicted future water scarcity in Burkina Faso, water management in the context of climate change is key to continuing agricultural sector growth. Careful coordination at the national level will help engage these multinational actors behind a climate-resilient Burkinabe agenda.

Table 18: Select current CSA projects in Burkina Faso (Note See Annex B, Section C for Past Projects)

Fund	Project	CSA Relevance	(USD \$M)	Term
World Bank	Burkina Faso Agriculture Resilience and Competitiveness Project	irrigation, land-tenure, input supply, advisory services, producer group services, institutional capacity, emergency response, market connections, finance services, natural resource management	261	2019-2025
	Local Forest Communities Support Project	strengthen capacity of local communities to participate in REDD; restore 2,000 ha of forest; increase benefit of 8,000 household from forest product and non-timber products; ensure 3,500 household have adopted sustainable land management practices	4.5	2015-2020
	First Fiscal Management, Sustainable Growth and Health Service Delivery Development Policy Operation	Increase livestock production and productivity through vaccination	50	2019-2020
	Livestock Sector Development Support Project	Improving livestock producers' access to animal health services and quality inputs to strengthen the efficiency and competitiveness	78.9	2017-2022
	Strengthening Climate Resilience in Burkina Faso	Institutional capacity building, infrastructure, and ICT development to deliver hydro-meteorological and climate information, early warning, and emergency response	33	2018-2024
	Financial Inclusion Support Project in Burkina Faso	Strengthening credit supply for farmers and small / medium-sized enterprises	100	2019-2025
Netherlands Development Organization, with World Bank, and Burkinabe National Government	National Biodigester Program	Clean energy course leveraging on-farm materials to reduce deforestation	6.6 from WB	2010-2024
United States Agency for International Development	Land Technology Solutions Project	Improve land and resources governance and strengthen property rights through Mobile Applications to Secure Tenure technology	3.99	2017-2020
African Development Bank	Support Project for Establishing an Agribusiness Bank	Enhance agricultural financing and insurance and support value chain development	10.86	2019-2021
United States Department of Agriculture	Sesame Marketing and Exports	enhancing the production, marketing and export of sesame	24	2016-2021
International Fund for Agricultural Development	Projet d'Appui à la Promotion des Filières Agricoles	develop the value chains of rice, vegetables, sesame, and cowpea and improve productivity for 57,000 households	71.7	2017-2024
International Fund for Agricultural Development	Agricultural Value Chains Support Project in the Southwest, Hauts-Bassins, Cascades and Boucle du Mouhoun Regions	improved access of certified seed and fertilizer, strengthen value chains of rice, horticulture, sesame, cowpeas, non-wood forest products, and fish.	123.93	2019-2025
Food and Agriculture Organization	Action Contre la Désertification	Improved productivity and land restoration	nd	2016-present
Fund	Project	CSA Relevance	(USD \$M)	Term

African Development Bank	Bagre Growth Pole Support Project	Extension of canal, surge tanks, and irrigation infrastructure; Construction of warehouse and post-harvest infrastructure; Train farmers on water and soil management	193	2011-2020
International Fund for Agricultural Development	Participatory Natural Resource Management and Rural Development Project in the North, Centre-North and East Regions	Reclamation of degraded lands using soil water conservation techniques such as stone bunds, vegetative barriers, filter dikes, zai micro-catchments.	117.45	2012-2021
GPRBA, with IFC and World Bank	Burkina Faso Smallholder Irrigation Project	Small-scale irrigation	5.85	2018-present

Design and Implementation Opportunities

103. **There are a number of favorable circumstances around each project that could support their success.** These may, in some cases, help offset the challenges outlined above. Eight of the nine proposed investments linked the strong knowledge base that exists in Burkina Faso as the greatest existing asset supporting implementation (Table 19). This was followed by policy alignment (7 investments), another important asset working in favor of these projects. Although policy is not, in all cases, consistently implemented, the all-important first step of establishing the regulation is already done. Availability of needed inputs (6), and highly climate resilient and strong demand (5 each) and aligned interests (4) will also support many of the projects proposed herein (Table 19). Existing assets for project success are described in greater detail in Annex A.

Table 19: Opportunities Supporting CSAIP Investment Design and Implementation

Opportunities	BIOGAS	FINANCE	ORGANIC	WATER RESOURCES	OIL-PROTEIN	SOIL	FOREST & GARDEN	CAPACITY	LIVESTOCK
Knowledge Base	✓	✓	✓	✓		✓	✓	✓	✓
Policy Alignment	✓	✓		✓	✓	✓		✓	✓
Inputs Available	✓	✓		✓		✓	✓	✓	
Highly Climate Resilient	✓	✓		✓	✓	✓			
Aligned Interests	✓	✓	✓	✓	✓				
Strong Demand	✓	✓	✓		✓		✓		
Uses Existing Technology	✓	✓	✓	✓					
Program Alignment	✓		✓		✓			✓	
Private Sector Support	✓		✓		✓				
Low Cost			✓				✓		
Funding Available	✓								

3.9 Assessing the Potential for Supporting Collaboration and Partnerships and Institutionalizing CSAIP Investments

104. **Partnerships, and the number of potential collaborators, can bring both strong benefits, as well as complexity, depending on what the arrangements are with specific partners.** Clarifying obligations up front is vital to strong partnerships, and to understand how well partners can support project design and implementation (e.g. what they bring) versus the obligations inherent in partnerships (what partners expect). There are also huge differences in the level of involvement of partners, whether providing support and expertise, funding, and whether legal requirements are involved. Projects with many partners need to prioritize and clarify roles and responsibilities of all partners at the outset of projects.

105. **Clarity on institutional arrangements will be important during further project design phases,** given that many of the proposed investments would, potentially, be involved with a high-number of ministries (Table 20). Table 20 only identifies potential ministries, not the myriad of other national-level Secretariats, Directorates, Federations, National Councils or other agencies; another 12 of these were identified in the prioritizing process. The Livestock investment recorded eleven different Ministries to engage with, while Finance and Insurance identified seven, and Biogas identified five. For public sector partners, this can bring particular challenges, with a need to identify lead agencies, funding flows and responsibilities, what Ministries or Agencies have authority, and how to resolve any policy conflicts. In terms of which ministries, virtually all envisioned coordination with the Ministry of Environment, Green Economy and Climate Change, and the Ministry of Agriculture and Food Security.

Table 20: Ministry Level Coordination and Proposed Investments

Key Public Institutional Collaborators	Biogas	Live-stock	Forest & Garden	Capacity	Water	Finance	Organic	Oil-Protein	Soils
MINISTRIES - total	5	11	2	3	3	7	4	1	2
Environment, G. Econ & CC	✓	✓	✓	✓	✓	✓	✓		✓
l'Agriculture & Aménagements Hydro-agricoles	✓	✓		✓	✓	✓	✓	✓	✓
Water Resources		✓		✓	✓				
Animal Resources	✓	✓				✓			
Higher Education, Scientific Research & Innovation	✓	✓							
l'industrie, du Commerce et de l'Artisanat		✓				✓	✓		
Health		✓	✓						
Economy and Finance		✓				✓			
l'Action Sociale et de la Solidarité Nationale		✓				✓			
l'Administration Territoriale et de la Décentralisation		✓				✓			
Energy	✓								
l'Urbanisme et de l'Habitat							✓		
Hydraulic Facilities and Sanitation		✓							

106. **Many of the proposed investments identified a large number of international and national collaborators, ranging from CGIAR international research institutions to Bukinabe NGOs to a variety of aid organizations.** The collaborators are identified in Annex A with each investment, and serve as a preliminary list, since others will be identified during the design and implementation process. Some of the investments, such as the Capacity Building Investment, identified large numbers (12) different NGOs or research organizations for collaboration. Again, this level of engagement has both benefits and costs, and the higher the number of collaborators, the more important it is to have a clear plan for what the collaboration entails. Yet overall, engaging both public and NGO collaborators offers strong potential mechanisms for institutionalizing CSAIP objectives, both for specific investment and for the overall CSAIP. Yet this will not happen spontaneously but must be purposefully planned.

3.10 Assessing Financing in CSAIP Priority Investments

107. **The Government of Burkina Faso will ultimately need to fully endorse and support this plan, and be active in identifying the financial resources to implement the suite of investments proposed in this CSAIP.** Additional funding streams are needed to take changes to scale and meet Burkina Faso's NDC commitments. At the COP 25 in Spain in 2019, Burkina's Minister announced that there was interest expressed by 10 different partners to support Burkina Faso's NDC implementation¹⁰¹.

108. **All of the investments in this portfolio help support Burkina Faso's objectives to improve food security and support rural livelihoods.** However, the interest and importance of climate change, agriculture, and environment may diminish as Burkina Faso deals with a humanitarian crisis and declining insecurity. One estimate is that nearly one-third of the population is affected by insecurity, and that in 2020, 2.2 million people will require humanitarian assistance. The UN is providing emergency food aid, and there is a 2020 Humanitarian Response Plan for Burkina Faso is requesting \$295 million to assist 1.8 million people¹⁰². This highlights the trade-offs between long-term investments that can improve food security and improve rural lives in the context of climate change, with the potential need to prioritize shorter term food aid over longer term agricultural development.

109. **Financing costs and needs are well within the scope of reasonable projects that are financed by a range of institutions and organizations.** There are a range of different approaches to funding for the nine investments, from multilateral and bilateral organizations, foundations and donors (e.g. Gates Foundation), national budgets, NGO support, and private sector financing (see Table 21 and potential funding sources for each investment in Annex A). All of the projects identified at least once source of government funding. Blended finance from the Government and the Environmental Intervention Fund (FIE) was proposed by three projects. In contrast, eight of the nine projects believed that financing could be obtained from the World Bank, with seven of the nine looking toward the African Development Bank, with six projects looking at financing from either (or both) the Global Environment Fund (GEF) or the Green Climate Fund. In all, the nine proposed investments identified more than 40 different potential sources of funding, ranging from international foundations (e.g. the Gates Foundation), bilateral assistance (Swiss

¹⁰¹ <https://sdg.iisd.org/news/six-countries-launch-ndc-partnership-plans-at-unfccc-cop-25/>

¹⁰² December 11, 2019 - <https://www.un.org/press/en/2019/db191211.doc.htm>

or Netherlands or USDA), regional African support, (e.g. ECOWAS), and a range of other initiatives shown in Table 21.

110. **There was also a strong role envisioned for the private sector, with many of the proposed investments identifying potential partners for private sector financing and collaboration (Table 21).** The Agriculture Bank of Burkina as the most likely source of private sector funding for: Forest and Garden, Water Management, Finance and Insurance, Biogas, and Soils. The Oil Protein and Organic farming Investments identified other potential funding private funding sources, including Sinergi Burkina, a new fund for small businesses (that might well be a partner, but the lending size is too small for the overall investment needed for the entire project). The Livestock Investment identified several potential funding streams from within that sector.

Table 21: Financing Needed and Key Identified Potential Sources

Proposed Investments	Estimated PROJECT BUDGET (millions of USD)	Total Beneficiaries	\$/ Beneficiary	Results of Maximizing Finance for Development Analysis (Preliminary)
Capacity Building	\$55.0	500,000	\$110	Multi & bilateral, donor organizations;
Finance & insurance	\$40.0	200,000	\$200	Multi & bilateral, and donor organizations; ABB;
Oil-protein Crops	\$55.0	240,000	\$228	Multi & bilateral, and donor organizations
Sustainable Livestock	\$37.5	150,000	\$250	Multi & bilateral, and donor organizations; private sector commercial producers/processors;
Integrated soil management	\$60.0	200,000	\$300	Multi & bilateral, and donor organizations; ABB;
Forest, agroforest, & garden	\$55.0	180,000	\$306	Multi & bilateral, and donor organizations; ABB; over 10 private sector;
Water resources & irrigation	\$65.0	100,000	\$650	Multi & bilateral, and donor organizations; ABB;
Organic farming	\$50.0	60,000	\$833	Multi & bilateral, and donor organizations; many private sector possibilities;
On-farm biogas	\$55.0	65,000	\$846	Multi & bilateral, and donor organizations; ABB;
Total	\$472.5	1,695,000		

ABB= Agricultural Bank of Burkina, a private bank

111. **Opportunities exist to greatly expand private sector investment, in ways that are consistent with Burkina Faso's private sector development plans and the World Bank's Country Partnership Framework (CPF)**¹⁰³. The World Bank and IFC recommend diversifying agricultural value chains, with key private sector entry points in agriculture and food processing. Many of the investments identified in this proposal support recommendations for private sector expansion and resilience. For example, cold-storage

¹⁰³ World Bank & International Finance Corporation, 2019. "[Creating Markets in Burkina Faso](#)," [World Bank Other Operational Studies](#) 32401, The World Bank.

units, irrigation systems, value-chain creation, and expansion of oilseed crops and livestock are all parts of this CSAIP that are components of private sector strengthening in Burkina Faso¹⁰⁴.

112. An operational framework to guide CSAIP programming into practice will help ensure that funding is focused on priorities crucial to project success. Effective frameworks support planning and implementation by producing concrete information. There are many potential private, public and international funders and financing instruments, as shown in Annex B. Developing financing strategies specifically for priority investments, or packages tailored to specific donors for financing can help secure funding. Specific targeting and proposed partnerships with the private sector, where appropriate, are also vital and recommended in recent analyses of improving private sector performance in Burkina. Once funding is obtained, strong operational mechanisms will be needed to assure that CSAIP bring together needed partners and stakeholders to plan for implementation.

3.11 Key Objectives of CSAIP Priority Investments

113. The Global Commission on Adaptation¹⁰⁵ identifies four key and high priority recommendations to help countries adapt to climate change, giving particular emphasis to supporting small-scale farmers.

- Improve small-holder productivity;
- Help small producers manage risks from increased variability and climate shocks;
- Address the challenges of the most climate affected and vulnerable;
- Achieve policy coherence by making agriculture interventions climate-smart.

114. The investments, and their objectives build on these recommendations and Burkina Faso's own policy priorities. Specifically, through its Plan National de Développement Economique et Social (PNDES), Burkina sees agriculture underpinning a major effort to drive economic growth and contribute to poverty reduction. The plan also supports a broad transformation within the agricultural sector. Other key Burkinabe policies include the National Policy on Food Security and Nutrition, and Burkina Faso's NDC and NAP. We have added another way to assess investments; whether it has the potential, as a demonstration project, to leverage change and begin to transform a sector. This is done by examining whether the investment strongly supports any of the following national priorities:

- Supporting Productivity, Food and Nutritional Security;
- Building Agricultural Sector Resilience;
- Creating Agricultural Value Chains;
- Improving Sustainable Management of Lands and Resources;
- Supporting Mitigation;
- Supporting Agricultural Sector Transformation;
- Supporting Women and Youth.

115. All of the investments support increasing agricultural productivity and improve sustainability (Table 22). Eight of the investment improve food security, and all support resilience to climate change directly, except the on-farm biogas, which does so indirectly. All of the investments but three directly

¹⁰⁴ World Bank & International Finance Corporation, 2019. "[Creating Markets in Burkina Faso](#)," [World Bank Other Operational Studies](#) 32401, The World Bank.

¹⁰⁵ Global Commission on Adaptation, WRI, 2019

support value chain creation, jobs, and or agro-processing, with Organic Farming, Livestock, and Forest having the largest potential impact. Mitigation is addressed in all of them, although in the livestock project, there is a reduction in GHG intensity, but not in overall emissions. Four of the proposed investments in this CSAIP have the potential to help transform the agricultural sector, catalysing new approaches and actions.

The Capacity Building and Finance and Insurance Investments are intended as transformational activities to catalyze knowledge and awareness of CSA across the country, and promote investment. Such information is critical within government, and within the private sector to appropriately support investments that are less “risky” given Burkina Faso’s vulnerability to climate, and to help investments for more resilient crops or areas. Seven of the investments directly support women and youth. Livestock production could be greatly increased, and there is a potential to increase value chains and jobs by improving production and processing of livestock products. The investments in Forests, Agro-Forests and Gardens protect forests from destruction or degradation, and promotes expansion of existing and new value chains. All of the investments could be designed in ways that more directly support policy objectives, although individually, and as a portfolio, this CSAIP strongly supports Burkina Faso’s agricultural objectives. Scenarios with and without the investment, and the key investment outcomes are shown in Table 23.

Table 22: Objectives of Investments

Proposed Investments	Productivity	Burkina Faso Agriculture & Rural Sector Objectives						
		Food Security	Support Ag. Sector Resilience	Ag. Value Chains	Sustainable Land/Farming	Mitigation	Support Women & Youth	Ag. Sector Transformation
CAPACITY Building	✓	✓	✓	✓	✓	✓	✓	✓
FINANCE and insurance	✓	✓	✓	✓	✓	✓	✓	✓
WATER resources & irrigation	✓	✓	✓	✓	✓	✓	✓	✓
Integrated SOIL Management	✓	✓	✓		✓	✓	✓	✓
ORGANIC Farming	✓	✓	✓	✓	✓	✓		
FORESTS, Agroforests, & GARDENS	✓	✓	✓	✓	✓	✓		
LIVESTOCK Intensification	✓	✓	✓	✓	✓		✓	
OIL-PROTEIN Crops	✓		✓	✓	✓	✓		
On-farm BIOGAS	✓				✓	✓	✓	

Table 23: Investments, Scenarios Without and Without the Investment, and Objectives, shown from most objectives to least

Climate Change	Project Importance	Scenario WITHOUT Investment	Scenario WITH investment	Investment objectives
Diversification through FORESTS & GARDENS				

Medium resiliency	Livelihoods & food sources for women & youth thru garden, agroforestry, & NTFPs	Low diversification increases smallholder vulnerability to economic & food insecurity under climate change; continued deforestation & environmental degradation; untapped international market opportunities	Smallholder food & economic resilience, particularly for women & youth; forest conservation & enhanced ecosystem services; national economic growth through value chain development of high-value export products.	• Food security • Resilience • Value chains • Sustainability • Mitigation • Livelihoods • Inclusivity
Sustainable management of WATER RESOURCES				
Medium resiliency	Protect & sustainably leverage water resources to improve climate resiliency.	Agriculture remains exposed to increasingly unpredictable precipitation. Crop & animal loss, food & economic insecurity. Conflict over scarce resources. Watershed degradation.	Efficient water capture allows storage & management. Irrigation stabilizes crop & animal production, food supplies, & market prices. Reduced erosion & flooding, watershed protection.	• Food security • Resilience • Value chains • Transformation • Sustainability • Environmental quality
Foster climate-smart agriculture with FINANCE SERVICES				
Foundational resilience through essential services	Enable smallholder investment in CSA through loan, insurance, & other basic services.	Smallholders have limited capacity to invest; cost of maintaining status quo increases with climate impacts; decreased food & economic security; downward poverty cycle	Credit & loan enable investment in system transformation; insurance & risk distribution instruments offer protection from loss, enabling farmers to take the risk of changing to new system	• Food security • Resilience • Value chains • Trade • Sustainability • Transformation • Mitigation
Building climate-smart CAPACITY				
Foundational resilience through essential services	Establish sustainable systems for developing & disseminating CSA innovation	Farmers make short-term decisions when facing uncertainty or high risk; perpetuates poverty cycle, degrades resources, & increases vulnerability.	Robust research, extension, & information services support a suite of site-tailored solutions that drive resilience, productivity, environmental management	• Food security • Resilience • Value chains • Sustainability • Transformation • Mitigation
Develop value chains to meet demand for ORGANIC				
Highly vulnerable	Leverage untapped domestic & international markets for diversification	Low sectoral diversity, dearth of livelihood & income opportunities, particularly for youth & women	Diversification improves economic resilience by creating livelihoods all along several organic production value chains; environmental impacts reduced	• Food security • Resilience • Value chains • Sustainability • Livelihoods • Economic resilience
Developing resilient value chains for OIL-PROTEIN CROPS				
Medium resiliency	Strengthen value chains to build livelihoods, reduce import dependency, & build export markets	Soybean yield plummets with climate change; soil health decline; income declines; reliance on imports	Sustainable productivity, reduced reliance on imports, improved national food security, export market development, new livelihood opportunities.	• Resilience • Value chains • Sustainability • Trade • Food security • Economic resilience
Agricultural productivity & environmental restoration through integrated SOIL MANAGEMENT				
Foundational resilience through essential services	Provide location-tailored soil information, recommendations, & tools & policy to support implementation	Agriculture & extreme weather events degrade soils; productivity decreases; cost of production rises with increased fertilizer inputs; increased GHG emissions exacerbate climate change; environmental degradation	Soils are resilient to climate stress; sustainable productivity; food & economic security; GHG storage; strong environmental services; optimum fertilizer usage	• Food security • Resilience • Sustainability • Transformation • Mitigation
Sustainable on-farm BIOGAS				
Resilient under climate change scenarios	Leverage farm waste to produce clean energy, support health & reduce GHGs	Continued deforestation, High GHG emissions, unreliable & costly cooking fuel & fertilizer purchases, Indoor air pollution & related diseases affecting especially, women & children	Reduce deforestation & GHG emissions; Clean cooking fuel & fertilizer produced on-farm for free; improved crop yields; Improved health outcomes.	• Sustainability • Mitigation • Soil fertility • Health
Climate-resilient LIVESTOCK				
Vulnerable to climate change	Create climate resiliency to ensure future food security & income	Reduced productivity as heat stress & mortality increase, feed sources decline. Increased pastoralist-smallholder conflict.	Improved health & productivity; food & income security; reduced farmer-pastoralist conflict; reduced GHG emission intensity; reduced environmental degradation	• Food security • Resilience • Mitigation

116. **All of the projects could be designed to directly support women and youth.** Care needs to be taken that investments for commodities and crops that are managed by women and youth do not have control pass to men's control when they are monetized. For this reason, three of the projects have been identified as potentially having mixed results. The projects where value chains are most likely to be taken over by men are the Forest and Garden, The Organic Farming, and Oil Protein Crops. In contrast, Building Capacity, Finance and Insurance, and Soil Management are potentially transformational for the impact that they could have on women.

117. **Taken together, these investments opportunities represent a well-balanced portfolio.** The investments span different sectors of the country, have different levels of risk, target divergent groups of beneficiaries, and introduce a wide range of well-demonstrated CSA technologies and practices. All of them support CSA pillars, and contribute to meeting Burkina Faso's objectives as identified in its national plans, and that are fundamental to addressing its future.

CHAPTER 4: SUMMARIES OF THE NINE PRIORITIZED CSAIP INVESTMENTS FOR BURKINA FASO

118. **This CSAIP includes five national-scale investments and five regional climate-smart crop and animal investments.** All of these are described in short summary form in this section, while more detailed information on each of the investments is provided in Annex A. The five national investments that build capacity, financing, resilience, support mitigation and offer human health benefits are:

- **Sustainable On-Farm Biogas Production:** increase access to and knowledge of sustainable domestic energy sources in order to conserve wood resources, reduce greenhouse gas emissions, and reduce poverty, food insecurity, and health threats to rural communities.
- **Sustainable Intensification of Livestock Production:** foster climate resilience in Burkinabe livestock systems for improved food and nutritional security and economic outcomes.
- **Financial and Insurance Products:** foster smallholder ability to invest in CSA innovations through good access to robust financial services, including credit, loan, insurance and risk instruments, savings and payment services.
- **Forest, Agroforest, And Garden Production for Climate-Smart Diversification:** foster climate-resilient livelihoods and food sources for women and youth smallholders through home gardens, agroforestry, and non-timber forest product harvesting and value-addition
- **Building Capacity in Climate Smart Agriculture:** fully integrate national climate-smart priorities into Burkina Faso's agricultural research and extension programs, and build robust extension mechanisms for delivering timely, practical climate-smart information to farmers and other stakeholders through highly accessible channels.

119. **Four climate-smart investments were prioritized to support adaptation of agricultural production systems by introducing a variety of climate-smart practices into the different investments.**

- **Sustainable Management of Water Resources and Irrigation:** Fully leverage Burkina Faso's water resources in sustainable ways to improve productivity, **food and** nutritional security, climate resiliency, and ecological health in the Nord Soudanian region for 100,000 farmers and their families.
- **Developing Climate-Smart Organic Value Chains:** strengthen the quality, yield efficiency, and value chains of oil-protein crops to sustainably meet domestic demands and support international market development in the Nord Soudanian and Sud Soudanian regions for 60,000 farmers and their families.
- **Developing Resilient Oil-Protein Value Chains:** strengthen the quality, yield efficiency, and value chains of oil-protein crops to sustainably meet domestic demands and support international market development in the Nord Sahelian, Nord & Sud Soudanian regions for 240,000 farmers and their families.
- **Integrated Soil Management for Agricultural Productivity & Environmental:** Provide producers and extension agents with location-tailored information on soil characteristics and best management practice recommendations, as well as the tools, products, partnerships, and policy environment to implement those recommendations for 200,000 farmers and their families in the North Sahel and North and South Sudan regions.

120. **While undertaking the technical analysis, points were raised for further consideration and analysis.** The prioritized investments and their locations are decisions made in-country and by stakeholders. Yet there are findings that emerge during the variety of different analyses that are could be reasons to modify components of the proposed investments. These are:

- **Infrastructure components within a CSAIP:** All agree that infrastructure (dams, irrigation, roads, processing and storage, energy systems) are vital components to supporting agriculture, including climate smart agriculture. Yet infrastructure requires special knowledge, analytical methods, and safeguards, which are not included within the scope of this CSAIP.
- **Climate Change Impact and Crops:** Several strong results emerged from the climate change analysis that potentially sway the selection of crops. Maize performs terribly and sorghum badly, in all scenarios. While these are key cereals in Burkina Faso, it is important to consider alternatives (e.g. other cereals and rice) and strongly support their adoption. Similarly, the Oil-Protein investment may greatly improve the resilience of oilseeds, but as a group (groundnuts, soybean and other oilseeds), they are not resilient. In contrast, yams and sweet potatoes do well, as do cowpeas, and these should be included for their nutritional value and resilience.
- **Consider Changing Investment Scope and Scale:**
 - The Sahel regions are the poorest and driest, yet the Sudanian regions have more investment on them. The productivity/poverty trade-offs of the location of projects should be assessed more specifically in future scoping activities, since this become a fundamental policy-related decision.
 - The soils project is currently limited in scope (regional), but is greatly needed at a national scale, and could potentially be transformational to the agricultural sector. It would likely enjoy economies of scale if implemented more broadly, and help Burkina reduce vulnerability to climate change.
 - Oil-Protein is proposed in three of four regions. While the CSA investment strongly increases yield, oil-crops are not generally climate resilient, so there may be achievable economies of scale with specific targeting as a national program, to be sure that anyone growing these crops benefits from CSA practices.

121. **It is fundamental that an understanding of what CSA means, and what potential climate impacts will be on agriculture be understood at different scales, from within government policy-making circles, at the Chamber of Commerce, in private banks, and among different international donor organizations.** During the analysis, we found many proposals or studies by donors and other groups, recommending that Burkina Faso expand crops such as maize, vegetables or mangos. Because these crops fare poorly given climate warming (although CSA improves their performance), it is vital that Burkina Faso itself has the capacity to understand the short and long term trade-offs associated with crops that are not climate resilient. More bluntly, Burkina Faso is getting policy and project advice that may be good in the short-term, but fails to understand longer term climate impacts and risks. The Capacity Building for CSA is one of the proposed investments intended to do this.

122. **If Burkina chooses to support investments in crops with low resilience, all efforts must be made to ensure that these investments are climate smart.** It is vital that all parties, government, donors, investors and farmers understand that these are high-risk crops, so that they can make an informed choice on whether to proceed or substitute more resilient crops. Given Burkina Faso's high poverty, the high vulnerability of its crops and people to climate change, and scarce financial resources, all donors, projects and policies related to the agricultural sector or value change **MUST** take climate resilience into account. Rather than short-term higher risk investments, it is better to be building resilient crops, commodities, and value chains that bridge Burkina to a resilient future.

4.1. SUSTAINABLE ON-FARM BIOGAS PRODUCTION

PROJECT SUMMARY

REGIONS: National

BENEFICIARIES: 65,000 farmers and their families

OBJECTIVE: Increase access to and knowledge of sustainable domestic energy sources in order to conserve wood resources, reduce greenhouse gas emissions, and reduce poverty, food insecurity, and health threats to rural communities.

HIGHLIGHTS:

- Generates fertilizer and clean cooking fuel from on-farm resources, protecting forests and reducing costs and labor

JUSTIFICATION AND KEY INVESTMENTS

Burkina Faso's energy supply is characterized by a dependence on fossil fuels and low and inequitable access to electricity.¹⁰⁶ As of 2017, about 20% of the national population had some degree of access to electricity services, and only about 15% had reliable electricity access. There is a notable dichotomy between urban areas, where over 66.5% of the population has some degree of electricity access, and rural areas, where just 3.2% of people have at least limited access.¹⁰⁷

Energy access may worsen.¹⁰⁸ Demand for electricity services has been growing as the economy diversifies. But supply has not kept pace with demand, and prices are consequently rising, further excluding the majority of households by pricing them out of the market.¹⁰⁹

Biomass remains the primary energy source for over 80% of Burkinabe households.¹¹⁰ The particulate matter released from open biomass fires can be extremely detrimental to human health, particularly when indoors. This is an especially great threat to women and children, who tend to spend the most time collecting fuel, tending fires, and preparing meals. Burkina Faso is one of the top 21 countries most heavily impacted by this issue.

Biodigesters are an alternative method of energy production that require minimal infrastructure, inputs, and maintenance. Biodigesters are highly complementary to farm production¹¹¹; they leverage on-farm resources, and they help reduce household fuel and fertilizer costs in terms of both time and money. Biodigesters also have tangible environmental benefits. Biogas does not produce particulate matter air pollution, and does not involve felling trees or clearing vegetation. As such, biogas does not carry the human health implications or high greenhouse gas emissions associated with biomass and fossil fuel use.

Some of the necessary enabling factors are already in place to support the success of biodigester projects. The Burkinabe national government is heavily invested in equitable energy access. There is also significant international support for biodigester programs in Burkina Faso. Synergy with other energy programs will help maximize the impacts of biodigester solutions. Remaining to be addressed are land tenure, robust extension, and finance services; these are essential to the success of sustainable energy production solutions.

Biodigester systems have limitations. Although biodigesters relieve the burden of purchasing or collecting fossil or biomass fuel, they add the burden of collecting fresh manure from primarily extensive livestock systems, or converting to intensive livestock management. Producers must also have consistent access to a fairly large amount of manure--the equivalent of at least 3 cows--in order to effectively employ a biodigester. This excludes the poorest

¹⁰⁶ Ministère des Mines et de l'Énergie, "Politique Sectorielle de l'Énergie 2014-2025."

¹⁰⁷ Direction générale des études et des statistiques sectorielles, "Annuaire Statistique 2016 Du Ministère de l'Énergie."

¹⁰⁸ World Bank, "The Power of Dung."

¹⁰⁹ Ministère des Mines et de l'Énergie, "Politique Sectorielle de l'Énergie 2014-2025"; Direction générale des études et des statistiques sectorielles, "Annuaire Statistique 2016 Du Ministère de l'Énergie."

¹¹⁰ Kampmann and Thiombiano, *Atlas de La Biodiversité de l'Afrique de l'Ouest, Tome II*.

¹¹¹ World Bank, "The Power of Dung."

and most vulnerable members of the community. Finally, biodigesters do not power in-home lighting, electro-domestics, water pumps, technology solutions, and other basic services made possible by electricity.

KEY PROJECT INVESTMENTS

Site-tailored CSA pillar promotion with a focus on:

- Create a sustainable source of on-farm energy and fertilizer production using readily available residues
- Bolster research and development capacity for sustainable on-farm energy production technologies
- Raise public awareness of the benefits of sustainable energy production and use

POTENTIAL PROJECT IMPACTS

PRODUCTION	will increase even as input costs decrease via on-farm production of high quality compost.
RESILIENCE	is increased through decreased reliance on unreliable inputs, decreased cost of crop production and fuel usage, improved health outcomes, and stable source of cooking fuel
MITIGATION	as a result of reduced GHG emissions from manure, felling trees, and burning biomass
COST	Of US \$846 per beneficiary, or a total cost of about USD \$ 54.99 million
YIELD	Increased by 45% for 65,000 small farming families

ECONOMIC AND FINANCIAL ANALYSIS

Cost Benefit Analysis with and without Climate Risks

With Climate & Pest Risks					
Mean No. Beneficiaries	Change in yield (%)	Mean NPV (million USD)	Chance Pos NPV (%)	ROI	BCR (SD)
65,000	45%	-1.5	32%	-0.09	-0.02 (0.66)
Without Climate & Pest Risks					
65,000	45%	16.9	51%	0.3	0.23 (0.83)

ENABLING ENVIRONMENT

Existing Potential Assets in Support of Project Success

- Strong demand for energy services
- Inputs (crop residues and manure) are already available
- Short- and long-term cost and labor savings
- Complementary to forestation and soil fertility programming
- Many sources of funding, foundational knowledge, and lessons learned
- Established carbon credits program for biodigesters
- Excellent alignment with national and international policy environment

Existing Potential Barriers to Project Success

- Labor burden of collecting manure from extensive livestock systems
- Livestock ownership requirement excludes poorest households
- Insufficient manure availability could exacerbate land clearing
- Potential lack of buy-in if electricity services become available by other means e.g. dam hydroelectricity

FINANCING

Public Financing Opportunities

- Environmental Intervention Fund
- National Biogas Program

Private Financing Opportunities

- Banque Agricole du Burkina (Agriculture Bank of Burkina Faso)
- Institut International d'Ingénierie de l'Eau et de l'Environnement (2iE)
-

International Financing Opportunities

- Global Environment Fund
- Green Climate Fund
- World Bank's Carbon Initiative for Development
- United Nations Clean Development Mechanism
- African Development Bank (AfDB)

4.2. SUSTAINABLE INTENSIFICATION OF LIVESTOCK PRODUCTION

PROJECT SUMMARY

REGIONS: National

BENEFICIARIES: 150,000 farmers and their families

OBJECTIVES: Foster climate resilience in Burkinabe livestock systems for improved nutritional security and economic outcomes

HIGHLIGHTS:

- 38% improvement in livestock productivity for 150,000 farm households
- Increased income and nutritional security and diversification
- Reduced conflict between pastoralists and sedentary producers
- Reduced greenhouse gas intensity and mitigation of agricultural expansion

JUSTIFICATION AND KEY INVESTMENTS

Burkina Faso is an agro-pastoral country. The livestock sector contributes about 18% of GNP, accounts for 26% of exports, and constitutes a source of incomes for nearly 80% of the population¹¹². Livestock typically free range in the dry season and are kept tethered, sometimes with feed supplementation, in the wet season.

The Burkinabe livestock sector has consistently low productivity. This comes as a result of extensive farming practices¹¹³ exacerbated by a burgeoning population¹¹⁴, suboptimal livestock feeding, recurrent animal diseases, unimproved breeds, and poor market access¹¹⁵. The government has responded to these issues with strong policy support for livestock intensification, and smallholders have also adopted crop-livestock integration to some extent. However, rapid intensification, inconsistent system integration, and climate change impacts have taken their toll, resulting in increasing soil and environmental degradation¹¹⁶.

Livestock producers are highly exposed to unpredictable climate extremes and variability as a result of climate change¹¹⁷. Precipitation variability makes the availability of forage and water unpredictable, affecting livestock productivity and pushing pastoralists to travel longer distances and exploit more land in order to sustain their herds¹¹⁸. Drought, flood, and extreme heat increase livestock mortality and destabilize markets¹¹⁹. Poor market access to inputs and financing, widespread animal disease despite the efforts of veterinary services¹²⁰, and weak infrastructure further challenge the livelihoods of livestock farmers¹²¹. These issues exacerbate existing tensions between pastoralists and sedentary producers for natural resources¹²².

Climate-smart agriculture reduces the environmental impacts of livestock systems, and makes livestock systems more resilience in the face of climate change¹²³. For example, improved varieties of ruminants offer resource use

¹¹² Ministère des Ressources Animales, "Politique Nationale de Développement Durable de l'Élevage (PNDEL) Au Burkina Faso."

¹¹³ Vall et al., "Co-Design of Innovative Mixed Crop-livestock Farming Systems in the Cotton Zone of Burkina Faso."

¹¹⁴ Food and Agriculture Organization of the United Nations, "Burkina Faso."

¹¹⁵ Ministère des Ressources Animales, "Politique Nationale de Développement Durable de l'Élevage (PNDEL) Au Burkina Faso"; Okhimamhe et al., "Climate Variability Adaptation Strategies."

¹¹⁶ Vall et al., "Co-Design of Innovative Mixed Crop-livestock Farming Systems in the Cotton Zone of Burkina Faso."

¹¹⁷ Government of Burkina Faso, "Burkina Faso National Climate Change Adaption Plan."

¹¹⁸ GiZ, "National Investment Plan for the Implementation of the Determined Contributions."

¹¹⁹ Doumbia, "Changements Climatiques et Agriculture Intelligente En Côte d'Ivoire: Diagnostic Du Contexte National et Recueil Des Résultats de La Recherche Sur Les Facteurs Socio-Économiques Favorisant l'adaptation et Les Technologies Appropriées de l'AIC Chez Les Petits Agriculteurs."

¹²⁰ Ministère des Ressources Animales, "Politique Nationale de Développement Durable de l'Élevage (PNDEL) Au Burkina Faso"; Okhimamhe et al., "Climate Variability Adaptation Strategies."

¹²¹ Williams et al., "Agriculture Climate Smart Facing the African Context."

¹²² Ministère des Ressources Animales et Halieutique, "Plan National d'Adaptation (PNA) Aux Changements Climatiques Du Secteur de l'Élevage Burkina Faso."

¹²³ van Wijk, Yameogo, and Ayantunde, "Assessment of the Potential Food Security Benefits of Increased Income from Crops, Livestock and off-Farm Employment in Burkina Faso."

efficiency and tolerance to adverse conditions¹²⁴. Water reservoirs help maintain livestock in dry areas. Livestock corridors support pastoralists in search of resources and protect smallholders. Fodder production supports improved feed quality, health outcomes, and productivity. Agroforestry-based forage production offers multiple production, adaption, and mitigation benefits. Integrated crop-livestock production systems are synergistically more productive and resilient than either system alone¹²⁵. Small ruminant are particularly advantageous¹²⁶ for their short gestation periods, high prolificacy, rapid growth rate, high feed conversion efficiency, high diseases resistance capacity, easy marketability¹²⁷, and the chemical similarity of their manure to synthetic fertilizer¹²⁸.

The opportunities for improvement in livestock production may have both positive and negative impacts on women farmers. Increases in livestock productivity and a shift toward small ruminants can increase women's workload per traditional gender roles¹²⁹. Careful community-based planning and prioritization is necessary to ensure that the responsibilities and benefits of such programs are equitably distributed across stakeholders¹³⁰.

KEY PROJECT INVESTMENTS

Site-tailored CSA pillar promotion with a focus on:

- Conservation and selection of local cattle breeds
- Control of emerging and re-emerging animal diseases related to climate change
- Establishment of agroforestry-based fodder stands
- Full crop-livestock system integration
- Pastoral infrastructure and equipment, establishing water reservoirs and livestock corridors
- Capacity building in livestock intensification

POTENTIAL PROJECT IMPACTS

PRODUCTION	is increased through improved feed, health care, and management to augment income and nutritional security
RESILIENCE	is increased through improved breeds, crop-livestock integration, and infrastructure development
MITIGATION	is a co-benefit of more efficient production, integration of livestock into crop systems, and reduced agricultural expansion
COST	US\$250 per beneficiary for a total of US\$37.5 million
YIELD	This project will increase livestock yield by 38%.

ECONOMIC AND FINANCIAL ANALYSIS

Cost Benefit Analysis with and without Climate Risks

With Climate & Pest Risks

¹²⁴ Crookston et al., "How Do You Know 'Resilience' When You See It?"

¹²⁵ Villano, Asante, and Bravo-Ureta, "Farming Systems and Productivity Gaps"; Asante et al., "Determinants of Farm Diversification in Integrated Crop-Livestock Farming Systems in Ghana."

¹²⁶ van Wijk, Yameogo, and Ayantunde, "Assessment of the Potential Food Security Benefits of Increased Income from Crops, Livestock and off-Farm Employment in Burkina Faso."

¹²⁷ Sejian et al., "Genes for Resilience to Heat Stress in Small Ruminants."

¹²⁸ Ansah et al., "Manure Characteristics of Small Ruminants Fed Agro By-Products in the Guinea Savannah Agro-Ecological Zone of Ghana."

¹²⁹ Expert Panel Workshop, Crop-Livestock Integration Project Components.

¹³⁰ Bayala et al., "Methodological Guide."

Mean No. Beneficiaries	Change in yield (%)	Mean NPV (million USD)	Chance Pos NPV (%)	ROI	BCR (SD)
150,000	38%	16.4	43%	0.67	0.32 (6.35)
Without Climate & Pest Risks					
150,000	38%	76.1	65%	2.57	1.49 (6.70)

ENABLING ENVIRONMENT

KEY ENABLING FACTORS

- Most smallholders already own livestock and practice some degree of crop-livestock integration
- Most smallholders already practice some degree of agroforestry
- Governmental policy support of livestock intensification¹³¹
- Close alignment with NDCs and many other national policies

KEY RISKS

- Competing interests of pastoralists and farmers
 - Labor intensity of using manure as fertilizer alternative under extensive grazing system
 - Land tenure insecurity dissuades farmers from investing in trees and soil
-

FINANCING

Public Financing Opportunities

- Action Plan and Investment Programme for the Livestock Sector

- African Development Bank
- West African Development Bank

International Financing Opportunities

- The Economic Community of the West African States
- Global Environment Fund
- Green Climate Fund
- World Bank

Private Financing Opportunities

- Commercial producers of by-products commonly used as supplemental feed
- Providers of animal health services & artificial insemination
- Animal product processors

¹³¹ Vall et al., "Co-Design of Innovative Mixed Crop-livestock Farming Systems in the Cotton Zone of Burkina Faso."

4.3. FINANCIAL AND INSURANCE SERVICES TO FOSTER CLIMATE-SMART AGRICULTURE

PROJECT SUMMARY

REGIONS: National

BENEFICIARIES: 200,000 farmers and their families

OBJECTIVE: Foster smallholder ability to invest in CSA innovations through good access to robust financial services, including credit and loan, insurance and risk instruments, and savings and payment services.

PROJECT HIGHLIGHTS:

- Cost-efficient program that will increase household yields by 19% across both crop and livestock systems, with concomitant increases in national GDP
- Significantly increased ability for farmers to implement advisory recommendations and take innovation risks to improve productivity
- Producer and sectoral resilience in the face of major events such as extreme weather

JUSTIFICATION AND KEY INVESTMENTS

Financial services are prerequisites to agricultural development¹³². Even the most robust advisory services only improve farmers outcomes¹³³ if they have the financial security to take the recommended risks in changing established practices¹³⁴. Financial services enable producers and agribusinesses to leverage collateral, decrease transaction costs, and reduce risk¹³⁵. Financial services include:

(1) credit and financing, including input loans and credit, crop loans, value-chain finance, equity investments, equipment loans or leases, warehouse receipts, and group loans ¹³⁶ to enable farmers to invest in crucial inputs, system transformations, equipment, and other technologies that increase productivity and resilience.

(2) insurance and risk instruments such as index-based insurance and disaster relief funds sustain producers' ability to invest in the face of costly unforeseen events. Farmers who are protected from major losses are much more likely to invest in higher-value crops, thus increasing their future income and climate resiliency¹³⁷.

(3) savings services ensure savings security, and low-risk investments prevent loss of value due to inflation.¹³⁸ In-home cash savings, on the other hand, exposes producers to loss from theft, misplacement, and currency devaluation.

¹³² WSBI-ESBG, "Africa Post Offices Give Migrants New Financial Network Access."

¹³³ Innovations for Poverty Action, "Evidence for Agriculture in Ghana."

¹³⁴ Barnett et al., "External Evaluation of Mobile Phone Technology-Based Nutrition and Agriculture Advisory Services in Africa and South Asia."

¹³⁵ {Citation}

^{136,136} Palmer, "Making Climate Finance Work in Agriculture"; Ruben, Wattel, and van Asseldonk, "Rural Finance to Support Climate Change Adaptation: Experiences, Lessons and Policy Perspectives"; Wattel and van Asseldonk, "Financial Service Supply with Potential for Supporting Climatesmart Agriculture Quick Scan around the CCAFS Climate-Smart Village AR4D Sites of the CGIAR-CCAFS Research Program in 20 Countries"; World Bank, "Atelier de Restitution, Diagnostic de Finance Agricole et Plan d'Action."

¹³⁷

¹³⁸ Ruben, Wattel, and van Asseldonk, "Rural Finance to Support Climate Change Adaptation: Experiences, Lessons and Policy Perspectives"; Wattel and van Asseldonk, "Financial Service Supply with Potential for Supporting Climatesmart Agriculture Quick Scan around the CCAFS Climate-Smart Village AR4D Sites of the CGIAR-CCAFS Research Program in 20 Countries."

(4) payment services such as mobile money and utility bill pay enable producers to make and receive payments without investing time and money in travel, and without the risk loss or theft¹³⁹. The resulting secure digital records lend transparency to local informal economies¹⁴⁰ and can serve as a line of credit for loan applications.

Secure land tenure, robust information services, reliable market access, and safe harvest and capital storage options further foster financial stability and enable investments.¹⁴¹

Financial services in Burkina Faso disproportionately underserve smallholder farmers. Rural residents, women, and low-income individuals—that is to say, smallholder farmers—represent an outsized portion of those with constrained financial service access. Microfinance institutions, cooperatives, and rural banks offer a good foundation to reach rural populations with diverse financial services¹⁴².

KEY PROJECT INVESTMENTS

Site-tailored CSA pillar promotion with a focus on:

- Address issues preventing penetration of financial institution services in rural areas:
 - Low economies of scale
 - Poor financial performance indicators and norms compliance
 - Low profitability and portfolio quality ratios
 - Barriers to digitalization
 - Lack of institutional capacity for consistent monitoring
- Expand digital services in rural areas:
 - Establish policy to ensure consumer protection and good practices
 - Optimize network coverage
 - Digitize producer records to support establishment of creditability
 - Digitize government records to support transparency and efficiency
- Ensure broad access to financial services by Burkinabe smallholders, including:
 - Crop and livestock insurance and risk instruments
 - Credit and financing
 - Savings services
 - Payment services
- Building financial literacy of smallholders

POTENTIAL PROJECT IMPACTS

PRODUCTION	supported through risk redistribution and up-front capital for innovation investments
RESILIENCE	achieved by enabling implementation of recommended CSA practices and reducing their risk
MITIGATION	as a co-benefit of sustainable increases in production and resilience
COST	US\$200 per beneficiary, total US\$40 million
YIELD	The project will increase benefits to farm households by 19%.

ECONOMIC AND FINANCIAL ANALYSIS

¹³⁹ CCAFS, “10 Best Bet Innovations for Adaptation in Agriculture: A Supplement to the UNFCCC NAP Technical Guidelines”; Wattel and van Asseldonk, “Financial Service Supply with Potential for Supporting Climatesmart Agriculture Quick Scan around the CCAFS Climate-Smart Village AR4D Sites of the CGIAR-CCAFS Research Program in 20 Countries.”

¹⁴⁰ Palmer, “Making Climate Finance Work in Agriculture.”

¹⁴¹ Andres et al., “Social Network to Inform and Prevent the Spread of Cocoa Swollen Shoot Virus Disease in Ghana”; Danso-Abbeam, Ehiakpor, and Aidoo, “Agricultural Extension and Its Effects on Farm Productivity and Income.”

¹⁴² World Bank, “Financial Inclusion Support Project.”

Cost Benefit Analysis with and without Climate Risks

With Climate & Pest Risks					
Mean No. Beneficiaries	Change in yield (%)	Mean NPV (million USD)	Chance Pos NPV (%)	ROI	BCR (SD)
200,000	19%	15.9	45%	0.43	0.30 (2.88)
Without Climate & Pest Risks					
200,000	19%	52.2	63%	1.51	0.98 (3.03)

ENABLING ENVIRONMENT

EXISTING POTENTIAL ASSETS FOR PROJECT SUCCESS

- Broad microfinance and mobile money networks
- Grassroots buy-in to mobile financial services and microfinance sector
- Established policy to regulate the financial industry
- Robust, profitable, stable banking sector
- Strong government support and investment in improved smallholder financial services access

EXISTING POTENTIAL BARRIERS TO PROJECT SUCCESS¹⁴³

- Relatively undeveloped information-communication technology networks
- Weaknesses in business operations of microfinance organizations
- Poor economies of scale of existing microfinance organizations
- Low financial literacy among smallholders

FINANCING

Public Financing Opportunities

- Burkina Social and Economic Development Fund
- Women's Entrepreneurial Activities Support Fund

International Financing Opportunities

- FinMark Trust
- World Bank
- International Monetary Fund
- International Finance Corporation
- United Nation Environment Program (UNEP)
- Fonds International de Développement Agricole (FIDA)
- World Food Program (WFP)
- Organisation Meteorologique Mondiale (OMM)

Private Financing Opportunities

- Réseau des Caisses Populaires du Burkina
- Première Agence de Microfinance Burkina Faso
- CreditInfo VoLo
- Banque Agricole du Burkina
- Réseau des Caisses Populaires du Burkina
- Première Agence de Microfinance Burkina Faso

¹⁴³

4.4. FOREST, AGROFOREST, AND GARDEN PRODUCTION FOR CLIMATE-SMART DIVERSIFICATION

PROJECT SUMMARY

REGIONS: National

BENEFICIARIES: 180,000 farmers and their families

OBJECTIVES: foster climate-resilient livelihoods and food sources for women and youth smallholders through home gardens, agroforestry, and non-timber forest product harvesting and value-addition.

HIGHLIGHTS:

- A 40% income increase for farm families through low-input, low-impact farming
- Income and nutritional diversification and security through high-value product chain development
- Greenhouse gas mitigation through soil and biomass sequestration

JUSTIFICATION AND KEY INVESTMENTS

Garden, forest, and agroforest systems are often more resilient than arable crops. Trees in particular play a key role in Burkinabe agricultural landscapes, and account for a significant portion of household incomes, particularly poor and women-led households.¹⁴⁴

NTFP, gardening, and agroforestry practices are common in Burkina Faso, and there is significant demand for these products in domestic and international markets. Examples include shea, gum arabic, honey, baobab, medicinal products, and fruits¹⁴⁵.

However, many farmers are unable to fully leverage these practices due to lack of training. For example, farmers trained in shea pruning, assisted natural regeneration, and grafting techniques have 44% higher revenue than those that are not trained. These techniques are low-cost and use tools that most farmers already have on hand.

Household returns are not currently sufficient to catalyze protection against environmental degradation. Value chains for NTFPs remain largely undeveloped, resulting in a low return on investment for harvested products, low forest resource valuation, and consequent overexploitation¹⁴⁶.

There are four synergistic opportunities to increase the value of agroforestry and NTFPs to this crucial threshold:

- Training in improved planting material, gardening, and NTFP and agroforestry management practices
- Technology innovation to reduce processing drudgery and augment income potential
- Enabling of land tenure, particularly for women¹⁴⁷, who are the primary owners of garden, agroforestry, and NTFP processes
- Enforcement of conservation laws and sustainable practice guideline

Projects activities addressing these four key opportunities must take into account the context of gender inequality¹⁴⁸ as well as geographic and ethnic variation in traditional uses and practices of forest and garden products¹⁴⁹.

KEY PROJECT INVESTMENTS

¹⁴⁴ Leßmeister et al., "The Contribution of Non-Timber Forest Products (NTFPs) to Rural Household Revenues in Two Villages in South-Eastern Burkina Faso."

¹⁴⁵ Altieri, Funes-Monzote, and Petersen, "Agroecologically Efficient Agricultural Systems for Smallholder Farmers."

¹⁴⁶ Shackleton and Shackleton, "The Importance of Non-Timber Forest Products in Rural Livelihood Security and as Safety Nets"; Ticktin, "The Ecological Implications of Harvesting Non-timber Forest Products."

¹⁴⁷ Simelton et al., *Multifunctional Land Uses in Africa Sustainable Food Security Solutions*.

¹⁴⁸ Westholm, "Fruits from the Forest and the Fields."

¹⁴⁹ Leßmeister et al., "The Contribution of Non-Timber Forest Products (NTFPs) to Rural Household Revenues in Two Villages in South-Eastern Burkina Faso."

Site-tailored CSA pillar promotion with a focus on:

- Non-timber forest products (NTFPs) for nutritional diversity and security
- NTFP processing enterprises, including shea, gum arabic, and baobab
- Gardens for nutritional diversity
- Soil fertility management
- Forest and agroforest management to support NTFP productivity

POTENTIAL PROJECT IMPACTS

PRODUCTION	of high-value, low-impact, low-investment products and highly nutritional foods for increased income and food security
RESILIENCE	through diversification and of income sources and nutritional sources, as well as environmental sustainability
MITIGATION	through profitable protection and management of forests and soils
COST	US\$306 per beneficiary for a total of US\$ 55.08 million
YIELD	This project will bring 40% increase in income to 180,000 farmers and their families

ECONOMIC AND FINANCIAL ANALYSIS

Cost Benefit Analysis with and without Climate Risks

With Climate & Pest Risks					
Mean No. Beneficiaries	Change in yield (%)	Mean NPV (million USD)	Chance Pos NPV (%)	ROI	BCR (SD)
180,000	40%	168.2	89%	3.47	2.28 (3.20)
Without Climate & Pest Risks					
180,000	40%	196.1	92%	4.06	2.66 (3.36)

ENABLING ENVIRONMENT

KEY ENABLING FACTORS

- Agroforestry and NTFPs are already widely practiced
- Requires very few inputs, training, investments, or technologies to get started
- Close alignment with NDCs and many other national policies

KEY RISKS

- Lack of training and capacity to optimize ROI
- Little or no access to improved planting materials
- Overexploitation of forest resources
- Land tenure insecurity, particularly for women

FINANCING

Public Financing Opportunities

- Environmental Intervention Fund

International Financing Opportunities

- Global Environment Fund
- Green Climate Fund
- World Bank
- African Development Bank

- Swiss government

- Organisation for Economic Co-operation and Development Scheme for the Certification of Forest Reproductive Material¹⁵⁰

Private Financing Opportunities

- Banque Agricole du Burkina

¹⁵⁰ Sacande and Parfondry, "Non-Timber Forest Products."

- ECOBANK
- Bank Of Africa
- CORIS BANK
- UBA
- BICIAB

- ORABANK
- Caisses populaires
- SOFIGIB
- Planet finance
- Terrafina

4.5. BUILDING CAPACITY IN CLIMATE SMART AGRICULTURE

PROJECT SUMMARY

REGIONS: National

BENEFICIARIES: 500,000 farmers and their families

OBJECTIVES: Fully integrate national climate-smart priorities into Burkina Faso's agricultural research and extension programs, and build robust extension mechanisms for delivering timely, practical climate-smart information to farmers and other stakeholders through highly accessible channels.

HIGHLIGHTS:

- Cost-efficient, high-impact project cutting across sectors and boosting the national economy
- Establishes robust research and extension services to augment farmer productivity, adaptivity, and mitigation in the face of climate change
- Empowers individuals to continue to advance Burkinabe climate adaptivity and innovation

JUSTIFICATION AND KEY INVESTMENTS

Strong agricultural research and extension networks are the primary predictor of agricultural productivity growth in Sub-Saharan Africa. Development and dissemination of new technologies accounts for 51% of productivity gains; improved trade and marketing policies account for 20%, and reductions in conflict for 18%.¹⁵¹ Climate information services are particularly helpful to farmers facing erratic weather patterns. Timing of advisory services¹⁵² and the integration of top-down/ value-chain / business approaches with bottom-up / smallholder / traditional approaches is also crucial to effectiveness.

Climate-smartness and research are both well-supported by the Burkinabe government, but they remain disjointed in practice. Burkina Faso has notable research and development capacity, and significant efforts have been made to support the dissemination of new research findings to farmers. This work is not yet fully aligned with national climate-smart policies, particularly in terms of climate and weather advisories.

Additionally, the country's waning extension workforce has created a gap that prevents new technologies and information from reaching farmers. The number of national extension agents on staff has been in decline since the 1990s, and now stands at approximately 1 agent per every 1000 farmers. In contrast, the World Bank's standard ratio is 1 agent per 800 farmers.¹⁵³ This degree of understaffing prevents the timely and effective dissemination of information and capacity-building to farmers.¹⁵⁴

Mobile phone network access is improving rapidly in Burkina Faso. Mobile-based solutions have been successful in providing services to rural Burkinabe populations, and hold promise in terms of providing actionable climate smart information to farmers and other stakeholders. However, existing digital agricultural information services are limited, and not well adapted to rural smallholders.

Significant access inequalities in the current advisory system perpetuate poverty and poor productivity. Just 19% of national agricultural researchers were women in 2014 (up from 13% in 2008).¹⁵⁵ Gender norms, patriarchal values, time poverty, and illiteracy also reduce farmer access to extension services, particularly for women farmers. Seniority, religion, class, and position within the household further reconfigure advisory services access.

Even farmers who have access to extension services often remain constrained from receiving and implementing advisory recommendation by external circumstances. Restrictive land tenure, unprofitable markets, poor storage, a dearth of finance services, and low household incomes prevent farmers from taking a risk to invest in changing

¹⁵¹ World Bank, "Transforming Agriculture for Economic Growth, Job Creation and Food Security."

¹⁵² Innovations for Poverty Action, "Evidence for Agriculture in Ghana."

¹⁵³ Feder, Ganguly, and Anderson, *The Rise And Fall Of Training And Visit Extension*.

¹⁵⁴ Bikienga, "Elaboration d'un Profil de l'Agriculture Climato-Intelligente Pour Le Burkina Faso."

¹⁵⁵ Magne Domgbo, Neya, and Stads, "Burkina Faso."

established practices¹⁵⁶. For these farmers, advisory services increases their knowledge, but does not improve their outcomes. In contrast, those with extensive social networks, many years of experience, large farm sizes, more secure land tenure, good access to financial services, group memberships, and greater general knowledge are much more able and likely to adopt new practices¹⁵⁷.

KEY PROJECT INVESTMENTS

Site-tailored CSA pillar promotion with a focus on:

- Training at national and regional levels for:
 - Transparency
 - CSA technical capacity
 - Production and management of information, particularly on water resources
 - Development of agricultural services and climate information services
- Establishment of evidence-based climate-smart research programs¹⁵⁸
- Timely, practical, and tailored extension services oriented toward climate-smart actions¹⁵⁹
- Timely, practical climate information services

POTENTIAL PROJECT IMPACTS

PRODUCTION	Across sectors is increased by demand-driven research and timely dissemination of climate-smart innovations
RESILIENCE	Is increased through adaptive, sustainable, site-tailored practices such that productivity is stable even under extreme and variable climate scenarios
MITIGATION	Is a co-benefit of climate-smart production and resilience practices
COST	US\$110 per beneficiary for a total of US\$55 million
YIELD	18% yield increase for 500,000 farmers and their families

ECONOMIC AND FINANCIAL ANALYSIS

Cost Benefit Analysis with and without Climate Risks

With Climate & Pest Risks					
Mean No. Beneficiaries	Change in yield (%)	Mean NPV (million USD)	Chance Pos NPV (%)	ROI	BCR (SD)
500,000	18%	52.5	62%	1.06	0.71 (4.80)
Without Climate & Pest Risks					
500,000	18%	140.4	80%	2.95	1.91 (5.05)

ENABLING ENVIRONMENT

Existing Potential Assets to Support Project Success

¹⁵⁶ Barnett et al., "External Evaluation of Mobile Phone Technology-Based Nutrition and Agriculture Advisory Services in Africa and South Asia."

¹⁵⁷ Danso-Abbeam, Ehiakpor, and Aidoo, "Agricultural Extension and Its Effects on Farm Productivity and Income."

¹⁵⁸ CIAT, "CSAIP Inception Report."

¹⁵⁹ World Bank, "Transforming Agriculture for Economic Growth, Job Creation and Food Security"; CIAT, "CSAIP Inception Report."

- University degree programs in agricultural studies to produce qualified agricultural extensionists (University of Ouagadougou among others)
- Well-established agricultural research institute networks, including:
 - Institut National de l'Environnement et de Recherches Agricoles (INERA)
 - Research Institute for Applied Sciences and Technologies (IRSAT)
 - National Forest Seed Center (CNSF)
 - University of Ouagadougou (UO)
- Strong governmental commitment to research and extension systems
- Strong farmer cooperative unions and Regional Chambers of Agriculture
- Strong alignment between NDCs and many national policies

Existing Potential Barriers to Project Success

- Lack of inter-sectoral coherence and coordination in regards to cross-cutting CSA topics
- Lack of high-level commitment to CSA
- Dearth of funding for National Adaption Program of Action¹⁶⁰
- Very low electricity penetration inhibits use of computer technologies and mobile devices to deliver services
- Persistent segregation and discrimination, particularly in terms of gender
- External circumstances that dissuade or prevent farmers from implementing recommendations:¹⁶¹
 - Restrictive land tenure
 - Unreliable access to profitable markets
 - Lack of safe harvest storage
 - Dearth of financial services such as loans and credit
- Strong social and cultural norms influence adoption of innovative practices

FINANCING

Public Financing Opportunities

- National Adaption Program of Action

International Financing Opportunities

- Netherlands Space Office
- ECOWAS Bank for Investment and Development (EBID)
- Regional Agency for Agriculture and Food (RAAF)
- UK Department for International Development (DFID)
- Netherlands Development Organization (SNV)
- Permanent Interstates Committee for Drought Control in the Sahel (CILSS)
- AGRICORA/ French Government
- West African Science Service Centre on Climate change and Adapted Land Use (WASCAL)
- World Bank

Private Financing Opportunities

- Orange Mobile
- Société d'Infrastructures Collectives (SICO)¹⁶²

¹⁶⁰ ECOWAS, "Intervention Framework for the Development of Climate Smart Agriculture under the West Agricultural Policy (ECOWAP/CAADP) Implementation Process," June 15, 2015.

¹⁶¹ ECOWAS, "Intervention Framework for the Development of Climate Smart Agriculture under the West Agricultural Policy (ECOWAP/CAADP) Implementation Process," June 15, 2015.

¹⁶² Orange, "Smart Metering Solution in Burkina Faso"; Ngounou, "BURKINA FASO."

- SOFITEX (for cotton) and other large processors and value-addition enterprises
- Private universities such as:
 - Institut de Gestion des Risques Industriel et du Développement Durable
 - Aube Nouvelle
 - Ecole Supérieur du Génie Rural et de l'Environnement (ESGRE)
 - Institut Panafricain pour le Développement (IPD)
 - Institut International d'Ingénierie de l'Eau et de l'Environnement (2iE)

4.6. SUSTAINABLE MANAGEMENT OF WATER RESOURCES & IRRIGATION

PROJECT SUMMARY

REGION: Nord Soudanien

BENEFICIARIES: 100,000 farmers and their families

OBJECTIVE: Fully leverage Burkina Faso's water resources in sustainable ways to improve productivity, food and nutritional security, climate resiliency, and ecological health.

HIGHLIGHTS:

- This project will increase benefits to farm households by 56%
- Taking risks into account, the project has a positive NPV of 21%, but over 50% if risks aren't realized.
- Increased yields and decreased flooding and erosion through water management infrastructure

JUSTIFICATION AND KEY INVESTMENTS

Water harvesting and irrigation improve yield and efficiency by partially unlinking production from the vagaries of rainfall. Worldwide, irrigated agriculture represents about 20% of agricultural land, and contributes about 40% of crop production.¹⁶³

The vast majority of Burkina Faso's agriculture depends on rainfall. Pollution, population growth, high evapotranspiration, and environmental degradation have also reduced water availability. In response to this, the Burkinabe government has committed to sustainable water management for agricultural production, and enjoys significant international commitment and support.

There is remarkable untapped opportunity to leverage water harvesting and irrigation to increase productivity in Burkina Faso. Nearly 4.4 million hectares of Burkinabe agriculture are rainfed, and just 25,000 hectares are irrigated. The country currently uses a tiny fraction of its 18.5 billion m³ renewable surface and groundwater water resources. The economy is strongly oriented toward agriculture, resulting in a low industrial demand for water.¹⁶⁴ As such, nearly all of the untapped water resources could be used to irrigate agricultural land.

Burkina Faso's diverse renewable water resources enable highly site-tailored water management solutions. This affords Burkina Faso significant flexibility in terms of resource options, and also necessitates careful situational analysis to ensure project solutions are sustainable. Where dams and reservoirs are appropriate, multiple-uses can be considered to ensure environmental and economic sustainability; for example, reservoirs and canals may be used for aquaculture¹⁶⁵, and hydroelectricity may be generated. In areas of low rainfall, high evapotranspiration, and rich groundwater resources, wells could be much more appropriate. About half of Burkina Faso's renewable water resources are groundwater.

Any given water management solution is not appropriate across all scenarios. The feasibility and potential impacts water projects vary widely across geography, time (given growing climate change impacts), the population served (economies of scale), and the type of system used. In some cases, improvements in existing infrastructure or improved use efficiency of existing water sources removes the need for new technologies.

Even after the appropriate site-tailored water management solution is identified, significant challenges remain. Land tenure, finance services, and robust extension are prerequisites to success of water management solutions¹⁶⁶.

KEY PROJECT INVESTMENTS

Site-tailored CSA pillar promotion with a focus on:

- Improvement or rehabilitation of existing infrastructure

¹⁶³ Signs and McCartney, "Fish in Irrigation Systems: An Ancient Practice May Be the Way of the Future!," September 2, 2019.

¹⁶⁴ Abou Zaki et al., "An Index-Based Approach to Assess the Water Availability for Irrigated Agriculture in Sub-Saharan Africa."

¹⁶⁵ Signs and McCartney, "Fish in Irrigation Systems: An Ancient Practice May Be the Way of the Future!," September 2, 2019.

¹⁶⁶ Akuffo-Bea-Essilfie et al., "Promoting Rainwater Harvesting for Improving Water Security."

- Protection of water bodies and associated ecosystems
- Establishment of irrigation and water infrastructure
- Capacity building for water use efficiency

POTENTIAL PROJECT IMPACTS

PRODUCTION	increases across crop and livestock systems
RESILIENCE	increases by sustainably unlinking production from precipitation
MITIGATION	as a co-benefit through intensification and reduced need for agricultural expansion into natural areas
COST	Of US \$650 per beneficiary, or a total cost of about USD \$ 65 million
YIELD	Increased by 56% for 100,000 small farming families

ECONOMIC AND FINANCIAL ANALYSIS

Cost Benefit Analysis with and without Climate Risks

With Climate & Pest Risks					
Mean No. Beneficiaries	Change in yield (%)	Mean NPV (million USD)	Chance Pos NPV (%)	ROI	BCR (SD)
100,000	56%	-15.8	21%	-0.35	-0.18 (0.67)
Without Climate & Pest Risks					
100,000	56%	22.9	54%	0.35	0.26 (0.98)

ENABLING ENVIRONMENT

Existing Potential Assets for Project Success

- Vast untapped water resources
- Minimal inter-sectoral competing interests for water resources
- Roughly equal division of water resources between ground and surface, enabling flexibility in solution design
- Aligned international agreements and national policies
- Good existing knowledge base through previous projects

Existing Potential Barriers to Project Success

- Low financial capacity
- Scant mechanisms for capacity building and extension
- Highly site-specific solution requiring significant upfront investment in diagnostics
- Lack of land tenure
- Divergence between government yield objectives and rural farmers' food security objectives

- Insufficient land access for farmers to leverage water resources in breaking poverty cycle¹⁶⁷

FINANCING

Public Financing Opportunities

- Blended finance opportunities via the government and the Environmental Intervention Fund

Private Financing Opportunities

- Banque Agricole du Burkina (Agriculture Bank of Burkina)
- Millennium Challenge Corporation¹⁶⁸
- Société burkinabè des fibres textiles (Sofitex)¹⁶⁹

International Financing Opportunities

- International Commission on Irrigation and Drainage (ICID)¹⁷⁰

- Comprehensive Africa Agriculture Development Programme (CAADP)
- Agricultural Water for Africa (AgWA)
- Economic Community of West African States Water Resources Coordination Centre¹⁷¹
 - Permanent Inter-State Committee to Fight Drought in the Sahel (CILSS)¹⁷²
- Global Environment Fund (GEF)
- Green Climate Fund
- World Bank
- African Development Bank (AfDB)
- West African Development Bank (BOAD)
- Global Partnership for Results Based Approaches (GPRBA)¹⁷³

¹⁶⁷ World Bank, "Burkina Faso Access to Irrigation for Cotton Farming."

¹⁶⁸ Millennium Challenge Corporation, "Improving Agriculture Through Irrigation in Burkina Faso."

¹⁶⁹ World Bank, "Burkina Faso Access to Irrigation for Cotton Farming."

¹⁷⁰ International Commission on Irrigation and Drainage, "Dakar Declaration on Irrigation."

¹⁷¹ Economic Community of West African States, "Guidelines for the Development of Water Infrastructure in West Africa."

¹⁷² Global Water Partnership West Africa, "GWP to Collaborate with CILSS on Sahel Irrigation Initiative."

¹⁷³ Global Partnership for Results Based Approaches, "Burkina Faso Smallholder Irrigation."

4.7. DEVELOPING CLIMATE-SMART ORGANIC VALUE CHAINS

PROJECT SUMMARY

REGIONS: Nord Soudanien and Sud Soudanien

BENEFICIARIES: 60,000 farmers and their families

OBJECTIVE: Develop a burgeoning niche market to create viable careers and shorten supply chains in peri-urban areas.

HIGHLIGHTS:

- Benefits to farm households will increase by 54%.
- Resilience to climate and pest risks is low, with a positive NPV of 29%.
- Improved varieties, integrated soil management, post-harvest technology, improved access inputs, and capacity building will create resiliency in a growing market subsector

JUSTIFICATION AND KEY INVESTMENTS

The risks of agricultural chemical exposure has catalyzed an interest in organic food products in Burkina Faso.¹⁷⁴

The benefits of organic agriculture in terms of both environmental impact and human health impact have been widely appreciated in Burkina Faso since the 1990s.¹⁷⁵

Organic labelling has also recently emerged in Burkina Faso. Smallholder producers of fresh fruits and vegetables have been prominent among those farms already certified by the label.¹⁷⁶

Producers are attempting to respond to the domestic and international demand for organic products¹⁷⁷. However, any producers remain severely limited by poor market linkages and a lack of capacity building. Organic certification demands significantly higher quality standards. It is incredibly challenging for producers to become familiar with and meet these quality standards without capacity building. Poor access to organic inputs and the higher costs of this inputs dissuades producers from investing in organic systems.¹⁷⁸ Transportation challenges force producers to work with traders who hold purchase prices artificially low¹⁷⁹. Consequently, despite strong consumer and producer interest, just 0.5% of Burkinabe land has been converted to agricultural production as of 2017.

Burkina Faso is becoming increasingly urbanized. The country has one of the highest birth rates in the world, and the burgeoning population is increasingly moving to urban areas in search of viable careers¹⁸⁰. Urban populations are heavily reliant on urban and peri-urban farmers for food. These farmers provide up to 36% of the city's total food demand, and up to 90% of its fresh vegetables.¹⁸¹

Climate change may put urban populations at significantly increased risk of food insecurity. Increasing frequent and severe droughts, downpours, and floods threaten the stability of regional food supplies, putting the entire urban population's food security at risk¹⁸²

Irrigation systems are key to unlinking farm productivity from the vagaries of climate change. In rural areas, feasible irrigation solution options are limited by low access to services e.g. electricity, as well as the tools and equipment necessary for system maintenance, operation, and repair. In urban and peri-urban areas, these access barriers are significantly lower, and concomitantly, the options for irrigation innovation are broader. Solar-powered

¹⁷⁴ CNABio, "L'association CNABio."

¹⁷⁵ CNABio.

¹⁷⁶ Capacity Development for Agricultural Innovation Systems, "Organic Certification Takes Root Thanks to Everyone Working Together."

¹⁷⁷ FiBL and IFOAM, "The World of Organic Agriculture - Statistics and Emerging Trends 2019."

¹⁷⁸ Karg and Drechsel, *Atlas of West African Urban Food Systems*.

¹⁷⁹ van Melle and Buschmann, "Comparative Analysis of Mango Value Chain Models in Benin, Burkina Faso and Ghana."

¹⁸⁰ International Finance Corporation, "Creating Markets in Burkina Faso."

¹⁸¹ International Finance Corporation; Gross and Jaubert, "Vegetable Gardening in Burkina Faso."

¹⁸² Karg and Drechsel, *Atlas of West African Urban Food Systems*.

irrigation is an excellent example of a solution that may be infeasible in most rural areas but could have strong potential in many urban and peri-urban areas.

Land tenure and finance services enable innovative irrigation solutions. finance services, and in particular those tailored to the unique characteristics of agricultural production, are scant.¹⁸³ Even farmers trained in a variety of irrigation solutions are unable to implement new knowledge without access to financing¹⁸⁴ and land security¹⁸⁵.

KEY PROJECT INVESTMENTS

Site-tailored CSA pillar promotion with a focus on:

- Organic cotton, mango, and vegetable production
 - Develop heat, drought, and disease resistant varieties
 - Integrated soil fertility management
 - Post-harvest technology and infrastructure development
 - Improve access to bio fertilizer and bio pesticide inputs
 - Build capacity and awareness of organic production and marketing
- Solar powered garden irrigation

POTENTIAL PROJECT IMPACTS

PRODUCTION	Increases for organic farmers in peri-urban and urban areas
RESILIENCE	of organic farming to climate and pest risks is moderate ; economic resilience is increased through sectoral diversification
MITIGATION	Is a co-benefit of improved soil quality
COST	Of US \$833 per beneficiary, or a total cost of about USD \$ 49.98 million
YIELD	Increased by 54% for 60,000 small farming families

ECONOMIC AND FINANCIAL ANALYSIS

Organic farming Cost Benefit Analysis with and without Climate Risks

With Climate & Pest Risks					
Mean No. Beneficiaries	Change in yield (%)	Mean NPV (million USD)	Chance Pos NPV (%)	ROI	BCR (SD)
60,000	54	-5.1	29%	-0.15	-0.08 (1.03)
Without Climate & Pest Risks					
60,000	54	21.8	54%	0.48	0.33 (1.27)

ENABLING ENVIRONMENT

Existing Potential Assets for Project Success

- Strong domestic interest in organic products
- Robust international market demand for organic products
- Rapidly growing year-round urban demand for garden products

¹⁸³ Yamegueu, Alokore, and Corso, "Potential of Microfinanced Solar Water Pumping Systems for Irrigation in Rural Areas of Burkina Faso."

¹⁸⁴ Anang and Awuni, "Effect of Training on Small Scale Rice Production in Northern Ghana."

¹⁸⁵ World Bank, "Burkina Faso Access to Irrigation for Cotton Farming."

- Established, innovative organic certification process¹⁸⁶
- Organic farming is already practiced with some farmers
- Requires relatively little investment in training and infrastructure
- Uses existing technologies

Existing Potential Barriers to Project Success

- Lack of access and availability of organic inputs¹⁸⁷
- Significant market control by powerful conventional companies
- Policy that supports market control by powerful conventional companies¹⁸⁸
- High dependency on external support mechanisms, and poor access to the same¹⁸⁹
- Current gap in extension service supply relative to demand¹⁹⁰

FINANCING

Public Financing Opportunities

- Blended finance opportunities via the government and the Environmental Intervention Fund
- National Council for Biological Agriculture

Private Financing Opportunities

- SINERGI BURKINA
- Textile Exchange
- Union of Vegetable and Fruit Producers (UFMB)
- Association des Professionnels de la Mangue au Burkina (APROMAB)
- Plateforme Nationale de Commerce Équitable Burkina
- Burkinabe Interprofessional Cotton Association (AICB)¹⁹¹
- Union Nationale des Producteurs de Coton du Burkina (UNPCB)
- Organic & Fairtrade Cotton Coalition West Africa
- Europe-Africa-Caribbean-Pacific Liaison Committee (COLEACP) for assistance in compliance with European import standards¹⁹²
- The three regional cotton groups providing inputs, value chain support, and advisory services:¹⁹³
 - SOFITEX (Western region, owned by GEOCOTON, the Union Nationale des Producteurs de Coton du Burkina (UNPCB), and the government)¹⁹⁴
 - SOCOMA (Eastern region, owned by GEOCOTON and UNPCB)¹⁹⁵
 - FASO Coton (Central region, part of the Industrial Promotion Services of West Africa S.A)
 - The Association Professionnelle des Sociétés Cotonnières du Burkina (APROCOB) jointly represents the three groups

International Financing Opportunities

- United States Department of Agriculture (USDA)

¹⁸⁶ Capacity Development for Agricultural Innovation Systems, “Organic Certification Takes Root Thanks to Everyone Working Together.”

¹⁸⁷ Métouolé Médà et al., “Institutional Factors and Farmers’ Adoption of Conventional, Organic and Genetically Modified Cotton in Burkina Faso.”

¹⁸⁸ Métouolé Médà et al.

¹⁸⁹ van Melle and Buschmann, “Comparative Analysis of Mango Value Chain Models in Benin, Burkina Faso and Ghana.”

¹⁹⁰ Consultations Nationales sur l’Eau, la sécurité alimentaire, “Rapport Final de La Consultation Nationale Sur l’eau, La Sécurité Alimentaire et La Nutrition.”

¹⁹¹ Babacar, “Cotton Prices Go up in Burkina Faso.”

¹⁹² Sahel and West Africa Club Secretariat, “Mali – Burkina Faso Cross-Border Co-Operation: Operational Framework Proposals and Policy Recommendations.”

¹⁹³ Métouolé Médà et al., “Institutional Factors and Farmers’ Adoption of Conventional, Organic and Genetically Modified Cotton in Burkina Faso.”

¹⁹⁴ “SOFITEX.”

¹⁹⁵ “SOCOMA.”

- World Bank
- African Development Bank
- United Nations
- Ecocert Burkina Faso ¹⁹⁶
- Organics International¹⁹⁷

¹⁹⁶ Ecocert, “Agir Pour Un Monde Durable.”

¹⁹⁷ Organics International and Its Action Group, “Cultivating Change.”

4.8. DEVELOPING RESILIENT OIL-PROTEIN VALUE CHAINS

PROJECT SUMMARY

REGIONS: Nord Sahelian, Nord Soudanien, Sud Soudanien

BENEFICIARIES: 240,000 farmers and their families

OBJECTIVES: Strengthen the quality, yield efficiency, and value chains of oil-protein crops to sustainably meet domestic demands and support international market development

HIGHLIGHTS:

- Raise income for 240,000 poor farming families by 39%
- With a NPV of 60%, the project is likely to be successful even if risks occur
- Improved quality seeds access, soil quality, soil fertility, and yields

JUSTIFICATION AND KEY INVESTMENTS

Oil-protein crops are well suited to Burkina Faso's climate and hold significant nutritional¹⁹⁸ and economics potential in Burkina Faso¹⁹⁹. However, the growth of the oil-protein crop markets has caused some producers to abandon their climate-smart production practices, increasing already-high vulnerability to climate change impacts.

COWPEA is a main source of both protein and income for most Burkinabe households.²⁰⁰ Burkina Faso now exceeds cowpea self-sufficiency and is beginning to export. There are important opportunities to increase cowpea productivity in Burkina Faso using existing technologies. For example, the shift from food crop to cash crop has implied a move away from traditional intercropping toward monocropped systems,²⁰¹ there is currently very little processing of cowpea crops, and prices are highly volatile.

SESAME offers significant opportunities for increased yield in Burkina Faso. In one study, farmer managed trials showed that 75 kg NPK per hectare increased yields by 75% and provided a 320% return on investment. Improved seed varieties returned 1,900% on investment as compared to recycled seed.²⁰² However, strong international demand has driven 1,500%²⁰³ growth in sesame production over the last decade, and to some extent has deterred improvements in production systems and value chains. Price and demand for crude sesame have been so high that local storage and processing facilities have not been commercially viable, and conventional sesame prices have been peaking above those of organic sesame.²⁰⁴

SOYBEAN sector development is a national priority. There is currently a 74% gap in animal feed demand that could met by domestic supplies, of which soybean is key.²⁰⁵ Soy is also becoming part of local diets, and regional export demand is growing.²⁰⁶ As a nitrogen-fixing legume, soybean requires relatively little fertilizer, and can be grown in rotation with other crops to help maintain soil quality. Soy also requires less pesticide than cotton, and is an excellent nutritional source.²⁰⁷

Developing oil-protein value chains could reduce Burkinabe dependency on imports. The nation has a 85,000 MT annual demand gap in edible oil that must be imported. Meanwhile, the amount of sesame and soybean oil produced

¹⁹⁸ Tarfa et al., "Groundnut and Soybean Response to Nutrient Application in West Africa."

¹⁹⁹ Tarfa et al.

²⁰⁰ Muñoz-Amatriáin et al., "Genome Resources for Climate-Resilient Cowpea, an Essential Crop for Food Security."

²⁰¹ Ouédraogo et al., "Farmers' Willingness to Pay for Climate Information Services."

²⁰² Common Fund for Commodities, "Development of Export-Oriented Sesame Production & Processing in Burkina Faso and Mali (CFC/FIGOOF/27) –."

²⁰³ Rivera, "Sesame Reaps Opportunity for West African Farmers."

²⁰⁴ Common Fund for Commodities, "Development of Export-Oriented Sesame Production & Processing in Burkina Faso and Mali (CFC/FIGOOF/27) –."

²⁰⁵ WASCAL, "Climate-Smart Oilseed Projects in Burkina Faso," 2019.

²⁰⁶ Byrne, "Soybean Production Is a National Priority for Burkina Faso."

²⁰⁷ WASCAL, "Climate-Smart Oilseed Projects in Burkina Faso," 2019.

remains small to negligible, in spite of the fact that the country has a competitive advantage in producing these crops.²⁰⁸

Women farmers are being pushed out of oil-protein crop production. Cowpea and sesame are traditionally women's crops, from production through processing. As these crops have gained economic value, they have become men's crops. Female-led households are now less likely to adopt sesame and cowpea production than male-led households, potentially as a result of women's more limited access to resources that would enable them to diversify into non-staple crops.

KEY PROJECT INVESTMENTS

For cowpea, soybean, and sesame:

- Improved seed varieties
 - Combining drought and pest resistance
 - Addressing standards and certification processes
- Value chains
 - Development
 - Integration
 - Infrastructure, particularly storage and oil processing
 - Organizing for economies of scale
- Optimized field management to maximize yield efficiency
 - Strengthen producer organizations
 - Build technical skills
 - Support robust basic farmer services (extension, climate, finance)

POTENTIAL PROJECT IMPACTS

PRODUCTION	of nutritionally and economically strategic crops increases
RESILIENCE	Increases as reliance on imports, synthetic fertilizer, and animal-based proteins decreases, and as agricultural economy diversifies to meet demand
MITIGATION	through intercropping, leguminous nitrogen fixing, shift from animal to plant -based proteins, and reduction in GHG intensity
COST	Of US \$228 per beneficiary, or a total cost of about USD \$ 54.72 million
YIELD	Increased by 39% for 240,000 small farming families

ECONOMIC AND FINANCIAL ANALYSIS

Cost Benefit Analysis with and without Climate Risks

With Climate & Pest Risks					
Mean No. Beneficiaries	Change in yield (%)	Mean NPV (million USD)	Chance Pos NPV (%)	ROI	BCR (SD)
240,000	39	32.5	60%	0.62	0.44 (2.02)
Without Climate & Pest Risks					
240,000	39	106.1	90%	2.19	1.44 (2.32)

ENABLING ENVIRONMENT

²⁰⁸ Chemonics International, "Burkina Faso Staple Food and Livestock Market Fundamentals."

EXISTING POTENTIAL ASSETS FOR PROJECT SUCCESS

- Burkina Faso's competitive advantage on the international sesame market
- High nutritional value of oil protein crops
- Rising domestic and international demand for oil protein crops
- Nitrogen-fixing properties of cowpea and soybean
- Natural drought resistance and resiliency

EXISTING POTENTIAL BARRIERS TO PROJECT SUCCESS

- Low accessibility to basic services (finance, extension, climate), particularly for women²⁰⁹
- Poor land tenure, particularly for women²¹⁰
- Removal of women from value chains as they become profitable
- Volatile pricing
- International demand and pricing for raw product may disincentivize domestic value chain development
- Difficult seed certification processes²¹¹

FINANCING**Public Financing Opportunities**

- Société Nationale de Gestion des Stocks de Sécurité Alimentaire du Burkina Faso

Private Financing Opportunities

- SINERGI BURKINA
- Seeds suppliers
- SIATOL
 - Entreprises de Services et Organisations de Producteurs, such as:
 - ESOP
 - HELVETAS
 - SIATOL

International Financing Opportunities

- United States Department of Agriculture
- Bill and Melinda Gates Foundation
- OPEC Fund for International Development
- International Fund for Agricultural Development

²⁰⁹ Kpadonou, Barbier, and Denton, Fatima, "New Crops for a New Climate: Understanding Farmer's Behavior towards Sesame and Cowpea Crops in the Sahel."

²¹⁰ Gildemacher et al., "Lessons Learned in Burkina Faso and Mali."

²¹¹ Bougma et al., "Overview of the Burkina Faso Seed System: Case of the Formal Seed System."

4.9. INTEGRATED SOIL MANAGEMENT FOR AGRICULTURAL PRODUCTIVITY AND ENVIRONMENTAL RESTORATION

PROJECT SUMMARY

REGIONS: North Sahel, North Sudan, South Sudan

BENEFICIARIES: 200,000 farmers and their families

OBJECTIVE: provide producers and extension agents with location-tailored information on soil characteristics and best management practice recommendations, as well as the tools, products, partnerships, and policy environment to implement those recommendations

HIGHLIGHTS:

- Improved agricultural soil fertility
- Concomitant increases in productivity, food and nutritional security, and economic growth that foster climate resilience
- Restoration of degraded lands, slowing of agricultural expansion, and greenhouse gas mitigation

JUSTIFICATION AND KEY INVESTMENTS

Soil health is essential to climate-smart agriculture, yet agriculture generally has negative effects on soils.

Burkina Faso's economy is threatened by soil degradation. Growing food demand as a result of rapid population growth has driven significant agricultural expansion²¹². Between 1971 and 2016, agriculture production expanded from 30% to 44% of total land area, and natural spaces fell from 83% to 57%²¹³. Today, 9 million hectares of productive land, or 33% of the total national territory, is degraded. Agricultural expansion is expected to accelerate²¹⁴, and degradation will be further exacerbated by extreme weather events such as drought, flooding, and high winds²¹⁵. The result is loss of productivity²¹⁶.

Integrated soil fertility management (ISFM) is an effective way to regenerate soils with limited inputs. Burkinabe soils can be very agriculturally productive under ISFM.

Burkinabe smallholder's degree of access to fertilizers creates demand for ISFM efforts. While current fertilizer application rates are insufficient for improving yields alone, they are sufficient for ISFM-based targeted dosing of inorganic fertilizers as informed by a national Soil Information System. Given current fertilizer access, such optimization of crop productivity via ISFM is crucial to achieving soil fertility and crop productivity.

Soil information systems enable ISFM on a large scale. The World Agroforestry Center (ICRAF) has developed spectral diagnostics²¹⁷ using infrared and x-ray technology. These systems enable rapid and low-cost analysis of soil properties and plant nutrients that can be applied at scale for digital mapping²¹⁸. The level of detail, accuracy, and geographic scale that this technology offers at low cost promises to transform the soil management paradigm²¹⁹.

Burkina Faso has committed to stop land degradation by 2030, and is signatory to several international initiatives to address degraded lands. Burkinabe stakeholders already practice some techniques to help maintain land and soil quality. However, a lack of equipment, materials, and sufficient labor force for soil restoration technologies are hindering the implementation of restoration and conservation management²²⁰.

²¹² Neya et al., "Trade-off of Tree Conservation and Crop Production on Agroforestry Parklands in Burkina Faso."

²¹³ World Bank, "Agricultural Land (% of Land Area) - Burkina Faso."

²¹⁴ "Rehabilitating Degraded Land Is a Priority for Burkina Faso", Says FAO Director General | Action Against Desertification | Food and Agriculture Organization of the United Nations."

²¹⁵ EAP, "Enquete Sur La Productivite Agricole Au Burkina Faso."

²¹⁶ Nyamekye et al., "Soil and Water Conservation in Burkina Faso, West Africa."

²¹⁷ Soil-Plant Spectral Diagnostics Lab, "Network of Dry Spectroscopy Laboratories."

²¹⁸ World Agroforestry Centre, "Soil-Plant Spectral Diagnostics Laboratory."

²¹⁹ World Agroforestry Centre, "Testimonials."

²²⁰ Nyamekye et al., "Soil and Water Conservation in Burkina Faso, West Africa."

KEY PROJECT INVESTMENTS

Site-tailored CSA pillar promotion with a focus on:

- Integrated soil fertility management capacity building
- Soil information system development and implementation

POTENTIAL PROJECT IMPACTS

PRODUCTION	Increases across all crops as a result of optimized soil fertility, resulting in improved incomes and food and nutritional security, as well as less drive for agricultural expansion
RESILIENCE	Is improved through leveraging readily available technologies and materials and reduced reliant on scant or inconsistently available inputs
MITIGATION	through slowed agricultural expansion, augmented soil and biomass sequestration
COST	US\$300 per beneficiary, totaling US\$60 million
YIELD	29% yield increase for 200,000 farmers and their families

ECONOMIC AND FINANCIAL ANALYSIS**Cost Benefit Analysis with and without Climate Risks**

With Climate & Pest Risks					
Mean No. Beneficiaries	Change in yield (%)	Mean NPV (million USD)	Chance Pos NPV (%)	ROI	BCR (SD)
200,000	29%	72.0	71%	1.35	0.9 (2.05)
Without Climate & Pest Risks					
200,000	29%	128.9	83%	2.46	1.61 (2.43)

ENABLING ENVIRONMENT**EXISTING POTENTIAL ASSETS FOR PROJECT SUCCESS**

- Close alignment with NDCs and many other national policies.
- Grass-roots buy-in; producers and extension agents are already familiar with soil conservation practices
- Robust governmental commitments and international support
- National fertilizer supply is sufficient for ISFM practices

EXISTING POTENTIAL BARRIERS TO PROJECT SUCCESS

- Fertilizer distribution may not reach all smallholders
- Limited organic matter soil inputs available
- Drought and flood events are increasing in frequency
- Land tenure insecurity curbs smallholder investments
- Low availability of quality planting material

FINANCING**Public Financing Opportunities**

- FIE (Environmental Intervention Fund)

International Financing Opportunities

- Global Environment Fund (GEF)
- Green Climate Fund
- World Bank
- African Development Bank (AfDB)

- West African Development Bank (BOAD)
- International Fund for Agriculture Development (IFAD)
- Global Soil Partnership
- Great Green Wall Initiative²²¹
- Land Degradation Neutrality (LDN) program²²²

Private Financing Opportunities

- Banque Agricole du Burkina (Agriculture Bank of Burkina)
- Irrigation companies
- Timber and forest product processors

²²¹ “Overview | Action Against Desertification | Food and Agriculture Organization of the United Nations.”

²²² “The LDN Target Setting Programme | UNCCD.”

CHAPTER 5: MONITORING AND EVALUATION: ASSESSING OUTCOMES AND IMPACTS

5.1 Context

123. **A robust and practical monitoring and evaluation (M&E) Plan is an integral component of the CSAIP for Burkina Faso.** These M&E guidelines aim to provide CSAIP implementors—particularly the Government—with a manageable system to facilitate generating, collecting and analyzing standardized data to assess the success of portfolio investments and inform operational and strategic decision-making. The M&E guide establishes and illustrates links between the plan’s objectives (expected outcomes and impacts) and the inputs and activities (projects). It also identifies key requirements for implementing the Plan (Box 5-1).

124. **M&E activities provide information on how project actions influence change and how investments are performing.** They create a mechanism for tracking progress against targets, learning lessons, increasing accountability, raising flags when adaptive action may be necessary, and telling data-driven stories of success by government agencies, financial institutions, subnational agencies, communities and other decision-makers. M&E plans and activities need to be dynamic and can be revised and adjusted as lessons emerge from its implementation.

Box 5-1: Defining key M&E terms

- **EVALUATION:** occasional and in-depth data collection for assessing outcomes and impact and the intervention strategy (e.g., effectiveness)
- **IMPACT:** high-level objectives identified by stakeholders during the development investment plan (i.e., the project development objective).
- **INDICATORS:** information documenting current state and changes of activities, outputs, outcomes or impact.
- **MONITORING:** continuous data collection tracking implementation of budgets and activities (planned vs. achieved).
- **OUTPUTS:** tangible products of project activities including trainings, publications, partnerships, new technologies, policies and infrastructure (e.g. weather stations).
- **OUTCOMES:** changes in behavior including knowledge, attitudes and skills of stakeholder groups that result from project activities and outputs.
- **RESULTS FRAMEWORK:** a graphic summary and management tool summarizing the expected results from particular interventions, such as investments, development plans or policies.
- **THEORY OF CHANGE:** a description or diagram of why and how the desired change and objectives will occur.

5.2 Theory of Change

125. **The CSAIP strongly focuses on improving the food and nutritional security of the population in Burkina Faso by sustainably increasing agricultural productivity and by building resilience of farms, farmers, ranches, landscapes and the food system generally.** The CSAIP emphasizes productivity and resilience because (i) the country's agriculture, forest and land-use sector (AFOLU) has been a low contributor to global carbon emissions; and (ii) the program is designed to address national food security and resilience priorities.

126. **This CSAIP also contributes to climate change mitigation, the third pillar of CSA and to the voluntary national mitigation commitments expressed in the NDC, via climate change mitigation co-benefits.** In its INDC, Burkina Faso committed to a 10% decrease in emissions associated with agriculture, land use and forestry by 2030 primarily through sustainable land management and reforestation²²³. Many projects reflected in this CSAIP—particularly the ones focused on sustainable livestock intensification—will contribute to decreases in GHG emissions per unit of product (known as GHG intensity). Investments in tree crops systems also sequester carbon and reduce emissions from farms and landscapes.

127. **These long-term benefits are embedded within the broader framework of the 2030 Agenda on Sustainable Development Goals (SDGs).** Improving productivity, food and nutrition security and increased farmers' income and resilience helps Burkina Faso achieve targets of at least six of the seventeen SDGs²²⁴, which are also reflected in the National Plan for Economic and Social Development (PNDES) and its 14 sectoral policies. This CSAIP increases impact in the six above-mentioned development areas, while the M&E Plan will support the Government in harmonizing reporting systems related to global voluntary commitments (SDGs, NDCs), and national goals.

128. **Four key action areas can foster change in the short, medium and long term:** (i) robust research and development (R&D); (ii) stakeholder engagement and partnerships, (iii) system-wide capacity to implement CSA and (iv) uptake of value chain-level practices and technologies. Robust R&D is needed to facilitate introduction of improved varieties and breeds adapted to new climate conditions (especially to droughts and floods), while reliable, timely knowledge and information is critical for risk management. Stakeholder engagement and partnerships are required to foster concerted action, particularly in the area of information exchange and learning, natural resource planning, management and use. System-wide capacity (human, technical, institutional, and financial) is essential for implementing CSA actions (technologies, services, policies) across the agriculture decision-making spectrum. On-farm actions that target production as well as value-addition and marketing technologies and strategies are important

²²³ Government of Burkina Faso, 2015 (<https://www4.unfccc.int>)

²²⁴ SDG 1 (No poverty), SDG 2 (Zero hunger), SDG 6 (Clean water and sanitation), SDG 12 (Responsible consumption and production), SDG 13 (Climate action), and SDG 15 (Life on land)

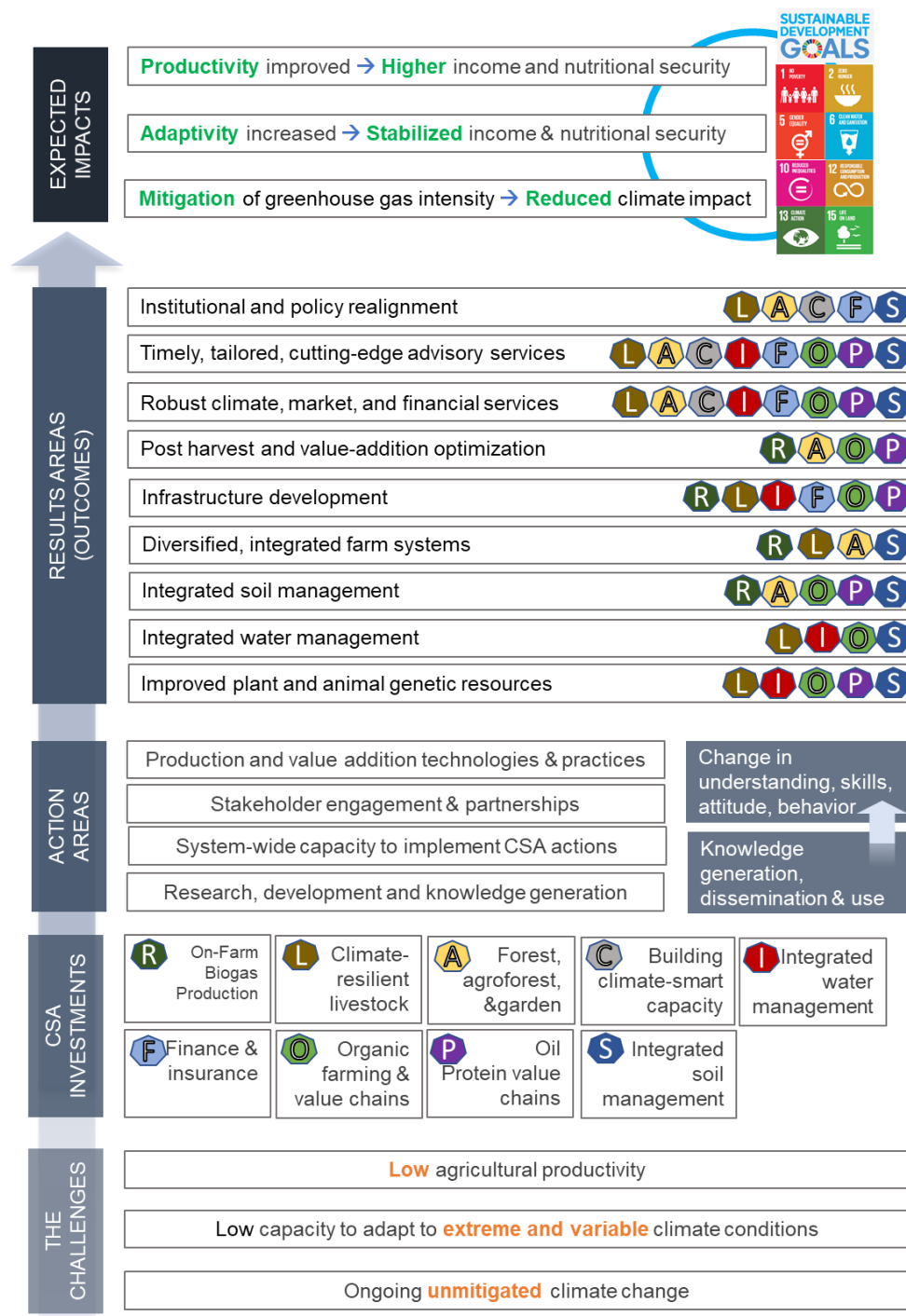
instruments for building resilience building of the food system. These four key actions build on each other and form the core of this CSAIP's theory of change.

129. **The theory of change (TOC) assumes that increased use of CSA will be facilitated by knowledge development and dissemination, but also by its accessibility and usability by the end-users (farmers, extension staff, private sector actors, decision makers, etc.).** According to the TOC, use of relevant knowledge and timely information will lead to changes in attitudes, awareness levels, skills, opinions and ultimately to changes in behavior (short term), to increases in productivity and incomes (medium term) and to improved food security, nutrition and resilience (long term).

130. **To build resilience of the agricultural system, change needs to happen at multiple scales, across sectors and landscapes—farm, policy (institutional), markets, private sector to meet development goals (Figure 20).** The CSAIP aims to work with diverse beneficiaries across the food system, located in different agroecological zones (AEZ) of the country (see chapter 3). Investments are directed toward crop farmers and livestock keepers, but also to smallholders engaged in aquaculture and fisheries. The CSAIP also plans activities that affect the functioning of markets and value chains with the private sector. Support to government institutions to establish the needed support framework for CSA is included, reaching policy setting and implementation, research, knowledge development, and capacity building. This broad support ensures that all the major types of actors in the food system will be engaged by and benefit from the CSAIP, catalyzing transformative change.

Figure 20: How CSAIP investments and implementation lead to benefits at multiple levels

The key action areas apply to all nine investments, enabling different types of outcomes. Outcomes are investment-specific. Medium-term and long-term changes (outcomes and impacts, respectively) contribute to the attainment of multiple SDGs.



5.3 Results Framework and Core Indicators

131. **The results framework forms a core component of the CSAIP M&E Plan, and is built from fundamental data characterizing farm, household and value-chain activities.** It will track productivity, resilience, adaptive capacity and to quantify the adaptation and mitigation benefits from the program investments. This approach allows it to be extended to agricultural interventions outside the CSAIP. It includes core indicators agreed upon by investment stakeholders that need to be collected and monitored at national and regional (investment zone) levels and reported on a regular basis. Each project will be monitored separately using guidelines that will be established at project inception, along with key performance indicators (KPIs) and targets. Various types of indicators should be reflected in the Results Framework, to suggest the different objectives of M&E activities—tracking implementation progress, adaptive project management, evaluation of outcomes and impacts. There are three types of indicators that track different scales:

- *Output indicators:* linked directly with portfolio investment activities and more easily tracked.
- *Outcome indicators:* closely—but not exclusively—related to project activities.
- *Impact indicators:* measuring change in the broader goals and provide insights toward the end of a five-year (or longer) project lifespan.

132. **A consultation process informed the development of this M&E guide.** Consultations (semi-structured interviews and focus groups) were carried out with staff at four CSAIP implementing ministries — Ministry of Agriculture and Water Resources (MAAH), Ministry of Animal and Fishery Resources (MRAH), Ministry of Water and Sanitation (MEA), Ministry of Environment, Green Economy and Climate Change (MEVCC)—, the World Bank, and the West African Science Service Center on Climate Change and Adapted Land Use (WASCAL), a key partner in the CSAIP design process. These stakeholders provided critical input and support in three main areas: i) establishing the scope of the M&E guidelines; ii) determining investment-level elements to be monitored and evaluated; and in iii) providing general direction on how M&E of the CSAIP will be carried out, based on current M&E system(s) available. Table 24 and Table 26 provide examples of output-level, outcome and impact indicators relevant for the nine portfolio investments.

133. **Output indicators are directly tied with the project activities.** Different investments may have similar project activities, so the indicators, and how they are measured, would be similar. A complete list is prepared when the investments are being developed and as part of the M&E system design. Data for output indicators is typically collected through the Project Management Information System (PMIS), which can be coordinated with existing information systems under the Ministry of Agriculture.

Table 24: Sample output-level indicators relevant for the CSAIP, by project

Project	Sample indicators
Sustainable management of water resources and irrigation	• Management committee(s) established for the dams, to coordinate and supervise planning and use of water and land resources in the area, # • Actions to protect water bodies and water banks (e.g., tree planting, agroforestry, livestock corridors, etc.) carried out, #, type • Production infrastructure for aquaculture and fisheries production established (e.g., basins, cages, etc.), #, type • Improved crop varieties released, #, type • Farmers receiving technical assistance on adequate use of inputs for aquaculture and crop production (fingerlings, improved feed, organic fertilizers, etc.), #, type • Farmers receiving technical assistance on improved CSA production practices and technologies (e.g., Systems of Rice Intensification, using water pumps and sprinklers, crop diversification, etc.), #
Sustainable on-farm biogas production	• Domestic biogas digesters installed, # • Farmers receiving technical assistance on management and use of organic residues for biogas and fertilizer production, including biogas digesters, #
Sustainable intensification of livestock production	• Farmers receiving technical assistance on sustainable livestock intensification, # • Equipment and facilities for sustainable livestock intensification installed (irrigation for fodder production, water ponds, fattening units), #, type • Livestock markets (physical infrastructure) established and/or improved, # • Animal vaccine production units established, # • Extension agents receiving training on adequate use of animal vaccines, #
Forest, agroforest, & garden production for climate-smart diversification	• Community-based park management committee established • Reforestation activities carried out, #, type • Nutritional/productive trees planted (Shea tree, baobab, gum Arabic, moringa), #, type • NTFP transformation units established, # • Communal nutritional gardens established, # • Communal harvest storage facility established, #
Building capacity in climate-smart agriculture	• Staff receiving training on agricultural impact evaluation (vegetation inventories, use of satellite images, early-warning systems using climate information), # • Equipment for measurement of water levels purchased, by type, value • Open-source platform that integrates climate and agricultural information established • Communication channels established to ensure weather, climate, agriculture information is accessible to users, #, type, # and type of users • Training module on CSA elaborated and integrated into school curricula (university and other specialized institutions) • CSA experimental areas established, # • International scholarships for regional knowledge exchange on CSA set up, # and value
Finance and insurance services to foster climate-smart agriculture	• Weather stations established, # • Financial products developed for smallholder farmers, #, type • Insurance products developed for smallholder farmers, #, type
Developing climate-smart organic value chains	• Farmers receiving technical assistance on organic production, #, by product type (cotton/mango/vegetable) • Product processing units established, #, type
Developing resilient oil-protein value chains	• Farmers receiving technical assistance on CSA practices and technologies for oil-protein crops, #, by crop type (sesame/leguminous) and practice type • Product processing units established, #, type

Integrated Soil Management for Agricultural Productivity and Environmental Restoration	- Farmers receiving technical assistance on CSA practices for soil conservation/restoration (stone bounds, zai pits, farmer assistant natural regeneration, agroforestry, etc.), #, type - Infrastructure and equipment (tractors) for soil restoration/recovery/conservation, #
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134. **Outcome-level indicators require M&E systems to be in place and special studies** as they are not reported on a routine basis. Special studies can include: baseline studies and follow-up longitudinal studies, national surveys, case studies introduced to routine M&E activities, and participatory beneficiary assessments (it is particularly important to evaluate farmers' opinions, changes in knowledge, behavior, etc.) (Table 25).

Table 25: Sample outcome-level indicators relevant for the CSAIP.

Investment	CSA Pillar	Indicator and measurement
Outcome: Improved and/or conserved breeds		
3	Productivity, Resilience	- Improved/Conserved breed stock (Percentage change, by type of breed) - Farmers using improved/local breeds (Number of/Percent, by type of breed) - Public/public-private funding allocated to R&D for breed improvement/conservation (Change in funding allocated by type of breed, %)
Outcome: Improved production practices, technologies, and infrastructure		
1,2,3,4,7,8, 9,10	Productivity, Resilience	- Farmers using CSA practices/technologies/infrastructure for production (Number of/Percent, by practice/technology/infrastructure type) - Area under CSA production practices/technology/infrastructure (Number of ha, % of agriculture area) - Farmers using production inputs as per extension agents' recommendation (Number of/Percent, by type of input)
Outcome: Improved post-harvest activities and facilities, enhanced marketing conditions		
3,4,7,8	Productivity, Resilience	- Farmers engaged in value addition activities (Number of/Percent, by activity type and value chain) - Unit cost for transportation of agriculture products (\$) - Distance to markets (km)
Outcome: Improved access to water for agriculture		
1,10	Productivity, Resilience	- Farmers using water harvesting technologies for crop/livestock/fish (Number of/Percent, by technology and purpose) - Increased crop area under irrigation (Percentage change) - Increased area under sustainable aquaculture (Percentage change)
Outcome: Improved access to clean, affordable energy sources		
2	Productivity, Mitigation	- Farmers using domestic biodigesters for biogas and/or fertilizer production (Number of/ Percent) - Decreased consumption of fossil fuels for farm and household energy demands (Percent change)

		• Money spent on farm and household energy consumption, \$/year
Outcome: Improved ecosystem service delivery		
3,4	Productivity, Resilience, Mitigation	• Decreased soil erosion rates (Percentage change) • Improved soils: Changes in soil physical indicators (texture, aggregation, moisture, porosity), chemical (C, N, mineral nutrients, organic matter), and biological indicators (microbial biomass C and N, biodiversity, soil enzymes, soil respiration, etc.) (Percent) • Area under sustainable communal grazing system (ha) • Farmers with access to sustainable communal grazing systems (Number of/Percent (#/%)) • Farmers with access to improved fodder/legume seeds (Number of/Percent) • Farmers with access to communal, nutritional gardens (Number of/Percent)
Outcome: Sustainably intensified, diversified, integrated farming systems		
1,3,4,8	Productivity, Resilience	• Area under sustainable livestock intensification (ha) • Area under agricultural diversification (by diversification type; products included) (ha) • Farmers using diversification practices on their farm (Number/Percent) • Farmers using integrated farm systems (crop-livestock; crop-trees; crop-livestock-trees; fish-poultry, etc.) (Number/Percent, by type of system)
Strengthened institutions and policies to enable CSA actions		
1,3,4,10	Productivity, Resilience	• Improved planning, supervision and coordination of water and land resources in the dam areas, # of meetings/year, # of decisions taken/year # of cases solved/year • Improved management of NTFP parks, # of committee meetings/year, # of decisions taken/year # of cases solved/year • National action plan for managing climate-related risks to animal health developed and implemented
Outcome: Strengthened capacity to implement CSA actions and manage climate risks		
1,2,3,5,6,7, 8,9	Productivity, Resilience	• Number of qualified extension staff (#); • Ratio of extension agents to farmers/farmer groups (by farmer type-crop, livestock- and value chain activity-production, post-harvest, storage, etc.); • Client contacts per extension agent (#); • Beneficiaries of digital-based extension services (by beneficiary type and digital tool) (#) • Quality rating of service/technical capacity by beneficiary (qualitative assessment); • Increase in funding for extension services (\$) • Quality rating of CSA training modules, by beneficiary (qualitative assessment) • Students with access to CSA experimental sites • Farmers using financial instruments and insurances to fund their farm activities (Number/Percentage)

Note: Investments: 1=Water management; 2=Organic residues; 3=Sustainable livestock intensification; 4=Non-timber forest products and nutritious gardens; 5=Capacity building for CSA; 6=Financial and insurance products; 7=Organic farming (cotton, mango, vegetables); 8=Oil protein crops (sesame, leguminous); 9=Restoration of degraded lands and soil conservation; 10=Surface water mobilization

135. **Impact indicators are broader in scope and refer to larger societal changes.** Various actors in Burkina Faso already collect relevant data for measuring and analyzing the impact of agricultural interventions. Much of the needed data for the sample impact indicators below can be found in agricultural censuses (French: Recensement Général de l'Agriculture), the National Nutritional Survey (French: Enquête Nutritionnelle Nationale), the Standardized Monitoring and Assessment of Relief and Transition Survey or donor reports (e.g., Nutrition Profiles of the United States Agency for International Development (USAID), FAO, etc.). Table 26 shows sample impact indicators.

Table 26: Sample impact indicators relevant to the CSAIP.

Impact Indicator	Suggested measurement(s)
Impact: Improved productivity, food and nutrition security	
Productivity	- Harvested crop yield per hectare (per CSAIP agricultural commodity) - Gap between actual and potential yields (Percentage, calculated as: $(1 - \text{actual} / \text{potential}) \times 100\%$) - Yield variability under varying conditions (Yield time series, kg/ha)
Nutrition of the population	- Population suffering from stunting (Number/Percentage, by age group, sex) - Population underweight (Number/Percentage, by age group, sex) - Population suffering from wasting (Number/Percentage, by age group, sex) - Population suffering from anemia (Number/Percentage, by age group, sex)
Food security of the population (availability, accessibility, utilization, stability)	- Prevalence of food insecurity, by type (Percentage); - Percent crop losses during storage (Percentage, calculated as: loss rate per time period x amount stored) - Dietary diversity and food frequency: Food Consumption Score (FCS); Household Dietary Diversity Scale (HDDS); Spending on food (\$/year); - Consumption behavior: Coping Strategy Index (CSI); Household Food Insecurity and Access Scale (HFIAS);
Impact: Increased farmers' incomes and climate resilience	
Income	- Average farm income (\$/year) - Value of crop/livestock/fish on local market (\$)
Farm resilience to food insecurity	- Resilience capacity index (FAO-RIMA II) - Increased representation of women in food and nutrition activities at community level (USAID Resilience goal for Burkina)

Box 5-2: Aspects to consider in the design of a comprehensive M&E Plan for the CSAIP.

WHY: Determine the scope of the M&E Plan. Describe how data will be used and by whom, how it will feed into policymaking and decision-making in general (including international reporting processes). Setting the scope is an essential first step, as it helps define the breadth of indicators required and the information system that will need to be put in place or adapted.

WHO: Clearly delineate roles and responsibilities. This refers to defining what institutions will be responsible for different M&E activities, how responsibilities are divided, staff and qualifications required to carry out M&E activities (from data collection to cleaning, analysis, reporting, dissemination, coordination, etc.). Establishing clear procedures for regular flow of information between decentralized implementation units and a central M&E unit will be essential for a smooth implementation of the Plan.

WHAT: Establish what information and data needs to be collected. This section of the CSAIP provides guidance on the initial design of the M&E, showing a first appraisal of potential opportunities for Burkina Faso based on current levels of elaboration in the project concept notes. In a next step, the list of indicators needs to be refined and aligned to the needs of information users and the objectives outlined in each project. To define these, additional consultations with investment stakeholders and M&E staff from existing governmental departments are required, to ensure participation, alignment and coordination.

HOW: Establish procedures and methodologies for M&E. Determine *what* methods and sources will be used to gather, analyze and report the data, *how* the Project Management Information System (PMIS) will manage data (information flow across levels and institutions) and *when* data needs to be collected and reported (frequency). Baseline data should ideally be available at project appraisal, but, if not, arrangements for collection of baseline data in the first year of the project should be spelled out in the M&E Plan. Tools for established procedures and guidelines may include: annual work plan and budget for M&E; planning and tracking tools; M&E calendar; and standard progress reports and quarterly reports, among others.

WITH WHAT RESOURCES: Carry out an initial capacity assessment with CSAIP stakeholders. Understand existing capacities at all levels (local, regional, national/coordination) in terms of established M&E systems, staffing, strengths, and weaknesses, and lay out additional training, technical assistance, and other inputs needed to strengthen M&E capacity for the CSAIP. This ensures a results-based reporting approach throughout the CSAIP implementation period.

5.4 Key considerations for next steps

136. **Several steps are needed to develop a comprehensive, nationally-integrated information system for measuring and reporting progress in implementing CSA**, based on a recent multi-country analysis²²⁵. Additional information is needed to outline the components of the M&E Plan, including: establishing the scope of the M&E activities, outlining roles and responsibilities of participating institutions, defining the tools for implementation, establishing data management protocols, refining logistics. These actions, adapted to Burkina's context, are summarized in Box 5-2. These suggested actions

²²⁵ Nowak et al., 2019.

inform assessments of resources required for M&E set up and implementation and define the space for the M&E system for the CSAIP and ensure long-term sustainability.

137. Refining indicators is a needed step. To make it actionable, the CSAIP M&E system should strike a balance between what is ideal to measure and what is practical (in terms of costs and time). The total number of indicators included needs to be reasonable and reflect the project's theory of change and also the resources available to track them periodically. Applying SMART criteria (Specific, Measurable, Attainable, Relevant and Timely) when selecting the indicators will help shorten the list, ensuring relevance and usability of the system. Ideally, the final list of prioritized impact, outcome, and output indicators should be developed by project component, contain clear targets (defined in terms of time, quantity, quality) and be gender-disaggregated, where possible.

138. Data from a variety of sources can be combined and used. The CSAIP's M&E system will also need to draw on data from various sources, including the project management information systems (PMIS), national data sets and other studies. PMIS are typically based on regular project implementation reports and provide data on inputs (e.g. number of extension staff), outputs (e.g., farmers trained) and initial outcomes (e.g., adoption rates of CSA technologies). National statistics are typically used for assessing long-term outcomes (5-10 years or more), while special studies and participatory evaluations are good sources for qualitative information (changes in attitudes, knowledge, behavior, wellbeing levels, etc.). Rapid rural appraisals and case studies are designed to provide targeted information in a cost- and time-effective way and can be an additional source of M&E data for the CSAIP projects.

139. Leveraging existing information systems can help quickly start the process. M&E systems embedded in different ministries and agencies working on agriculture and climate change in the country already contain a wealth of information relevant to CSA investments. All major policies²²⁶ lay down objectives and targets concerning aspects of productivity, food security and nutrition, and sustainable natural resource management. Information systems are embedded within sectoral ministries but also within larger coordination bodies, such as the information system for food security under the Executive Secretariat of the National Council for Food Security (SE/CSNA) or the M&E framework under the Permanent Secretariat for the Coordination of Agricultural Sector Policies (SP/CPSA)²²⁷. Table 5-2 illustrates how the M&E of CSAIP is relevant and aligned to key national strategies, plans, and programs and their results frameworks. Exploring synergies between the CSAIP and existing information systems will be a key step for improved coordination, alignment and effective resource allocation, but also for harmonizing reporting processes (national, regional, international).

²²⁶ For example, the Sectoral Policy for Agro-silvo-pastoral Production, the Rural Development Strategy, the National Policy for Food and Nutrition Security, the National Policy for Sustainable Development, the National Environmental Policy, to mention a few.

²²⁷ SP/CPSA is a key actor for M&E of the rural sector. It is an inter-ministerial structure mandated to ensure coordination, coherence, implementation and M&E of agriculture sector policies and strategies and to promote multi-stakeholder dialogue on rural development, among others. It is formed of representatives in the ministries whose work falls within the scope of rural development, including: agriculture, livestock and fisheries, water resources, environment and climate change.

140. **Existing policies and institutional set ups provide opportunities for basing the CSAIP's M&E on established, legitimate structures, rather than creating new ones.** The National Programme for Rural Development (PNSR) is the country's reference framework for agricultural and food policies and the main vehicle for operationalizing CAADP commitments. It also establishes a governmental mechanism for M&E of food and agriculture investments under PNSR, laying out key performance indicators, designating M&E roles and responsibilities for the main institutions that produce the bulk of agriculture and food data in the country and recommends M&E methods and tools to use for data collection, analysis, reporting, and dissemination²²⁸. Embedding the CSAIP's M&E into existing infrastructure like this would help avoid multiplication of structures and mechanisms, guarantee continuity and sustainability of efforts and combining available human, material and financial resources for greater efficiency.

141. **Clearly defining roles and responsibilities is integral to CSAIP M&E.** Even when using already established institutional structures and information flow mechanisms, roles and responsibilities need to be spelled out. Ideally, the M&E system for the CSAIP would be embedded within the SP/CPSA, with relevant directorates within line ministries and field offices carrying out routine monitoring. An M&E coordinating team would operate through the decentralized framework of the CSAIP, support and coordinate M&E staff at the project and district levels and assist with their training. The coordinating team would also provide the necessary technical support and be responsible for regular reporting to stakeholders, for tracking and documenting the implementation progress, and performance and outcome evaluation. Regular data collection would mostly fall under extension officers' mandates, while district-level data cleaning, analysis and reporting to the M&E coordination team would be undertaken by district-level M&E officers.

142. **Developing/strengthening permanent M&E capacity.** A robust, functioning M&E system will require adequate investment in staffing, both in terms of human resources and their training (skills). A study assessing capacities and gaps in investment planning, implementation and M&E of PNSR programs revealed that the low quality (skills) and quantity of human resources, as well as inadequate funding at all levels (local, regional, national/coordination), prevents the smooth functioning of the current system (Table 27). Moreover, more than 50% of the PNSR institutions do not have a formal M&E system, even if they produce periodic M&E reports, while many respondents highlighted that efforts to disseminate the data and information especially to non-state actors were insufficient, due to absence of funding for such activities²²⁹. The M&E system of the CSAIP needs to take stock of lessons learned from similar processes and build on them, in order to ensure its operability and sustainability.

²²⁸ For instance, there is an established multi-level information flow between: PNSR beneficiaries (population); M&E systems of PNSR sub-programs and line ministries which organize data collection and analysis and prepare initial reports; General Directorates of Studies and Statistics (DGESs) of line ministries which ensure production and dissemination of all internal M&E documents within the ministry; the coordination body that summarizes all reports (SP/CPSA), and the higher decision-making bodies (The National Committee for Guidance and Piloting, COP and the Inter-ministerial Technical Committee, CTI).

²²⁹ Yameogo and Bako, 2014 (<https://www.ifpri.org/node/15569>)

Table 27: Selected national strategies, plans and programs relevant to CSA in Burkina Faso and main relevant expected outcomes.

Selected policy objectives (PO) based on relevance to the main CSAIP outcomes and investments	Existence of M&E Plan with indicators
National Programme for the Rural Sector II, 2016-2020 (Programme National du Secteur Rural, PNSR)	
Focus Area 1: Food and nutrition security, resilience of vulnerable populations 1.1: Sustainable development of agricultural production 1.2: Food and nutritional crisis risk management and mitigation 1.3: Irrigation systems 1.4: Animal health and public safety 1.5: Development of aquaculture and fishery systems Focus Area 2: Access to markets, competitiveness of agro-silvo-pastoral and aquaculture products 2.1: Agricultural economy 2.2: Productivity and competitiveness of livestock products Focus Area 3: Environmental governance, promotion of sustainable development and management of natural resources 3.1: Environmental governance and sustainable development 3.2: Sustainable management of forest resources and fauna 3.3: Green economy and climate change 3.4 Sustainable management of pastures Focus Area 4: Water, sanitation and living environment 4.1: Water resource mobilization 4.2: Integrated water resource management Focus Area 5: Land security and strengthened human capital in the rural sector 5.1: Land security, agricultural vocational training and rural organization 5.2: Scientific and technological research	M&E Framework with output, outcome and impact indicators and baseline values. PNSR I (2011-2015) lays out an implementation guide with defined roles and responsibilities for M&E across the concerned ministries, with SP/CPSA spearheading M&E activities at national level
National Climate Change Adaptation Plan, 2015 (NAP)	
<i>Agriculture sector:</i> Overall objective: increase the resilience of family farms through climate change adaptations O1: recover and restore the fertility of degraded soils O2: Improve access for farmers to high-quality agricultural production factors (equipment, inputs, land, agricultural research outcomes etc.) O3: increase stakeholder resilience to climate change O4: develop early warning systems to ensure efficient management of variability and climate change <i>Livestock farming sector:</i> O1: Improve the security of pastoralism through better dissemination and promotion of information on pastoral resources and the relevant access conditions. O2: Improve the protection of animal capital in order to provide sustainable support for the pastoralist economy and improve stakeholder resilience with a view to sustainable food security in Burkina Faso O3: Reduce the climate vulnerability of pastoralist farmers and promote local economic development	Yes, a results-framework for the agriculture and livestock sector. Responsibilities rest with the Permanent Secretariat of the National Council for the Environment and Sustainable Development (SP/CONEDD), which is tasked to review the overall implementation progress and to run a database with performance indicators, and M&E focal points in the ministries covered by the Action Plan, which are expected to work closely with SP/CONEDD in updating the database periodically.

<i>Environment and natural resources sector:</i> O1: Increase ecosystem productivity and resilience O2: Improve biodiversity Conservation O3: Improve ecological research and monitoring O4: Mitigate greenhouse gas emissions <i>Energy sector:</i> O3: reduce the use of wood energy by promoting alternative energies such as butane and biogas and by promoting the use of agricultural residue biomass	
National Plan for Economic and Social Development, 2016-2020 (Plan National de Developpement Economique et Social, PNDES)	
EA 2.1.2: The nutritional status of the population, especially women and children, is improving EA 3.1.2: The resilience of agro-pastoral, wildlife and fishery households at risk is strengthened EA 3.5.1: The environment and natural resources are managed sustainably EA 3.5.2: Mitigating and adaption to the adverse effects of climate change are reinforced with a view to transitioning to the green economy	Yes, including baseline value for indicators. The M&E framework provides reporting formats at various levels: global (national), regional, sectoral.
National Priorities for Resilience, 2016-2020 (Priorites Resilience Pays, PRP)	
General objective: By 2035, poverty rates of the population and vulnerability of their livelihoods will be reduced visibly and their food and nutrition security sustainably improved Pillar 1: Improve social protection of communities and vulnerable households for securing their livelihoods Pillar 2: Improve nutrition of vulnerable households Pillar 3: Sustainably improve agricultural productivity, incomes of the most vulnerable, and their access to food 3.1. Intensification of agro silvo-pastoral and aquaculture production 3.2. Sustainable development of agricultural and pastoral water management systems 3.3. Strengthen links between producers and markets 3.4. Improve the income of vulnerable populations 3.5. Land security in rural areas 3.6 Combating desertification and adapting to the effects of climate change Pillar 4: Strengthen governance of food and nutrition security	Yes, with clear targets and baseline values for the suggested indicators. The technical and administrative supervision of the PRP is led by the ministries linked to the rural sector (agriculture, water and sanitation, livestock and fishery resources, environment and climate change), ministries of health, education, and infrastructure.

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